Charm Industrial Bio-Oil Geologic Storage + Range & Plains Biochar Design Document

v1.11 – 2025/10/21

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Section A – Key Project Data

Title
Please provide the title of your Project. This will be displayed as your Project's name on the Isometric Registry and all related documentation.
Charm Industrial Great Plains Bio-Oil Sequestration

Description
Please provide a brief description (50-100 words) of your carbon removal Project. A more detailed written technical description must be provided later.
Charm produces bio-oil at our pyrolysis facility in Fort Lupton, CO from non-merchantable woody biomass sourced from government-funded wildfire mitigation/management projects. Charm procures pyrolysis bio-oil produced by AE Cote-Nord Canada Bioenergy Inc (AECN) from waste material produced by a nearby lumber mill. We are using a combination of rail and truck to transport the bio-oil to our pre-processing site in Kansas where it is treated to fit the local regulatory requirements for injection before being transported to one of two injection sites: Vaulted in Hutchinson, KS or Charm's own injection site in Evangeline Parish, LA.

Project Location(s)
● *Please submit at least one Address and/or specific geo-coordinates for the project.* ● *You may submit multiple Project locations – please specify what role each location plays in the Project.*
Locations of Pyrolysis: Charm Industrial, 2360 S Rollie Ave, Fort Lupton, CO 80621 40.076570, -104.804680 Lot 3, Thomas Subdivision, Fort Lupton, CO 80621 (ancillary biomass storage site for Charm pyrolysis) 40.043436, -104.800078 Produits Forestiers Arbec S.E.N.C., 175 Bd du Portage des Mousses, Port-Cartier, QC G0W 1X0, Canada 50.025323, -66.818910 Location of Bio Oil Pre-Processing: 838 SW Purity Springs Rd El Dorado, KS 67042

37.813906, -96.914982 Location of Injection: Underground Cavern Stabilization, 7513 KS-14, Hutchinson, KS 67501 37.966035, -97.947845 Location of Injection: Evangeline Parish, Louisiana 30.916251, -92.297908

Project Participants
Please provide a complete list of organizations participating in the project, with a contact person for each organization. (*Please duplicate the below rows for each additional organization you wish to add)*
Company registration number (*Unique business identification number in your country of registration*): 82-4423905
Organization Name: Charm Industrial
Organization Address: 2575 Marin Street San Francisco, CA 94124
Contact Person: Max Lavine
Contact Email Address: max@charmindustrial.com
Contact Phone Number: 800-778-7879
Organization role in Project: The carbon removal company who will fund and manage the offtake, transportation and treatment of the bio-oil and sell the CDR credits.

Company registration number (*Unique business identification number in your country of registration*): 92-2524153
Organization Name: Carbon Removal Co., Inc. dba Vaulted Deep (Vaulted)
Organization Address: 11000 Richmond Avenue, Suite 191, Houston, TX 77042
Contact Person: Julia Reichelstein
Contact Email Address: info@vaulteddeep.com
Contact Phone Number:
Organization role in Project: Injection well operator

Company registration number (*Unique business identification number in your country of registration*): 1168427111
Organization Name: AE Cote-Nord Canada Bioenergy Inc.
Organization Address: 210-8000 Boulevard Langelier Sant-Leonard, QC H1P3K2 Canada
Contact Person:
Contact Email Address: pct.rh@arbec.co
Contact Phone Number: 418-766-2299
Organization role in Project: Bio-Oil Vendor

Legal ownership of carbon removal claims
Please provide reasoning and evidence for legal ownership over the rights to all removals that will be claimed from the Project and refer to Section 3.1 "Ownership" of the Isometric Standard.
Charm Industrial is the only organization that will have ownership of the carbon removal claims resulting from this project. Our relationship with all other organizations involved in the project is limited to contracting for specific goods and/or services required to complete the project. Confirmation that AECN is unable to claim carbon credits for its production once sold to Charm is included in the documentation for this PDD.

Technical description of Project activity
Please provide a brief technical description of your carbon removal Project activity in accessible language. This should include information on facilities and equipment, the age and average lifespan of equipment, and all further information essential to understanding how carbon removal is achieved by the Project.
Charm Bio-Oil Charm produces bio-oil at its pyrolysis facility in Fort Lupton, CO. The facility has operated in an R+D capacity since December 2022 in order to develop Charm's internal pyrolysis technology. Production of bio-oil for sequestration and carbon removal credit generation began on 8/12/24, at which point the facility became shared-use Charm uses woody biomass sourced from government-funded wildfire mitigation and restoration projects in Colorado. Charm works with foresters to source the non-merchantable logs removed from the forest as a part of this mitigation work. As a result, woody biomass without a clear alternative disposal pathway is removed from fire-prone collection areas and pyrolyzed so that its biogenic carbon can be permanently sequestered as bio-oil.

AECN Bio-Oil
Charm offtakes finished bio-oil from AE Cote-Nord Canada Bioenergy Inc. (AECN), a Quebec manufacturer who operates a pyrolysis facility. Construction began in 2016 and the facility was commissioned in 2018. It is expected to operate through 2042 and produce approximately 871,200 metric tonnes of pyrolysis oil during that time. AECN's pyrolysis plant is located next to the Arbec lumber mill, which provides the pyrolysis feedstock in the form of waste sawdust and shavings from lumber processing.

The production of this waste is incidental to the production of wood boards for construction, which is the mill's primary source of revenue. Absent an offtaker, the historical fate of the mill residue has been on-site piling, as is evidenced by the fact that the mill has already accumulated a large pile of sawdust that is years old. Piled, rotting wood waste is a source of both CO2 and methane emissions, with a significant global warming potential. By sequestering AECN bio oil, Charm enables AECN to divert more of this ongoing waste stream away from creating emissions, instead permanently sequestering the biogenic gasses underground and interrupting the emissions associated with the input biomass.

Charm contracts with transport providers to deliver bio-oil to Charm's pre-processing facility in El Dorado, KS. At this facility, the oil is introduced to our material tanks and treatment system, which was constructed/commissioned in 2023-4. This site allows Charm to treat the bio-oil prior to injection in order to ensure that it is fully compliant with the requirements for underground injection.

Injection at Vaulted
Once successfully pre-treated to ensure UIC/KDHE compliance, the oil may be trucked to Vaulted's Great Plains Facility in Hutchinson, KS for injection into an underground salt cavern. Following treatment, samples are taken for CHN, TAN, and Density testing and the oil is transported to the Vaulted injection facility. At that point, the oil is emplaced into the storage cavern. Vaulted is also contracted to monitor the emplacement cavern and provide Charm with MRV data from their monitoring equipment. The details of Vaulted's injection infrastructure and monitoring equipment is detailed in their own PDD and recorded in the relevant sections below.

Injection at Basco 6
Once successfully pre-treated to ensure UIC/LDENR compliance, the oil may be trucked to Charm's Basco 6 injection well in Evangeline Parish, LA for injection into an orphaned oil and gas reservoir. The Bio-Oil may be stored in tanks prior to injection into the sub-surface or injected directly from tanker trucks. When a batch of Bio-Oil is ready for injection, Charm will inject a small amount of brine into the formation to ensure pressure and formation access is achieved. Once injection into the subsurface has been established with brine, Charm will switch to the injection of Bio-Oil. As per regulations, Charm will continuously monitor the injection rate and surface injection pressures at the wellhead. Automated shut-downs are in place to ensure compliance with regulation.

Bio-Oil will continue to be pumped through the injection pump until the tank(s) containing the batch are empty. After all pre-processed Bio-Oil is injected, Charm will switch back to the brine to flush all residual oil from the injection lines, pump, and wellbore to ensure all Bio-Oil is delivered into the formation. After this brine flush is complete, the wellbore will

be shut-in and pressure will be monitored in preparation for the next Bio-Oil injection. The ongoing monitoring program for this injection well is detailed in Supplemental Doc 3 and Appendix 6.

Declaration of exclusive registration	
Please confirm that your Project may only claim credits for activities that are exclusively registered with the Isometric Registry.	
X	I confirm the Project for which I aim to generate credits under the Isometric Standard is not registered with any other voluntary or compliance scheme.

Public funding
Please describe briefly whether your Project has received any public funding, e.g., grants or subsidies
This project has received no public funding

Estimated carbon removal capacity

Please give an estimate of the net carbon removal capacity of this project in the coming years (metric tonnes)

The carbon removal capacity of the project is estimated based on a number of factors. The gross carbon sequestered via conversion of the waste biomass to bio-oil is calculated based on repeated carbon testing of the oil in order to establish confidence in a consistent baseline. The emissions generated via production, transportation and injection are measured, calculated or estimated depending on the type of parameter and the variability or opportunities for improvement (e.g. transitioning transportation from fossil fuels to renewable energy sources).

The net carbon sequestered is then estimated and forecasted into future years based on known and estimated capacity and operating constraints (e.g. pyrolysis capacity, injection capacity). Additionally, these figures are based on the assumption that Charm will be injecting bio-oil produced by multiple providers, including Charm's own pyrolysis operations, as well as additional third-party producers. Additional information regarding these sources will be added to this PDD in the future.

Given the above, the following estimates represent Charm's overall forecast for injections at this site as our operations and supply chains expand. As a startup operating in an emerging market, there are necessarily many sources of uncertainty in these estimates, as operating parameters, feedstocks, transport methods, etc. will be both diverse and subject to change as operations expand and market conditions evolve. In order to ensure conservative forecasting, the totals for 2026-2028 are marked as TBD and will be updated in the future as third-party offtake volumes and Charm's own production levels are more defined.

Year	Estimated carbon removal capacity (metric tonnes)
2023	N/A
2024	10,000
2025	25,000
2026	TBD – 25,000 +
2027	TBD – 25,000 +
2028	TBD – 25,000 +

Section B – Protocol and Monitoring Data

Selected Certified Protocol	
Please select the Isometric Certified Protocol you wish to use for this Project. *Please note that, as per the Isometric Standard, you must use the latest available version of a Certified Protocol, unless a grace period has been explicitly specified by Isometric, whereby a former version of a Protocol may continue to be used for a defined time period.*	
Protocol Name: Bio Oil Geological Storage	
Protocol Version Number: v1.0	
X	I confirm that I am using the most recent available Certified Protocol version, or that a grace period has been explicitly specified allowing the use of an earlier protocol version.
Protocol Name: Biochar Production and Storage	
Protocol Version Number: v1.0	
X	I confirm that I am using the most recent available Certified Protocol version, or that a grace period has been explicitly specified allowing the use of an earlier protocol version.
Protocol Name: Biochar Storage in Agricultural Soils	
Protocol Version Number: v1.0	
X	I confirm that I am using the most recent available Certified Protocol version, or that a grace period has been explicitly specified allowing the use of an earlier protocol version.

Project Eligibility
Please explain why this Project is eligible under the selected Protocol.
1. Biomass feedstocks are eligible in accordance with the Biomass Feedstock Accounting module. Please see the Biomass Feedstock Information appendices for full details on each feedstock and demonstration of eligibility. 2. The project is injecting bio-oil produced from eligible feedstocks into geological formations (salt caverns and/or permeable reservoirs) for long-term storage using permitted underground injection wells 3. The project is additional – See the below sections demonstrating Additionality for further details.

Acknowledgement of responsibility for providing notification of changes to operations	
Please confirm that you acknowledge responsibility for notifying Isometric of changes to operations which deviate from this submitted PDD	
X	I acknowledge responsibility for notifying Isometric of any changes to operations

Project Boundary
Complete the below table detailing the Project boundary, including all GHGs considered across all Sources, Sinks and Reservoirs (SSRs) in both the Project and Baseline scenario. *Additionally, please give a description of the Project boundary. You may also optionally provide a diagram of the Project boundary.*
Description of Project boundary: All project activities undertaken to permanently sequester carbon are considered within the system boundary including: transportation of biomass feedstock to pyrolysis, the process of pyrolysis itself, transportation of injectate to and from pre-processing, the pre-processing itself, injection and monitoring activities. Embodied emissions associated with equipment procured or contracted by Charm for the purposes of project execution are included. The emissions resulting from the direct and induced use of energy (fuel, electricity) and the embodied emissions of non-waste consumables are included. *Charm Feedstock:* The biomass feedstock used by Charm also has no counterfactual use, so the replacement emissions are also 0. See Appendix 2 and 3 for details. The biomass feedstock consists of non-merchantable wood cleared from the forest as part of government-funded wildfire mitigation/fuels reduction projects. The harvesting of the biomass and transportation to the collection site is not contingent on the project, so upstream emissions associated with these processes are not included in the project boundary.

AECN Feedstock:
Any counterfactual use case of organic waste biomass, including any relevant replacement emissions, are also included – However, because the biomass feedstock used by AECN has no counterfactual use, the replacement emissions are 0. See Appendix 1 for details.

Because the biomass feedstock is a waste product (sawdust) produced by a lumber mill whose function and production of waste is not contingent on this project, the upstream emissions associated with growing, harvesting, transporting and producing the biomass are not included within the boundary.

Geographic Boundary:

Charm
Biomass is collected from the forestry project collection site(s) and transported to Charm's pyrolysis facility in Fort Lupton Colorado.
Injectate is transported from the Pyrolysis location by a combination of truck and rail to the pre-processing facility in El Dorado, KS. Once pre-processing is complete, injectate is transported to either the Vaulted injection site in Hutchinson, KS or Basco 6 injection site in Evangeline Parish, LA for emplacement.

AECN
Pyrolysis occurs in Port Cartier, QC. Injectate is transported from the Pyrolysis location by a combination of truck and rail to the pre-processing facility in El Dorado, KS. Once pre-processing is complete, injectate is transported to either the Vaulted injection site in Hutchinson, KS or Basco 6 injection site in Evangeline Parish, LA for emplacement.

[Optional] Project boundary diagram:

Baseline / Project	**Carbon Flux / Emission Source**	**Included/ excluded from project boundary?**	**Greenhouse Gas(es)**	**Justification/description**
Baseline	Counterfactual Emissions: Fuel Use from Feedstock Sourcing and Creation – AECN	Excluded	CO_2, CH_4, NO_2	There is no additional fuel usage on the part of the feedstock supplier due to project activities. The feedstock is a byproduct of commercial activities that would have been carried out absent the project, and are therefore

				considered as zero for the purposes of this document. Full details can be found in Appendix 1.
Baseline	Counterfactual Emissions: Fuel Use from Feedstock Sourcing and Creation – Charm	Excluded	CO_2, CH_4, NO_2	There is no additional fuel usage on the part of the feedstock supplier due to project activities. The feedstock is a byproduct of government-funded wildfire mitigation activities that would have been carried out absent the project. Full details can be found in Appendix 2 and 3.
Baseline	Counterfactual Emissions: Electricity Use from Feedstock Sourcing and Creation – AECN	Excluded	CO_2, CH_4, NO_2	There is no additional electricity usage on the part of the feedstock supplier due to project activities. The feedstock is a byproduct of commercial activities that would have been carried out absent the project, and are therefore considered as zero for the purposes of this document. Full details can be found in Appendix 1.
Baseline	Counterfactual Emissions: Electricity Use from Feedstock Sourcing and Creation – Charm	Excluded	CO_2, CH_4, NO_2	There is no additional electricity usage on the part of the feedstock supplier due to project activities. The feedstock is a byproduct of government-funded wildfire mitigation activities that would have been carried out absent the project. Full details can be found in Appendix 2 and 3.
Baseline	Counterfactual Emissions: Replacement of Feedstock Function – AECN	Excluded	CO_2, CH_4, NO_2	Feedstock is a waste product from commercial activities that would have been carried out absent the project. The supplier developed pyrolysis capacity as a solution to excessive residue accumulation. Therefore replacement emissions are considered 0 based on Biomass Feedstock Sourcing criterion C1: The feedstock currently serves no purpose. Full details can be found in Appendix 1.
Baseline	Counterfactual Emissions:	Excluded	CO_2, CH_4, NO_2	Feedstock is a non-merchantable product from government-funded

	Replacement of Feedstock Function – Charm			wildfire mitigation activities that would have been carried out absent the project. There are no clear alternative pathways to utilize this material. Therefore replacement emissions are considered 0 based on Biomass Feedstock Sourcing criterion C1: The feedstock currently serves no purpose. Full details can be found in Appendix 2 and 3.
Baseline	Counterfactual Emissions: Temporary Carbon Storage from Feedstock – AECN	Included	CO_2, CH_4	The Arbec sawmill accumulated a large pile of sawdust and wood shavings as a result of the shutdown of local businesses who had purchased these residues in the past. AECN was developed as a solution to this issue of waste accumulation. Research on sawdust piles (Pier and Kelly, 1997) has shown that due to anaerobic conditions within the pile, the total global warming potential of pile emissions within 15 years would be greater than the emissions from a total release of biogenic carbon due to the production of methane. Charm is only calculating removal credits in terms of the release of biogenic carbon. Full details can be found in Appendix 1.
Baseline	Counterfactual Emissions: Temporary Carbon Storage from Feedstock – Charm	Included	CO_2, CH_4	The biomass feedstock is sourced through government-funded wildfire mitigation activities. As such, it has a counterfactual storage value of 0 per EC14 in the Isometric Biomass Feedstock Accounting module. Full details can be found in Appendix 2 and 3.
Project	Feedstock Sustainable Sourcing – AECN	Included	N/A	Feedstock for the project is harvested by Groupe Remabec from provincially-owned forests under rights granted by the Province of Quebec as part of the Provincial Management Plan for

				forests. In addition, Groupe Remabec's harvest activities have been granted SFI certification by the Bureau de Normalisation du Québec.
Project	Feedstock Sustainable Sourcing – Charm	Included	N/A	Feedstock for the project is harvested under the auspices of the Colorado State Forest Service's wildfire management program.
Project	Fuel Use from Feedstock Transport – AECN	Included	CO_2, CH_4, NO_2	Fuel use from feedstock transport is provided as part of bio oil vendor LCA and included in their calculation of CO2e/ton of bio oil produced.
Project	Fuel Use from Feedstock Transport – Charm	Included	CO_2, CH_4, NO_2	Transport emissions are calculated using the Distance Based Method outlined in the Isometric Transportation Emissions Accounting module, using scale tickets to document the mass transported. Additional fuel used in feedstock handling is calculated per ton of feedstock based on an average fuel consumption factor for lumber handling
Project	Embodied Emissions from Feedstock Transport–Charm	Included	CO_2, CH_4, NO_2	Per-mile embodied emissions are calculated based on GREET emissions levels for the manufacture of medium-and-heavy-duty trucks and trailers and divided by EPA expected useful life in mileage for heavy-duty diesel engines. This calculation yields a per-mile embodied emissions level that can be applied to Charm-specific journeys using vehicles owned by the vendor
Project	Fuel Use from Biomass Pyrolysis – AECN	Included	CO_2, CH_4, NO_2	Fuel use from feedstock pyrolysis is provided as part of bio oil vendor LCA and included in their calculation of CO2e/tonne of bio oil produced
Project	Fuel Use from	Included	CO_2, CH_4,	Emissions impact of fuel usage is

	Biomass Pyrolysis – Charm		NO_2	calculated based on GLEC emission factors for fuel combustion in terms of TCO2e/kg fuel multiplied by the quantity of fuel purchased for production on site as documented by purchase invoices. This quantity is adjusted for fuel pump uncertainty as detailed in the Uncertainty Analysis
Project	Electricity Use from Biomass Pyrolysis – AECN	Included	CO_2, CH_4, NO_2	Electricity use from feedstock pyrolysis is provided as part of bio oil vendor LCA and included in their calculation of CO2e/ton of bio oil produced
Project	Electricity Use from Biomass Pyrolysis – Charm	Included	CO_2, CH_4, NO_2	Electricity emissions are calculated by kWh usage multiplied by the per kWh emissions for the RMPA grid subregion as reported by the EPA eGrid emissions factors, modified by Green-e to show residual grid mix. This is supplemented by a per-kWh addition to account for the embodied emissions associated with grid infrastructure as quantified by a weighted average of power generation sources based on the fuel mix for the subregion reported by the EPA and the applicable electricity infrastructure emissions factors reported by GREET. This quantity is adjusted for meter uncertainty as detailed in the Uncertainty Analysis.
Project	Stack Emissions from Biomass Pyrolysis – AECN	Included	CH_4, NO_2	AECN has not directly analyzed the emissions from their process. Pyrolysis stack emissions are routed to heat the biomass dryer, which is the sole source external system exhaust. In lieu of a direct analysis, Charm is using an analysis of the same technology pyrolyzing woody biomass as reported by their LCA software provider (GHGenius) to the Canadian Government to quantify

				emissions CO2e.
Project	Stack Emissions from Biomass Pyrolysis – Charm	Included	CH_4, NO_2	For each type of feedstock processed, Charm has a tail gas composition testing performed by a qualified practitioner. This data is used by Charm's engineering team to calculate a quantity of tailgas emitted per kg of biomass processed. This quantity is then scaled based on the biomass throughput recorded for production.
Project	Embodied Emissions from Facilities and for Equipment Biomass Pyrolysis – AECN	Included	CO_2, CH_4, NO_2	The bio oil vendor has not performed an evaluation of the embodied emissions associated with the construction of their building or manufacture of their equipment. Therefore, the best option available to Charm for determining these impacts is to use a cost-based calculation based on costs incurred by AECN. This is achieved by applying per-dollar-spent supply chain emissions factors to the cost of construction and equipment and dividing that total by the expected lifetime production of the facility to arrive at a quantity of CO2e per MT of oil produced associated with facility embodied emissions.
Project	Embodied Emissions from Facilities and for Equipment Biomass Pyrolysis – Charm	Included	CO_2, CH_4, NO_2	Charm has compiled and periodically updates an inventory of equipment and facilities improvements undertaken in order to support production activities at the pyrolysis facility. The embodied emissions of each line item in the inventory are quantified using LCA based on emission factors from reputable sources such as GREET database, the EcoInvent database, NAICS supply Chain Emission Factors, USLCI database, ICE database, and/or the Athena Impact Estimator. These values are

				amortized over the expected useful life of the equipment and improvements used, which is estimated in terms of total expected facility biomass throughput
Project	Fuel Use from Injectate Transport to Pre-Processing	Included	CO_2, CH_4, NO_2	Emissions impact of transport calculated based on GLEC emission factors using cargo weight, vehicle type, and distance traveled with a correction for truck scale uncertainty detailed in the Uncertainty Analysis
Project	Embodied Emissions from Injectate Transport to Pre-Processing	Included	CO_2, CH_4, NO_2	Per-mile embodied emissions are calculated based on GREET emissions data for the manufacture of medium-and-heavy-duty trucks and trailers and divided by EPA expected useful life in mileage for heavy-duty diesel engines. This calculation yields a per-mile embodied emissions level that can be applied to Charm-specific journeys using vehicles owned by a transport contractor. Because GREET does not provide comparable emission factors for the manufacturer of railcars, a cost-based calculation is used based on the average cost and expected useful life of a DOT-111 railcar
Project	Electricity Use from Injectate Pre-Processing	Included	CO_2, CH_4, NO_2	Electricity emissions are calculated by kWh usage multiplied by the per kWh emissions for the SPP North grid subregion as reported by the EPA eGrid emissions factors, modified by Green-e to show residual grid mix. This is supplemented by a per-kWh addition to account for the embodied emissions associated with grid infrastructure as quantified by a weighted average of power generation sources based on the fuel mix for the subregion reported by the EPA and the

				applicable electricity infrastructure emissions factors reported by GREET
Project	Fuel Use from Injectate Pre-Processing	Included	CO_2, CH_4, NO_2	Emissions impact of fuel usage is calculated based on GLEC emission factors for fuel combustion in terms of TCO2e/kg fuel multiplied by the quantity of fuel purchased for use on site as documented by purchase invoices. This quantity is corrected for fuel pump uncertainty as detailed in the Uncertainty Analysis
Project	Embodied Emissions from Facilities and Equipment for Injectate Pre-Processing	Included	CO_2, CH_4, NO_2	Charm has compiled and periodically updates an inventory of equipment and facilities improvements undertaken in order to support production activities at the pyrolysis facility. The embodied emissions of each line item in the inventory are quantified using LCA based on emission factors from reputable sources such as GREET database, the EcoInvent database, NAICS supply Chain Emission Factors, USLCI database, ICE database, and/or the Athena Impact Estimator. These values are amortized over the expected useful life if the equipment and improvements used.
Project	Embodied Emissions from Consumables for Injectate Pre-Processing	Included	CO_2, CH_4, NO_2	When liquid caustic soda (LCS) is added to the bio oil, the volume is measured by on-site staff using a flow-meter. The rate of LCS addition is measured for each injection batch and multiplied by the GREET emissions factor for LCS to calculate the associated embodied emissions. The salt added to the oil is not included because it is an incidental waste-product (off-spec/non-saleable salt) from a salt production facility

Project	Fuel Use from Consumables Transport to Pre-Processing Site	Included	CO_2, CH_4, NO_2	Emissions impact of transport calculated based on GLEC models using cargo weight associated with material delivered to the site, vehicle type, and distance traveled
Project	Embodied Emissions from Consumables Transport to Pre-Processing Site	Included	CO_2, CH_4, NO_2	Per-mile embodied emissions are calculated based on GREET emissions data for the manufacture of medium-and-heavy-duty trucks and trailers and divided by EPA expected useful life in mileage for heavy-duty diesel engines. This calculation yields a per-mile embodied emissions level that can be applied to Charm-specific journeys using vehicles owned by a transport contractor.
Project	Fuel Use from Processed Injectate Transport to Injection Site	Included	CO_2, CH_4, NO_2	Emissions impact of transport calculated based on GLEC models using cargo weight, vehicle type, and distance traveled
Project	Embodied Emissions from Injectate Transport to Pre-Processing	Included	CO_2, CH_4, NO_2	Per-mile embodied emissions are calculated based on GREET emissions data for the manufacture of medium-and-heavy-duty trucks and trailers and divided by EPA expected useful life in mileage for heavy-duty diesel engines. This calculation yields a per-mile embodied emissions level that can be applied to Charm-specific journeys using vehicles owned by a transport contractor.
Project	Electricity Use from Injection (Vaulted)	Included	CO_2, CH_4, NO_2	The Well Operator is required to provide a report on energy usage specific to Charm emplacements. Emissions impact of electricity usage is calculated based on the per kWh emissions for the SPP North grid subregion as reported by the EPA eGrid emissions factors. This is supplemented by a per-kWh addition to account for the emissions associated with grid infrastructure as quantified by a weighted average of power

				generation sources based on the fuel mix for the subregion reported by the EPA and the associated electricity infrastructure emissions factors reported by GREET
	Electricity Use from Injection (Charm)			See the Combined Emissions Monitoring and Accounting Plan for the site, included in Supplemental Doc 3
Project	Fuel Use from Injection (Vaulted)	Included	CO_2, CH_4, NO_2	The Well Operator is required to provide a monthly report on energy usage specific to Charm emplacements. Emissions impact of fuel usage is calculated based on GLEC models for fuel combustion in terms of TCO2e/kg fuel multiplied by the quantity of fuel reported. This quantity is corrected for fuel pump uncertainty as detailed in the Uncertainty Analysis
	Fuel Use from Injection (Charm)			See the Combined Emissions Monitoring and Accounting Plan for the site, included in Supplemental Doc 3
Project	Embodied Emissions from Injection Equipment (Vaulted)	Included	CO_2, CH_4, NO_2	In the completion of their own PDD, Vaulted, the well operator for this project, reported the embodied emissions associated with the injection equipment on their site divided by the estimated well capacity to arrive at a quantity of embodied emissions per emplaced tonne of material. This value is deducted from the net CDR associated with each tonne of bio oil emplaced by Charm.
	Embodied Emissions from Injection Equipment (Charm)			See the Combined Emissions Monitoring and Accounting Plan for the site, included in Supplemental Doc 3
	Water use from Injection (Charm)			See the Combined Emissions Monitoring and Accounting Plan for the site, included in

				Supplemental Doc 3
	Fuel Use from Consumables Transport to Injection Site (Charm)			See the Combined Emissions Monitoring and Accounting Plan for the site, included in Supplemental Doc 3
	Embodied Emissions from Consumables Transport to Injection Site (Charm)			See the Combined Emissions Monitoring and Accounting Plan for the site, included in Supplemental Doc 3
Project	Carbon Content of Injected Bio Oil (Vaulted)	Included	CO_2	Samples are taken from each injection batch prior to the addition of salt and liquid caustic soda to modify injectate salinity and pH. These samples are sent to an analytical laboratory to quantify the carbon content of that batch. The mass of consumables added to each batch is monitored by the site operators. The Total Injectate Mass is measured using a calibrated truck scale at the well site and the Mass of Bio Oil Injected is computed by subtracting the mass of consumables added from the Total Injectate Mass. The Mass of Bio Oil Injected is multiplied by the % C content to compute the gross C sequestered by an injection. See the Monitoring Plan for additional details.
	Carbon Content of Injected Bio Oil (Charm)			See the Combined Emissions Monitoring and Accounting Plan for the site, included in Supplemental Doc 3
Project	Transport Emissions from Transport of Test Samples to Lab	Included	CO_2, CH_4, NO_2	The number of tests, samples, and sample weights are internally standardized for each injection batch and calculated as reported by

				the Charm Chemistry and Research teams. Emissions are calculated for air shipment of samples based on a conservative estimate of the point-to-point distance between the sampling site and labs in Google maps. Weight and distance are used to calculate emissions using the GLEC emissions factor for air freight.

Baseline scenario
Please describe the baseline scenario of what would have happened if your Project did not take place (refer to Section 2.5.2 "Baselines" of the Isometric Standard and the requirements outlined in the relevant Protocol). Projects will only be credited for Removals above this counterfactual baseline.
Charm Bio-Oil The biomass Charm sources are forest residues (green logs) harvested from government regulated and funded forest management projects with the goal of wildfire mitigation, amongst other co-benefits. Given the region's history of recent, catastrophic wildfires and the State of Colorado's prioritization of mitigation of wildfire hazards, this project would have taken place irrespective of Charm's involvement According to the contractor and the forester from the CSFS, the forest residues that Charm sources from this project are not suitable for use in sawmills and have very limited disposal options in the area. This can be partly due to the quality, type of tree species, and inability to access disposal options due to the lack of economic attractiveness. The forester in charge of this project has provided an affidavit acknowledging the lack of merchantability for this material in the fire-prone ecosystem. As such, the biomass is considered to 1) serve no purpose and 2) have 0 counterfactual storage per Condition C1 in Section 3.2.2 and EC 14 in Section 2.2 of the Isometric Biomass Feedstock Accounting Module v1.2. A complete characterization of the feedstock and assessment of the GHG storage counterfactual can be found in the Biomass Feedstock Information appendix 2 and 3 below. Charm's bio-oil production facility is located in an area of Fort Lupton, CO that is zoned I2 for heavy industrial activity. The immediate area hosts other, comparable heavy industrial applications including a large Halliburton location, a crane rental company (Maxim Crane Works), and a manufacturer of truck and van bodies (Summit Truck & Van Bodies). Given the uses of neighboring properties, it is reasonable to presume that Charm's location would be used for comparable commercial activity. For this reason, only facility development work and equipment installation specific to Charm's production operations are considered above the baseline for the property. Emissions impacts of operational activities are captured in the accounting of aggregate fuel usage and equipment embodied emissions for pyrolysis

operations.

AECN Bio-Oil
The bio-oil Charm sources from AECN is produced using residues, primarily sawdust, generated by the Arbec lumber mill adjacent to AECN's pyrolysis facility. By purchasing the oil, Charm creates demand that AECN meets by offtaking additional residues from the mill. AECN is jointly owned by the sawmill, Arbec, and wood harvester, Groupe Remabec.The pyrolysis facility was developed as a solution to the ongoing buildup of residues.

This development was, in part, triggered by the closure of other businesses in the region who would have otherwise purchased this byproduct as an input for their own production. Due to the remote location of the mill, the development of economically viable offtake relationships with businesses outside the region is a challenge. This is evidenced by Arbec and Groupe Remabec's decision to invest over $100 million CAD to build out an onsite pyrolysis facility, as well as the accumulation of a large pile of sawdust as sawmill operations continued without an offtaker for residues for a significant period of time. The AECN facility was the solution to this problem.

Because its primary revenue stream is the production of wood boards for construction, the mill has operated, creating sawdust and wood shavings absent AECN or other offtakers. There is clear evidence that the historical practice given this condition has been to create the sawdust pile that still exists onsite, so it is reasonable to conclude that these waste products would continue to accumulate in this fashion given no other intervention.

In such a scenario, existing literature suggests that the anaerobic conditions in the center of a pile of sawmill residues would result in the emission of methane, alongside CO2. The high global warming potential (GWP) of methane would likely result in total emissions with a higher GWP over a 15-year period than if all biogenic carbon were released (Pier and Kelly, 1997). Charm's sequestration of the oil made by pyrolyzing this feedstock removes what would otherwise act as the source material for these emissions and permanently sequesters it underground after processing into pyrolysis oil.

A complete characterization of the feedstock and assessment of the GHG storage counterfactual can be found in the Biomass Feedstock Information appendix below.

Pre-Processing and Injection
The project also involves pre-processing of bio-oil at a rented site at AJ's Services in El Dorado, KS and injection of the processed oil into either a salt cavern managed by Vaulted Deep in Hutchinson, KS or Charm's injection site in Evangeline Parish, LA.

Both AJ's and Vaulted are fully operational commercial sites whose footprint and use are not significantly changed by project activities. AJ's is an active oil field services company primarily servicing the local oil and natural gas industry. Project activities have no significant effect on their operations. In order to ensure a conservative estimate of removals, the embodied emissions of all project equipment and work on the site, as well as emissions associated with consumables and energy consumption, is accounted for and amortized against net removals associated with oil processed by Charm at the facility.

Vaulted uses its salt caverns and associated infrastructure for underground waste emplacement irrespective of its relationship with Charm and participation in this project, including their own sequestration work certified under the Isometric Protocol. Charm accounts for all project-specific emissions as a result of emplacement at Vaulted, including a fraction of their equipment embodied emissions in proportion with the use of the equipment for project emplacements.

The Basco 6 EW Well is a former saltwater disposal well located in Evangeline Parish, LA. It was drilled in 2011 and, while active, injected nearly 20,000 bbls of produced water before transitioning to orphan well status in 2016. Despite efforts to plug orphaned wells in Louisiana, their number has continued to grow. Between October 2023 and April 2024, the LDENR recorded a 12.4% increase in the number of orphaned wells statewide, with a total count of 4,785 wells. Given this state of affairs, the likely fate of the Basco 6 EW Well absent remediation by Charm to support bio-oil storage would be continued deterioration as an unplugged, orphaned oil and gas asset. Left in such a condition, these wells pose the risk of long-term environmental and economic harm to the surrounding ecosystems and communities.

Details regarding the accounting discussed above can be found in the GHG Statement Appendix 4 and, for the Basco 6 injection well, Supplemental Doc 3.

Leakage Assessment

Please give a robust assessment of how you have considered potential increases in GHG emissions outside the defined Project boundary that occur as a result of the Project activity. Where the potential for such Leakage is identified, it must be quantified and deducted from the CO_2 Removals in accordance with the relevant Protocol. Please refer to Section 2.5.4 "Leakage" of the Isometric Standard.

Charm accounts for all potential leakage across all stages of the project. Based on these assessments we have concluded that there is no identifiable leakage related to this project. In general, the highest risk for leakage in bio-oil sequestration projects comes from replacement or land-use change stemming from the feedstock.

Charm
The feedstock supplier, Altitude Forestry, has been contracted to treat the areas and harvest wood used by Charm for wildfire mitigation projects funded by the Colorado State Forest Service (CSFS). An official project description from the CSFS, and the bid provided by the contractor to the State Forest Service are provided as supporting documentation for this work. As a result, offtake of these products does not present a leakage risk as defined by the Isometric Biomass Feedstock Accounting module.

AECN
This feedstock is a byproduct from a sawmill whose primary source of revenue is the production of boards which creates residues–sawdust and shavings–that are used in the pyrolysis process. In addition, all wood processed by the mill is harvested from Provincially-owned lands under the Provincial Management Plan for Quebec forests. As a result, offtake of these products does not present a leakage risk as defined by the Isometric

Biomass Feedstock Accounting protocol.

Demonstration of Additionality: Financial Additionality
Please describe and provide evidence for how your Project is financially additional. Refer to Section 2.5.3. "Additionality" of the Isometric Standard and the requirements outlined in the relevant Protol.
In order to occur, this project must generate revenue in order to recoup the costs incurred by its activities. Charm may be contracted to sell the biochar that is produced as a co-product of its bio-oil pyrolysis to a third party, or use the char to generate additional credits under a different protocol. However, the revenue those activities will generate is far below the cost to produce the associated bio-oil. In addition, the sequestration of third-party produced bio oil will only generate revenue through the sale of carbon credits. Absent the ability to sell these credits, none of the project activities would be financially viable. The project does not qualify for a 45Q tax credit. It therefore qualifies as financially additional as defined by Section 2.5.3 of the Isometric Standard.

Demonstration of Additionality: Environmental Additionality
Please describe and provide evidence for how your Project is environmentally additional. Refer to Section 2.5.3. "Additionality" of the Isometric Standard and the requirements outlined in the relevant Protocol.
This project sequesters carbon-rich bio oil produced from a feedstock that would have otherwise produced emissions equivalent to at least the GWP attributable to 100% of its biogenic carbon content. As a result of this activity, the project results in a net removal of atmospheric carbon when considered against the counterfactual scenario and given all related project emissions. It therefore qualifies as environmentally additional as defined by Section 2.5.3 of the Isometric Standard.

Demonstration of Additionality: Regulatory Additionality
Please describe and provide evidence for how your Project is regulatorily additional. Refer to Section 2.5.3. "Additionality" of the Isometric Standard and the requirements outlined in the relevant Protocol.
This project is fully compliant with KDHE Class V UIC regulations, however it is not required by any government regulations. The purpose of the project is to sequester carbon that would have otherwise been returned to the atmosphere in order to generate credits that can be sold to participants in voluntary carbon markets. It therefore qualifies as regulatorily additional as defined by Section 2.5.3 of the Isometric Standard.

Durability Assessment
Please provide justification for how the Project adheres to the durability requirements outlined in the selected Protocol, which may include references to published literature or internal research. You may further expand on the Monitoring approach used to support the claimed Durability assessment in the "Overview of monitoring for durability" section of the next section.
All bio oil injection resulting from this project will occur at the Vaulted Great Plains Facility in Hutchinson, KS or Charm's Basco 6 injection well in Evangeline Parish, LA. Vaulted has completed their own PDD under the Isometric Biomass Geological Storage Protocol and their durability assessment is copied below: "Vaulted's Great Plains Facility stores the waste in sealed salt caverns 500+ feet underground for permanent (10,000+ year) geologic storage. Neither leakage nor re-emissions of the waste and its carbon content is expected. Durability is expected to exceed much further than 10,000 years. This expectation is based on a combination of direct measurement, and modeling. Vaulted employs a monitoring program both during and post operation which includes regular testing for mechanical integrity of the wells and cavern integrity. This monitoring program enables the confirmation that the cavern is stable and does not have any subsurface leak pathways. This testing is done when a new cavern is opened, during pre-injection, injection, and post-injection operations. Vaulted monitors the displaced brine returned from the cavern for any waste, which is then filtered out and recycled back into the injection stream. The Great Plains Facility employs slurry injection into salt caverns. Salt is an impermeable formation, thus making it an effective method for securely storing waste materials without risk of leakage into surrounding environments. The caverns at the facility exist at depths where waste is no longer buoyant; if the wells have integrity (confirmed using the same techniques as described above) and are property plugged, waste is permanently sequestered. Safe and durable sequestration of waste in salt caverns has been confirmed in literature, including here, here, and here. To further reduce any risks, Vaulted injects at minimal pressure (80 psig at the wellhead, which maintains total pressure – injection plus hydrostatic – below 0.75 psi / ft to the cavern top). This pressure threshold maintains cavern integrity (the triaxial stress capacity of salt is 1 psi/ft of depth). In addition to direct measurement, the broader literature models durability of Vaulted's sequestration approach. Salt caverns have been identified in the literature as a viable location for permanent storage of CO2." Charm's durability assessment of Basco 6 is as follows: The Lower Miocene sandstone formation ~3500' below the Basco 6 wellhead has been characterized and assessed for the performance required to ensure durable storage of injected bio-oil. The primary considerations in assessing durability were the sequestration reservoir thickness and reservoir quality relative to adjacent confining layers, demonstration of wellbore integrity, and lack of offset activity in the sequestration zone in the region.

In the course of this assessment Charm determined that the pressure required to pump bio-oil into the formation would be consistent with a typical injection formation (normally pressured or below) without exceeding maximum allowable injection pressure, and that the formation brine density is sufficiently lower than that of bio-oil. This ensures that Charm can pump bio-oil under the regulatory requirements and that upon entry into the formation, the bio-oil will sink as predicted. In addition, Charm performed a mechanical integrity test to ensure casing integrity and performed an advanced cement analysis to ensure hydraulic seal from the formation and the wellbore. The wellbore is proven to be in good mechanical condition and the formation is of sufficient injectivity and storage size.

Given the above, we are confident in our assessment that the Lower Miocene zone penetrated by the Basco 6 EW possesses the characteristics required for bio-oil storage on a 1000+ year timeline. In addition, measures taken to monitor the durable storage of bio-oil are detailed in the Permanence Monitoring and Permit Compliance Plan in Appendix 6

Monitoring Plan

Please specify whether this Project will follow the default Monitoring Plan, following the requirements specified in the "Monitoring Plan Requirements" appendix of the relevant Protocol and its associated modules (Option A), or whether a Monitoring Plan will be submitted which may contain additional or modified monitored parameters (Option B).

Additionally, please give an overview of how monitored parameters and/or models will be used to support the assessment of durability. You may also provide any additional information here about how Project monitoring will be conducted.

	A – I confirm this Project will follow the default Monitoring Plan, following the requirements specified in the "Monitoring Plan Requirements" appendix of the relevant Protocol and its associated modules
X	B – I have submitted a Monitoring Plan, which meets the requirements specified in the "Monitoring Plan Requirements" appendix of the relevant Protocol and its associated modules, and which describes any additional or modified parameters monitored which deviate from the default Monitoring Plan

Overview of how monitored parameters or models will be used to support assessment of durability. If Option B was selected, please give details of how the monitoring plan adds to or differs from the default monitoring requirements outlined in the Protocol:

Monitoring Plan parameters relevant to the Bio-Oil Geologic Storage module:

1. **Biomass Feedstock Transport [Charm Bio-oil]**
    a. Transport Emissions are calculated in terms of ton-miles as detailed in the GHG Statement
        i. Mass is monitored by Scale Tickets provided by a certified truck scale for each delivery

            1. The gross weight is subtracted from the tare weight of each the delivery vehicle, which is established by taking the empty weight of the vehicle using a certified truck scale
        ii. For Truck Transport, the origin and all destinations are entered into Google Maps to obtain a distance value. When multiple routes are presented, the longest route is assumed to ensure a conservative estimate of removals
        iii. Ton-miles are multiplied by the appropriate GLEC emission factor based on vehicle type used (flatbed truck)
    b. In situations where the transportation vehicle has an integrated grapple that is also used for loading and unloading(such as with self-loading trucks), the additional emissions are accounted for using a standard emission factor for lumber handling scaled by lumber mass
    c. Depending on operating circumstances, biomass may be transported either directly to the pyrolysis facility, or to a nearby storage site. Biomass at the storage site is chipped and transported to the pyrolysis facility. Chipping and transport use Charm equipment that uses fuel that is invoiced to Charm. The monitoring of energy inputs, specifically fuel use, outlined below extends to the fuel used by this equipment.
    d. Biomass transport emissions may be allocated to bio-oil based on the quantity of raw biomass consumed during a given period, or as it is delivered – If allocated, usage rates are monitored by measuring the linear feet of raw logs before and after processing. The appropriate proportion of total transport emissions will then be allocated to the oil produced during that period
    e. Biomass may be shared between Production and R+D, which is outside of the system boundary. Usage will be allocated appropriately as evidenced by pyrolyzer up-time
2. **Pyrolysis Emissions**
    a. Process Emissions [AECN]
        i. Energy Inputs (Fuel, Electricity – Feedstock Transport Included)
            1. Our pyrolysis oil vendor has provided an LCA summarizing the emissions associated with energy inputs, including feedstock transport, consumed by their pyrolysis process.
                a. The initial LCA was based on 3 months of operations (Apr-June) in 2023. The vendor is expected to update the LCA at least annually, or in light of major operational changes that would be presumed to increase the carbon intensity of production, whichever comes first
                b. This is monitored by ongoing communication with the vendor
    b. Process Emissions [Charm Bio-oil]
        i. Energy Inputs and consumables
            1. Charm will provide documentation summarizing the emissions associated with energy inputs consumed by its biomass processing and pyrolysis process with each verification. Emissions sources and documentation will include:

a. Diesel fuel
    i. Vendor invoices
b. Propane
    i. Vendor invoices
c. Electricity usage
    i. Total facility usage reported by power utility less estimated G+A usage calculated based on historical usage rates
d. Gasoline
    i. Vendor invoices/transaction receipts
e. Water
    i. Water usage is monitored using a flow meter

ii. Where consumables are shared between Production and other functions outside of the system boundary, usage will be allocated appropriately as evidenced by pyrolyzer up-time

iii. Note – Monitoring processes will be progressively automated as Charm operations develop. Relevant changes will be reflected in GHG Statements and PDD updates as appropriate

c. Tailgas [AECN]
    i. The vendor has not analyzed stack emissions from their pyrolysis process. In absence of direct data, we were provided a modeling of process emissions from the technology in use (Ensyn RTP) processing woody biomass prepared for Natural Resources Canada detailing the addition of pyrolysis oil pathways to GHGenius software.
        1. The vendor routes all non-condensable gasses to their biomass dryer. Because some portion of the gas is converted to energy at this step, it is likely that actual emissions are below the estimate provided. However, Charm assumes the full quantity of estimated emissions in order to ensure a conservative estimate of removals.
d. Tailgas [Charm Bio-oil]
    i. For each type of feedstock processed, Charm has a tailgas composition testing performed by a qualified practitioner. This data is used by Charm's engineering team to calculate a quantity of tailgas emitted per kg of biomass processed. This quantity is then scaled based on the biomass throughput recorded for production.

ii. Production throughput is logged by operators during the course of operations

iii. Stack emissions testing will be updated for each type of feedstock processed as required by Section 7.4.3.5 of the Bio Oil Geological Storage module

3. Bio Oil Transport Emissions

a. Bio Oil Transport from Production to Pre-Processing
    i. Transport Emissions are calculated in terms of ton-miles as detailed in the GHG Statement section
        1. Mass may be monitored by Bill of Lading, Scale Ticket and Invoice documents provided by the bio-oil producer, or by

weighing each container of bio-oil and providing a list of containers transported.
        a. Values from a truck scale are measured by weighing the offtake truck prior to and after loading on a certified scale
    2. Distance is monitored differently for Truck and Rail transport
        a. For Truck Transport, the origin and destination are entered into Google Maps to obtain a distance value. When multiple routes are presented, the longest route is assumed to ensure a conservative estimate of removals
        b. For Rail Transport, the specific route distance is provided by the transport provider
    3. Ton-miles are multiplied by the appropriate GLEC emission factor based on vehicle type used (tanker truck, railcar)

b. Injectate Transport from Pre-Processing to Injection (Vaulted)
    i. Transport Emissions are calculated in terms of ton-miles as detailed in the GHG Statement section
        1. Mass is monitored by Scale Tickets provided by the injection well personnel.
            a. These values are measured by weighing the delivery truck prior to and after injection on a certified scale
        2. To monitor distance, the origin and destination are entered into Google Maps to obtain a distance value. When multiple routes are presented, the longest route is assumed to ensure a conservative estimate of removals
    ii. Ton-miles are multiplied by the appropriate GLEC emission factor based on vehicle type used (tanker truck)

c. Injectate Transport from Pre-Processing to Injection (Charm)
    i. Please see the Combined Emissions Accounting and Monitoring Plan in the Basco 6 Supplementary Document for details

4. Pre Injection Processing

a. Process Emissions
    i. Electricity Use
        1. Electricity use will be monitored using utility invoices
    ii. Fuel Use (Diesel)
        1. All diesel is delivered into a single reservoir on site that is used by both power generators and on-site mobile equipment
        2. The quantity of diesel delivered is monitored by the invoices the vendor provides
    iii. Liquid Caustic Soda (LCS) Use
        1. LCS Use is monitored by onsite staff using a flow meter
    iv. Consumables Deliveries
        1. Transport Emissions are calculated in terms of ton-miles as detailed in the GHG Statement section
            a. The mass of product delivered is monitored by invoiced quantities, which are either measured in mass directly, or in volume which is converted to mass based on product density.

b. Distance of the product journey is monitored based on the supplier location. Given this information, the origin and destination are entered into Google Maps to obtain a distance value. When multiple routes are presented, the longest route is assumed to ensure a conservative estimate of removals

2. Ton-miles are multiplied by the appropriate GLEC emission factor based on vehicle type used (flatbed truck, tanker truck)

v. Thermal Oxidizer Emissions

1. Based on manufacturer specifications and the expert opinion of both internal and external engineers, at the current maximum production levels the site can accommodate, thermal oxidizer exhaust levels are
   a. Far below the levels that would require an air permit
   b. De minimis as a potential source of GHG emissions
2. This parameter is monitored indirectly in terms of site production capacity. Currently, the maximum site production level is gated by sparging tank availability with a maximum processing capacity of approximately 1 tank of bio-oil every 24 hours.
   a. Site changes that increase this capacity, specifically the installation of additional sparging tank(s), will trigger additional measurement of thermal oxidizer exhaust in order to determine air permitting requirements and GHG emissions impact

vi. Other Parameters

1. The addition of salt to the bio-oil is monitored by site operators using a digital scale. The addition of salt does not contribute to process emissions because the salt is a waste product and the transport emissions associated with its use are already accounted for by the procedure outlined in the Consumables Deliveries section of this plan. However, the quantity is monitored in order to ensure 1) adequate quantity is added to achieve required injectate salinity, and 2) as one of the mass measurements required to compute the Mass of Bio-Oil Injected based on the Injectate Mass

5. Injection (Vaulted)

a. Process Emissions

i. Energy Inputs

1. Electricity is the only energy input used during the injection process. The site office and injection equipment are on two different meters. The reading on both meters is noted by site staff prior to and following injection in order to measure the electricity use specific to Project activities
2. Electricity use for each injection is monitored by onsite staff and reported in a monthly MRV Report that is provided by the injection site operator as required in their service contract

b. Injectate Characterization

i. Injectate Mass

1. Mass is monitored by Scale Tickets provided by the injection well personnel.
    a. These values are measured by weighing the delivery truck prior to and after injection on a certified scale

ii. Mass of Bio Oil Injected
1. During bio-oil processing, the mass of consumables added to the bio oil (LCS, salt) are monitored by the site operators for each batch
2. The mass of consumables added to the bio-oil being processed are subtracted from the Total Injectate Mass to compute the Mass of Bio-Oil Injected

iii. Injectate C content (%)
1. Bio-oil is sampled after the sparging process, but prior to the addition of consumables, and samples are sent to an ISO accredited analytical laboratory for the quantification of C content
    a. Bio-oil is sampled at this point in the process because the addition of consumables causes phase separation, which precludes samples from being the "well-mixed and representative aliquot" required by section 7.4.1.1.1 of the Bio-Oil Geological Storage protocol
    b. Per section 7.4.1.1.1 of the Bio-Oil Geological Storage protocol, a sample is taken for each injection batch and analyzed for C content to quantify the gross removal
2. The injectate C content is computed by multiplying the Mass of Bio Oil Injected by the average C content of the sample as established by lab testing. This is then multiplied by 44/12 to convert the C content to the equivalent in CO2e.

iv. Other Injectate Characteristics
1. Each injection batch is sampled after the blending process, prior to injection, samples are sent to an ISO accredited analytical laboratory for the quantification of the Total Acid Number (TAN) and density of the injectate
2. Upon arrival to the injection site, injectate is tested to establish neutral pH (4.0 or greater) and temperature prior to injection.
    a. Results are provided to Charm on a monthly basis
    b. Injectate will not be injected if the pH reading is below 4.0

c. Reversals

i. In the event that a gaseous release at the wellhead should trigger a gas monitoring alarm, the well operator has affirmed that they will contact Charm within a 2-hour period.
1. Charm, Isometric, and the VVB have agreed that it is more appropriate for the well operator to notify Charm regarding a wellhead gas alarm being triggered rather than the VVB as prescribed in Salt Cavern Storage module Section 3.1.1. This

is because 1) additional context-specific investigation is necessary to establish whether the triggering alarm is or is not indicative of a reversal, and 2) the well operator has their alarms set to conservative trigger points below levels that would indicate a reportable event so that any anomaly is investigated before it becomes a larger issue.

2. Any identified reversal will be promptly reported to Isometric and the VVB and investigated by Charm – Reports will be made between 1 and 3 business days from the identification of a gaseous release per Section 5.6.1 of the Isometric Standard
3. Relevant monitoring data will be included for the next Verification to be assessed by the VVB per Section 5.6.1 of the Isometric Standard and Section 3.1.1 of the Salt Cavern Storage module

ii. The well operator has installed a detection system for hydrocarbons and methane at the wellhead

6. **Ongoing Regulatory Compliance**
   a. All third-party project participants are queried annually to affirm ongoing regulatory compliance
      i. Participants are required to return a signed affirmation of ongoing compliance
      ii. In the event that there is a compliance issue, participants must
         1. Identify the issue, its time of discovery, and a plan for remediation including timeline
   b. Ongoing monitoring of regulatory compliance requirements at the El Dorado pre-processing site are described above in Section 3 Part iv and is dependent on physical modifications to the site in order to increase processing capacity

Monitoring Plan parameters relevant to the Biomass or Bio-oil Storage in Salt Caverns module:

c. Pre-Injection (Baseline) Testing

i. Brine pH
   1. Who is Testing?
      a. Vaulted team
   2. How Frequently?
      a. Once
   3. How is Testing Performed?
      a. pH meter
   4. When is Testing Performed?
      a. Prior to first emplacement
   5. How are Results Recorded?
      a. Results are reported to and stored by Charm

ii. Brine Temperature
   1. Temperature
      a. Who is Testing?
         i. Vaulted team

b. How Frequently?
    i. Once
c. How is Testing Performed?
    i. Thermometer
d. When is Testing Performed?
    i. Prior to first emplacement
e. How are Results Recorded?
    i. Results are reported to and stored by Charm

iii. Brine Conductivity/Chloride Content
1. Who is Testing?
    a. Vaulted team
2. How Frequently?
    a. Once
3. How is Testing Performed?
    a. Electrical Conductivity meter
4. When is Testing Performed?
    a. Prior to first emplacement
5. How are Results Recorded?
    a. Results are reported to and stored by Charm

iv. Brine TOC Level
1. Who is Testing?
    a. Vaulted team
2. How Frequently?
    a. Once
3. How is Testing Performed?
    a. On-site use of a TOC analyser or shipment to an ISO-certified analytical laboratory are both acceptable at Vaulted's discretion
4. When is Testing Performed?
    a. Prior to first emplacement
5. How are Results Recorded?
    a. Results are reported to and stored by Charm

v. Wellhead Gas Composition
1. Who is Testing?
    a. Charm team
2. How Frequently?
    a. Once
3. How is Testing Performed?
    a. Gas samples collected at wellhead and sent to ISO-certified analytical laboratory
4. When is Testing Performed?
    a. Prior to first emplacement
5. How are Results Recorded?
    a. Results are reported to and stored by Charm

d. Injectate and Operational Monitoring

i. Injectate pH
1. Who is Testing?
    a. Vaulted team
2. How Frequently?

        a. One sample per day of injection
    3. How is Testing Performed?
        a. pH meter
    4. When is Testing Performed?
        a. Upon injectate delivery
    5. How are Results Recorded?
        a. Results are recorded on the inspection report/weight scale ticket form that is completed upon receipt of the injectate by Vaulted and returned to Charm
        b. Values are input into the Charm Ledger corresponding with the injection batch for storage

ii. Injectate Temperature
    1. Who is Testing?
        a. Vaulted team
    2. How Frequently?
        a. One sample per day of injection
    3. How is Testing Performed?
        a. Thermometer
    4. When is Testing Performed?
        a. Upon injectate delivery
    5. How are Results Recorded?
        a. Results are recorded on the inspection report/weight scale ticket form that is completed upon receipt of the injectate by Vaulted and returned to Charm
        b. Values are input into the Charm Ledger corresponding with the injection batch for storage

iii. Injectate Conductivity/Chloride Content
    1. Who is Testing?
        a. Charm Team
    2. How Frequently?
        a. For each injection batch
    3. How is Testing Performed?
        a. Charm will maintain field notes and records of how much salt is added to each injection batch to ensure compliance with the salinity requirements elaborated by KDHE in the comfort letter issued for bio oil sequestration activities at Vaulted
    4. When is Testing Performed?
        a. During injectate pre-processing
    5. How are Results Recorded?
        a. Results are recorded by the Charm pre-processing team as a standard part part of production
        b. Values are input into the Charm Ledger corresponding with the injection batch for storage

iv. Injectate Total Acid Number
    1. Who is Testing?
        a. Charm team
    2. How Frequently?

            a. Every injection batch – Frequency may decrease if data shows significant consistency across batches
        3. How is Testing Performed?
            a. Sample is sent to an ISO-accredited analytical laboratory
        4. When is Testing Performed?
            a. Samples are pulled prior to injectate delivery and mailed to lab
        5. How are Results Recorded?
            a. Lab results are reported to Charm and input into the Charm Ledger corresponding with the injection batch for storage
    v. Injectate Density
        1. Who is Testing?
            a. Charm team
        2. How Frequently?
            a. Every injection batch
        3. How is Testing Performed?
            a. Sample is sent to an ISO-accredited analytical laboratory; lab to report results relative to specific temperature
        4. When is Testing Performed?
            a. Samples are pulled prior to injectate delivery and mailed to lab
        5. How are Results Recorded?
            a. Lab results are reported to Charm and input into the Charm Ledger corresponding with the injection batch for storage

e. Displaced Brine Monitoring

i. % by wt of Carbon
    1. Who is Testing?
        a. Vaulted team
    2. How Frequently?
        a. Every two weeks for 3 months. If measurements are consistent this can be changed to every month for 6 months, then quarterly for 2 years, and finally every 6 months if results remain consistent.
    3. How is Testing Performed?
        a. On-site use of a TOC analyser or shipment to an ISO-certified analytical laboratory are both acceptable at Vaulted's discretion
    4. When is Testing Performed?
        a. At Vaulted's discretion, provided minimum frequency requirements are met
    5. How are Results Recorded?
        a. Lab results are reported to Charm
        b. On-site measurements will be included in a monthly report provided to Charm by Vaulted per the service agreement between the parties

c. Documentation of results will be stored by Charm

ii. Displaced Brine Temperature
1. Who is Testing?
   a. Vaulted team
2. How Frequently?
   a. Initially, after each injection. Frequency can be reduced once consistency between samples is statistically proven
3. How is Testing Performed?
   a. Thermometer
4. When is Testing Performed?
   a. After injection
5. How are Results Recorded?
   a. Results are recorded on the inspection report/weight scale ticket form that is completed upon receipt of the injectate by Vaulted and returned to Charm
   b. Values are input into the Charm Ledger corresponding with the injection batch for storage

iii. Displaced Brine pH
1. Who is Testing?
   a. Vaulted team
2. How Frequently?
   a. Initially, after each injection. Frequency can be reduced once consistency between samples is statistically proven
3. How is Testing Performed?
   a. pH Meter
4. When is Testing Performed?
   a. After injection
5. How are Results Recorded?
   a. Results are recorded on the inspection report/weight scale ticket form that is completed upon receipt of the injectate by Vaulted and returned to Charm
   b. Values are input into the Charm Ledger corresponding with the injection batch for storage

iv. Displaced Brine Chloride Content/Conductivity
1. Who is Testing?
   a. Vaulted team
2. How Frequently?
   a. Initially, once per day of emplacement. Frequency can be reduced once consistency between samples is statistically proven
3. How is Testing Performed?
   a. Electrical Conductivity meter
4. When is Testing Performed?
   a. After injection
5. How are Results Recorded?

a. Results are recorded on the inspection report/weight scale ticket form that is completed upon receipt of the injectate by Vaulted and returned to Charm
b. Values are input into the Charm Ledger corresponding with the injection batch for storage

f. Cavern Integrity

i. *As the Well Operator, Vaulted is responsible for monitoring all of the following parameters. In addition, Vaulted is responsible for all monitoring and reporting required to maintain their UIC permit.*

1. Internal Well Integrity
2. External Well Integrity
3. Cavern Temperature
4. Cavern Pressure
5. Cavern Volume and Fill (Sonar Survey)
6. Wellhead Gas Composition
    a. Gas Isotope Stability if Wellhead Gas Composition is outside of normal average baseline range
7. Surface Subsidence
8. USDW Quality
9. Salt permeability
10. Porosity and Permeability of caprock and any interbeds
11. Existing subsurface features such as faults

7. Injection (Charm)

a. For details regarding emissions monitoring at Charm's Basco 6 well site, please see the Combined Emission Monitoring and Accounting Plan in the Basco 6 supplemental doc

b. For well integrity and permanence monitoring details please see Appendix 6

[Optional] Additional information on Project monitoring:

Data collection and storage

Please describe the data collection and storage approach taken, including:

- *How data is transmitted and collected*
- *How data is stored*
- *Length of time for which records are archived*
- *Backup procedures and strategies for identifying and coping with lost or poor-quality data.*
- *Person(s) / organization(s) responsible for measurement and data collection*

1. Data Transmission and Storage – *Note: Additional details on data collection and usage are in the GHG Statement Data section*
    a. Offtake Tonnage
        i. At AECN, oil is weighed by vendor using an on-site truck scale. The vendor provides a Bill of Lading, scale ticket and invoice document detailing the total weight. At Charm, individual containers are weighed when filled added to an inventory for transport documentation
        ii. Mass value is entered into the Ledger using the User Interface (UI) and the document is uploaded with the entry
    b. Process and Embodied Production Emissions – AECN
        i. Process and embodied emissions are entered into the Ledger as a static value based on the Vendor LCA (process emissions) and Charm calculations (embodied emissions) and applied to each MT of bio oil purchased
    c. Process and Embodied Production Emissions – Charm
        i. Process and Embodied Emissions are determined on a periodic basis, referred to as a Production Period, for the bio-oil produced during that period. This requires collecting data for biomass handling and delivery emissions, biomass throughput, pyrolysis machine up-time, energy usage, usage of non-energy consumables and pyrolysis outputs. Many manual recording processes will be progressively automated
            1. Biomass handling fuel use automatically calculated based on biomass tonnage and a standard emission factor for lumber handling
            2. Biomass delivery emissions are logged in the manner described in the “Transport Emissions” subsection below
            3. Biomass throughput to pyrolysis is logged by operators who record and audit the mass of biomass processed and consumed during each production period
            4. Pyrolysis machine up-time is logged automatically using telemetry
            5. Energy usage data is collected by vendor delivery invoices in the case of fuel delivered to site, or by the use of utility power metering data
            6. Use of water in the pyrolysis process is recorded using a flow meter
        ii. Embodied emissions and estimated useful lives of equipment and facility improvements are recorded in an inventory that is updated as new equipment is commissioned into use and facility improvements are completed. New inventory items are submitted for LCA and the associated emissions impacts are added to the Ledger and amortized over their expected useful life, which is estimated in terms of total expected facility biomass throughput over time
            1. The use of smaller, regularly exchanged components (e.g. nuts and bolts) is estimated on a per-hotflow hour basis. This

estimate is periodically evaluated and updated by the operations team

d. Transport Emissions
    i. Fuel Emissions
        1. Point-to-point distances are stored in the Ledger based on the distance provided in Google Maps, in the case of trucks, or the rail carrier, in the case of trains
            a. In order to ensure a conservative estimate of removals, the longest route is selected when multiple routes are provided
        2. The cargo weight (documented by certified scale tickets, BOLs, and/or an inventory of pre-weighed containers), vehicle type and point-to-point distance for the journey is used in conjunction with GLEC emissions factors to calculate transport Fuel Emissions
    ii. Embodied Emissions
        1. Trucking
            a. Per-mile embodied emissions for the vehicle type (Tanker and non-Tanker trucks) are entered into the ledger as a static value and applied to each journey based on mileage accounted for by stored point-to-point distances between sites
        2. Rail
            a. Per-mile embodied emissions for the vehicle type (Railroad Rolling Stock) are entered into the ledger as a static value and applied to each journey based on mileage accounted for by stored point-to-point distances between sites

e. Pre-Processing Site Emissions
    i. Fuel Use
        1. Fuel use is measured based on amounts purchased (delivery invoices) and entered into the ledger via User Interface.
    ii. Electricity Emissions
        1. Electricity usage as quantified by utility data is entered into the Ledger via User Interface
    iii. Equipment Embodied Emissions
        1. Embodied emissions are calculated based on the expected useful life of the equipment and amortized against removals as a straight-line depreciation for each reporting period.
    iv. Consumables Embodied Emissions
        1. Liquid caustic soda (LCS) is added to the bio oil at a volume measured by on-site staff using a flow-meter. The quantity added to each injection batch is recorded by site staff and entered into the Ledger via User Interface.
    v. Consumables Delivery Emissions
        1. Origin and weight of delivery are recorded by site staff and entered into the Ledger via User Interface. This provides the data necessary to calculate transport emissions as detailed in the GHG Statement

vi. Salt Addition to Bio-Oil
    1. Salt is added to the bio oil at a volume measured by on-site staff using a scale. The quantity added to each injection batch is recorded by site staff and entered into the Ledger via User Interface.

f. Injection Emissions – Vaulted
    i. Energy Use Emissions (Fuel and Electricity)
        1. The Well Operator is required to provide a monthly report on energy usage specific to Charm emplacements. These values are input into the Ledger as received via User Interface
    ii. Injection Equipment Embodied Emissions
        1. Embodied emissions are entered into the ledger as a static value based on Vaulted's PDD and relevant GREET emission factors and applied to each MT of bio oil emplaced
    iii. Sampling Emissions
        1. Pre-injection sampling is executed at a pre-determined cadence for each injection batch. Emissions are calculated according to the number of tests and associated transport emissions required to test each batch. This value is entered into the Ledger as a static value and applied to each injection batch of bio oil emplaced

g. Injection Emissions – Charm
    i. For specifics on Charm injection site data collection, see the Combined Emissions Monitoring and Accounting Plan in the Basco 6 supplemental doc

2. How Data is Stored
    a. Ledger data is stored in an AWS relational database. It is backed up in S3, a service for storing files.
3. Length of Time for Which Records are Archived
    a. Most data is stored indefinitely in the AWS database. In the unusual case where data is deleted, it will be stored in backups on S3 which enables retrieval even after deletion.
4. Backup Procedures and Strategies for Identifying and Coping with Lost or Poor-Quality Data
    a. Charm maintains daily snapshots of the relational database for the past 30 days.
    b. All objects are backed up on S3, enabling retrieval even after an update or deletion
5. Person(s)/Organization(s) Responsible for Measurement and Data Collection
    a. Charm Industrial
        i. Variable data is entered by Charm staff based on relevant documentation such as bills of lading, receipts for fuel purchase, etc.
    b. Vaulted
        i. Vaulted is required by their service agreement to provide to Charm with a monthly report including
            1. Weight of product delivered by Charm as documented by weight scale tickets
            2. Energy consumed on site associated with the processing and injection of product delivered by Charm

ii. Vaulted will also provide inspection/receipt reports for each injectate delivery including
    1. Weight of injectate delivered
    2. Injectate pH
        a. 1 sample per day of injection
    3. Injectate temperature
        a. 1 sample per day of injection

c. AECN
    i. AECN is responsible for taking measuring and recording the weight of bio oil to be delivered to Charm as documented by a Bill of Lading
    ii. AECN has also provided an updated production LCA measuring the process emissions associated with their production of Bio Oil

Notes on Charm's Ledger software

In this section and elsewhere in the PDD, reference is made to the Ledger. This is a piece of proprietary software developed internally by Charm's software team. Once a given batch of bio-oil comes into Charm's custody, as defined by issuance of a Bill of Lading, scale ticket and invoice document as discussed earlier in this section, the batch becomes a "Lot" in the Ledger.

The software is designed to measure and track Lots as they are split and combined, maintaining a clear mass balance and chain of custody as they move between transport modes (e.g. truck to railcar to truck) and are stored, batched, and treated along the pathway to injection. Once a given Lot, or batched combination of several split Lots, is injected it is then logged as a Removal. The data for a given Removal includes the process emissions associated with the Lot components, calculated on a mass balance basis which traces back to the original Lot.

The Ledger also captures data from "site emissions", described in Appendix 2, which are facility emissions that cannot be neatly mapped to a specific Lot (e.g. site electricity usage). When a Removal is completed, the CHN testing associated with that removal assigns it a C and CO2e content, from which both process emissions and either all or a portion of site emissions for the reporting period in which the Removal is claimed, are deducted in order to compute the Net CDR for a given Removal

Data Quality Assurance Measures
Note: Data quality assurance is an ongoing process. The following is a list of measures currently place, which is sure to grow as operational experience demonstrates optimization opportunities:

1. Removals are calculated and removal data is managed within the Charm Ledger. This specialized piece of software improves data quality by
    1. Ensuring that emissions factors and measurements are used consistently across all estimates
    2. Ensuring that calculation methods are used consistently across all estimates
    3. Ensuring consistent amortization of embodied and site emissions

4. Reducing human error by automating calculation work such as unit conversions
5. Reducing human error by minimizing the amount of manual entry required to generate estimates
6. Allowing for multiple layers of manual testing and review of functions by both internal and external parties
7. Requiring specific documentation such as bills of lading, mass documents, routing documents, documentation for emission factors/carbon intensities, etc. for entries
    1. Documents are also archived separately for ease of reference

2. Where possible, data is sourced from documents that provide an additional layer of oversight/quality control (e.g. official bills of lading, official invoices, direct-printed scale tickets from calibrated scales)
    1. When data from such documents is entered manually, the document must be provided as a check against entry errors
3. Datapoints requiring manual entry by operators are minimized. Where they are necessary, operators are trained on specific entry procedures. Opportunities for error are further minimized by eliminating cumbersome/complex entry requirements wherever possible.
4. Datapoints are double-checked by Charm's MRV team prior to Ledger entry to catch and investigate potential indicators of errors and nonconformities
5. Because data quality can never be fully guaranteed, high-sensitivity parameters are specifically modified for uncertainty as detailed in the Uncertainty Assessment and Sensitivity Analysis

Reversals

Please complete a reversal risk assessment for your Project, including consideration of the guidance in the Risk of Reversal Questionnaire below.

Overall risk of reversal (based on risk reversal questionnaire score; please select one):

Very low (0)
Low (1-3)
Medium (4-5)
High (>6)

[Optional]
Charm's use of pyrolysis oil significantly reduces the risk of methane production arising from anaerobic decomposition of organic materials. Because pyrolysis is a reaction that occurs in a high-heat, low oxygen environment, the oil does not contain compounds that would support the microbial life required for anaerobic decay. This is confirmed by a

peer-reviewed chemical characterization of pyrolysis bio-oil from multiple feedstocks (Negahdar et al., 2016) and a specific analysis of pyrolysis oil commissioned by Charm that are included in the supporting documents for this PDD

#	Question	If answered "Yes"	If answered "No"	Response
1	Is a reversal directly observable with a physical or chemical measurement as opposed to a modeled result?	Proceed to questions 2-8	Proceed to questions 7-8	Yes
2	Is the carbon being stored in a closed or impermeable system? (e.g., salt cavern)	Proceed to questions 10-11	Proceed to questions 3-11	Yes
3	Is the carbon being stored organic?	Add 1 to Risk Score		N/A
4	Does scientific consensus suggest that the carbon storage reservoir has a less than 10,000 year durability?	Add 1 to Risk Score		N/A
5	Is methane production a Project risk?	Add 1 to Risk Score		N/A
6	Does this approach have a material risk of reversal due to natural disasters including, but not limited to, floods, storms, earthquakes, fires, etc.?	Add 1 to Risk Score		N/A
7	Does this approach have a material risk of reversal due to human-induced events from outside actors, such as change in farming practices, change in ownership and management of Project sites, or similar?	Add 1 to Risk Score		N/A

8	[Applicable only for subsurface storage] Is the carbon being stored in the deep subsurface with multiple trapping mechanisms preventing reversals? (e.g., multiple confining layers, CO_2 dissolves or solidifies)	Minus 1 to risk (unless 0)		N/A
9	Is there 10+ years of monitoring and/or lab data demonstrating low project risk?	Minus 1 to risk (unless 0)		N/A
10	Does this pathway have a documented history of reversals?	Please consider the frequency and severity of previous reversals, and the shared characteristics between documented reversals and the present project. For pathways with no documented history of reversals, add 0 to the Risk Score. For pathways with a history of frequent reversal, add 2 to Risk Score. For pathways with a limited history of reversals, add 1 to Risk Score.		0
11	Is there one or more Project-specific factors that merit a high risk level?	Please consider the number and severity of risks identified. If one low or medium severity risk is identified, add 1 to Risk Score. If multiple risks are identified, or if any high severity risks are identified, add 2 to Risk Score		0
Total Risk Score				0

Uncertainty assessment	
Specify how uncertainty is considered, and how removals are to be conservatively estimated, in accordance with Section 2.5.7 "Uncertainty" of the Isometric Standard.	
Please specify which option(s) were used in consideration of uncertainty (*one or multiple options*).	
	Conservative estimate of input parameters
X	Variance propagation
	Monte Carlo Simulations
Please provide a sensitivity analysis. A sensitivity analysis is provided in the GHG Statement doc appended to this PDD	
●	I have uploaded a sensitivity analysis in accordance with the guidance in the Isometric Standard

Justification for your choice of uncertainty approach:
A Sensitivity Analysis has been carried out for each measured emissions source/parameter based on a Sample Calculation of one truckload of bio oil originated at AECN and transported to Kansas for Injection as well as the first Charm bio-oil injection in Kansas.

In the Sensitivity Analysis, the value of each source was arbitrarily increased by 20%, a value far higher than a reasonable expectation of measurement variance for any given parameter. These changes are compared against an estimated "Sample Removal" without this variance. Any parameter that, when increased by 20%, contributed to a change in the estimated removal that was greater than 1% of the original estimation was then subject to an Uncertainty Analysis, as specified in Section 2.5.7 of the Isometric Standard.

Note: What follows is in reference to AECN and Charm Kansas injections only. Sample calculations, sensitivity analysis and uncertainty assessment for injection operations in Louisiana can be found in the Basco 6 supplemental document

Parameters Identified as requiring Uncertainty Analysis are listed below, along with a brief summary of the approach to identifying and measuring the associated measurement uncertainty. All calculations are included in the Uncertainty Analysis sheet included in the GHG Statement Supplemental Doc appended to this PDD.

1. Bio Oil Process Emissions – AECN
    a. Process Emissions from the Vendor LCA are treated as an emissions factor for the process – Like other emissions factors, they are based on a study of the production data and emissions associated with the production of bio oil by AECN
        i. An Uncertainty Factor has been calculated for this emissions factor based on process and data quality considerations as outlined in Section 7.4 of the Isometric Standard.

ii. This Uncertainty Factor is included as an increase in the baseline emissions factor to ensure a conservative estimate of removals. This is shown in the Emissions Factors sheet in the GHG Statement Supplemental Doc appended to this PDD

2. Bio-Oil Transportation Fuel Emissions
    a. Transportation Fuel Emissions are calculated on a ton-mile basis using GLEC emissions factors. There is no Uncertainty Factor calculated for this value. This is due to previous consultations with Isometric confirming that the GLEC factors are based on millions of miles of data, driving uncertainty down to a negligible value
    b. Routing is based on the point-to-point distance provided by Google Maps (for trucks) or the rail service provider (for rail) for a given route. When multiple routes are given, Charm will use the longest option in order to ensure a conservative estimate of removals.
    c. The maximum allowable error for a legal-for-trade truck scale is 160 lbs for a fully-loaded truck. Bio oil is weighed twice: once when loaded for delivery at AECN, and once prior to injection at Vaulted. In order to ensure a conservative estimate of removals, Charm reports 1 standard deviation of uncertainty for each scale measurement, using 160 lbs as the half-width of a uniform distribution of scale error. The uncertainty in the bio oil weight is propagated to the calculation of transport emissions and all downstream calculations.
3. Bio Oil Pre-Processing Fuel Emissions
    a. Bio Oil Pre-Processing fuel emissions are calculated on a per-gallon basis using fuel purchase records and GLEC emissions factors. There is no Uncertainty Factor calculated for this value. This is due to previous consultations with Isometric confirming that the GLEC factors are based on millions of miles of data, driving uncertainty down to a negligible value
    b. The maximum allowable error for a legal-for-trade fuel meter is .28%. In order to ensure a conservative estimate of removals, Charm reports 1 standard deviation of uncertainty for each fuel delivery, using .28% as the half-width of a uniform distribution of meter error. The uncertainty in the fuel quantity is propagated to the calculation of fuel use emissions for bio oil pre-processing operations.
4. Bio Oil Pre-Processing Consumables Emissions
    a. The consumables used in pre-processing are liquid caustic soda (50%) (LCS) and salt.
        i. Liquid Caustic Soda
            1. The volume of LCS addition to each injection batch is measured by Charm site staff using a flow meter confirming volume measurements made with standardized marks on the container in which the LCS is delivered
            2. Embodied emissions are calculated based on the GREET emissions factor for the product
            3. An Uncertainty Factor has been calculated for this emissions factor based on process and data quality considerations as outlined in Section 7.4 of the Isometric Standard.

4. This Uncertainty Factor is included as an increase in the baseline emissions factor to ensure a conservative estimate of removals. This is shown in the Emissions Factors sheet in the GHG Statement Supplemental Doc appended to this PDD
5. In their study of water flow meter measurement uncertainties, Dias, Filho, and de Lucca (2013) identified 2.045% as the uncertainty for a calibrated flow meter. Because the site operations team is in the process of identifying which model of flow meter best fits their needs for the long term, this value will be used as the uncertainty assumption for LCS flow meter measurement until a manufacturer's specific uncertainty rating can be used for the long term. In order to ensure a conservative estimate of removals, Charm reports 1 standard deviation of uncertainty for each addition of LCS to a batch of bio-oil, using 2.045% as the half-width of a uniform distribution of meter error. The uncertainty in the fuel quantity is propagated to the calculation of fuel use emissions for bio oil pre-processing operations.

ii. Salt

1. Embodied emissions for salt are not included in the PDD due to the fact that the salt is a waste product. It is off-spec/non-saleable salt produced as a waste product incidental to the salt production process at a nearby Morton salt factory

5. Gross CO2e Sequestration
   a. Gross CO2e sequestration is calculated using the following procedure described in the Monitoring Plan. The Bio-Oil is sampled for CHN testing by an ISO accredited laboratory after the completion of Sparging. The % C reported by the laboratory is multiplied by the Mass of Bio-Oil Injected. That value is multiplied by 3.67 or 44/12 to convert the carbon content to CO2e.
   b. The maximum allowable error for a legal-for-trade truck scale is 160 lbs for a fully-loaded truck. The relative standard deviation reported by the manufacturer of the LECO instrument used to measure carbon is .4%. Each of these values are used as the half-width of a uniform distribution of error for their respective associated measurements. These uncertainties are propagated to the calculation of the gross quantity of CO2e sequestered for each injection batch.

Uncertainty for Charm Bio-Oil

A sample LCA and sensitivity analysis matching the methodology described above are included in the Production Period 1 summary for the CO facility. Parameters identified for uncertainty modification and the nature of the modifications are noted below:

1. Biomass Transportation Fuel Use
   a. Transportation Fuel Emissions are calculated on a ton-mile basis using GLEC emissions factors. There is no Uncertainty Factor calculated for this value. This is due to previous consultations with Isometric confirming that

the GLEC factors are based on millions of miles of data, driving uncertainty down to a negligible value

b. Routing is based on the point-to-point distance provided by Google Maps (for trucks) or the rail service provider (for rail) for a given route. When multiple routes are given, Charm will use the longest option in order to ensure a conservative estimate of removals.
c. The maximum allowable error for a legal-for-trade truck scale is 160 lbs for a fully-loaded truck. In order to ensure a conservative estimate of removals, Charm reports 1 standard deviation of uncertainty for each scale measurement, using 160 lbs as the half-width of a uniform distribution of scale error. The uncertainty in the bio oil weight is propagated to the calculation of transport emissions and all downstream calculations.

2. Processing and Pyrolysis Fuel Use
    a. Processing and pyrolysis fuel emissions are calculated on a per-gallon basis using fuel purchase records and GLEC emissions factors. There is no Uncertainty Factor calculated for this value. This is due to previous consultations with Isometric confirming that the GLEC factors are based on millions of miles of data, driving uncertainty down to a negligible value
    b. The maximum allowable error for a legal-for-trade fuel meter is .28%. In order to ensure a conservative estimate of removals, Charm reports 1 standard deviation of uncertainty for each fuel delivery, using .28% as the half-width of a uniform distribution of meter error. The uncertainty in the fuel quantity is propagated to the calculation of fuel use emissions for bio oil pre-processing operations.
3. Processing and Pyrolysis Electricity Use
    a. Electricity use is measured based on the total facility usage as recorded by the utility-provided electricity meter.
    b. The Colorado Code of Regulations (page 41) requires that the maximum measurement error for a service meter is 2%. In order to ensure a conservative estimate of removals, Charm reports 1 standard deviation of uncertainty for measured electricity consumption levels using these meters.
4. Pyrolysis Tailgas Emissions
    a. Tailgas emissions are measured based on composition and flow-rate data from a qualified emissions testing company approved by a state regulatory authority to perform compliance emissions testing.
    b. Pyrolysis tailgas emissions are scaled by the quantity of biomass processed during a production period. The measured value of processed biomass is modified for uncertainty as described below, which is then used as an input for computing the estimate of pyrolysis tailgas emissions for the production period
5. Mass of Processed Biomass
    a. Processed biomass is measured using a calibrated Class III scale with a tolerance of +/- 25 lbs. In order to ensure a conservative estimate of removals, Charm reports 1 standard deviation of uncertainty for the quantity of biomass processed computed based on this measurement

6. Disposal of Process Wastes
    a. Emissions accounted for in the disposal of process wastes includes both induced transport and the disposal of the material itself. Both of these are scaled by the mass of the waste being disposed.
    b. The maximum allowable error for a legal-for-trade truck scale is 160 lbs for a fully-loaded truck. In order to ensure a conservative estimate of removals, Charm reports 1 standard deviation of uncertainty for manifested load of waste measurement, using 160 lbs as the half-width of a uniform distribution of scale error. The uncertainty in the bio oil weight is propagated to the calculation of transport emissions and all downstream calculations.
7. Amortized Capital Embodied Emissions
    a. The amortization of capital embodied emissions is scaled by the quantity of biomass processed during a production period. The measured value of processed biomass is modified for uncertainty as described above, which is then used as an input for computing the quantity of capital embodied emissions to be amortized during the production period
8. Spare Parts and Consumables Embodied Emissions
    a. The rate of use for spare parts and consumables is estimated in terms of hotflow hours – The estimate of emissions for a given production period is scaled by the number of hotflow hours during that period
    b. Time in hotflow is tracked using telemetry that captures reactor temperature. The equipment manufacturer has not provided a specific measurement uncertainty. However, Charm's automation engineering manager has performed extensive system testing and has noted that in his experience and professional judgement the measurement uncertainty for time in hotflow is no greater than 1%
    c. In order to ensure a conservative estimate of removals, Charm reports 1 standard deviation of uncertainty for reported production period hotflow hours, using 1% of the measured quantity as the half-width of a uniform distribution of scale error. The uncertainty in hotflow hours is propagated to the calculation of spare parts and consumables embodied emissions and all downstream calculations.
9. Mass of Bio-Oil Produced
    a. Finished bio-oil is measured using a calibrated Class III scale with a tolerance of +/- 2.5 lb. In order to ensure a conservative estimate of removals, Charm reports 1 standard deviation of uncertainty for the quantity of bio-oil produced computed based on this measurement

Uncertainty for Basco 6 Louisiana Injection Operations

Sensitivity analysis and uncertainty modifications for Basco 6 Louisiana injection operations are included in the Sensitivity Analysis and Comprehensive Emissions

Monitoring and Accounting Plan sections of the Basco 6 Supplemental Doc included as Supplemental Doc 3 of this PDD

Use of Models

Please describe your use of models (if any) for quantification, monitoring, and meeting specified Protocol requirements. Describe the specific model and simulations used, with enough detail so that the work could be replicated.

Please provide model validation results to demonstrate model accuracy, and include an assessment of model uncertainty.

The vendor has not analyzed stack emissions from their pyrolysis process. In absence of direct data, we were provided a modeling of process emissions from the technology in use (Ensyn RTP) processing woody biomass prepared for Natural Resources Canada detailing the addition of pyrolysis oil pathways to GHGenius software. These values have been provided as documentation alongside this PDD.

Values associated with this model are provided in the GHG Statement Supplemental Doc and are incorporated into the Process Emissions emission factor for pyrolysis.

[Storage Module] Storage well overview

Please describe the storage well used and complete the following information.

KS
Charm is emplacing bio oil at Vaulted's Geat Plains storage well for this project. For that reason, what follows in this section is a reproduction of the Storage Well Overview from their PDD:

Storage well description:
"Vaulted operates a built and permitted injection well site in Hutchinson, Kansas. This site is a network of 60 salt caverns, with the total capacity to hold 2-3 million metric tons of organic waste. The wells are permitted under Class UIC V well permits by KDHE."

LA
Storage well/site description:
Charm has converted the formerly orphaned Basco 6 Class II SWD to a permitted Class V for the purpose of injecting the bio-oil into subsurface geologic formations. The wellbore configuration, depicted here in the wellbore diagram, is completed in the Lower Miocene formation from 3,492-3,534' KB. The LDENR Order for Class V injection permits the zone from 1,600 to 3,685' KB. For additional details on the physical characterization of the

subsurface features of the well, please refer to Appendix 9: Basco 6 Supplementary Subsurface Characterization.

Charm will also construct and install surface bio-oil receiving facilities, temporary storage, and an injectate circulation and pump delivery system, all of which will be located largely within the existing boundaries of the existing improved well pad area at the injection well site.

KS
Monitoring overview (*please summarise the current monitoring in place, as required by the well permit and in accordance with the protocol monitoring requirements. You may provide more information on individual monitoring parameters in the Project Monitoring Plan):*
"Charm has contracted with Vaulted for well operations related to this project. The following is quoted from the Storage Well Overview section of the Vaulted PDD:

Vaulted is responsible for ongoing monitoring of the injection site per our service agreement and the UIC regulatory requirements incumbent on them as well operators.

At the Great Plains Facility, Vaulted maintains a robust monitoring program. Vaulted employs a variety of direct (logging, monitoring, wireline, analysis of well returns, pressure testing) and indirect methods (simulation studies) to confirm containment of injected materials and their decomposition products if any during operations and during project Decommissioning.

This includes real time data acquisition of the injection and post injection pressure data. Vaulted also takes periodic measurements, such as depth checks, and sonar or other surveys, which provide a second method for confirming the same containment.
At the Great Plains Facility, pressure gauges at the injection wells and monitoring wells on the facility boundary allow Vaulted to ensure no material escapes the caverns. Regular surveys and depth checks are conducted within the cavern wells to understand remaining capacity and to ensure injected materials are accounted for.

Vaulted regularly interfaces with the Kansas Department of Health and Environment (KDHE), particularly their Underground Injection Control Division. Monthly and quarterly reports are submitted to KDHE showing that the site is running safely, including data on groundwater quality, emplaced material spec, volume of injected material, and pressures and stability in the subsurface caverns. Vaulted has a strong ongoing relationship with KDHE, with monthly calls to ensure they're comfortable with the facility.

Additionally, the US EPA issued a comfort letter on June 19, 2018 attesting to the safety of Vaulted's slurry injection technique and the appropriateness of using Class V wells for organic waste injection."

LA
For details regarding Charm's well monitoring plan please refer to Appendix 7 of this PDD

KS
Storage well location (Address / GPS coordinates): 37.966, -97.941
EPA Well Class: V
Permit number: KS-05-155-003
Permitting authority: Kansas Department of Health and Environment (KDHE)
Permit validity start date: 2022/08/02
Permit validity end date: 2027/08/02
Well storage capacity (total): 2-3M metric tons of wet organic waste
Well storage capacity (used): <10,000 metric tons of wet organic waste
Well storage capacity (available): 2-3M metric tons of wet organic waste
LA
Storage well/site location (Address / GPS coordinates): 30.916251, -92.297908
Regulatory Authority / Jurisdiction: Evangeline Parish, Louisiana
Well Class (if applicable): Class V
Permit number: IMD 2025-05 EW
Permitting authority: Louisiana Department of Environment and Natural Resources
Permit validity start date: 8/8/2025
Permit validity end date: 8/7/2030
Well/site storage capacity (total): Estimated range: 465,000 to 840,000 MT
Well/site storage capacity (used): 0
Well/site storage capacity (available): estimated range, 465,000 to 840,000 MT

Section C – Duration & Crediting Period

Project timeline
Please indicate the projected start date of your Project and, if applicable, its expected operational closure date.

Start date of Project: 2024/03/01
[Optional] Expected operational closure date of Project:

Project closure
Please describe the conditions under which the Project will be considered closed, and describe the Project Closure Plan – outlining any post-cessation actions that will be undertaken upon Closure of the Project.
Definition of Project cessation: **KS** Once Vaulted indicates that the cavern is at capacity, or Charm decides to cease injection at Vaulted's Great Plains Cavern facility, whichever comes first. **LA** Once Charm determines that reservoir is at capacity, or Charm decides to cease injection at Basco 6, whichever comes first.
KS Closure Plan: In either case, Vaulted is responsible for the closure and/or ongoing monitoring of the cavern after Charm has ceased operations at the facility. Vaulted has accounted for their closure methods: "As a cavern reaches its capacity, Vaulted shuts the well in and places the cavern on monitoring status (post-injection monitoring) for a period of time to confirm the well is static and to comply with KDHE rules. Once the appropriate post monitoring period is complete, each at-capacity cavern will be plugged with cement in a manner that prevents the movement of fluids either into or between underground sources of drinking water (USDW)." **LA** Please see Appendix 8 for details regarding Charm's well closure plan.

Crediting Period
Please indicate the planned start date and duration of your crediting period. The crediting start date may either be your Project's start date or up to two years prior to design submission, whichever is later. Unless otherwise specified in the relevant Protocol, the maximum crediting period is 5 years.
Start date of crediting period: 2024/03/01
Total length of crediting period: 5 years

Section D – Environmental and social impacts

Analysis of environmental and social impacts	
Please provide an assessment of the environmental and social impacts of the Project, in accordance with Section 3.7 "Environmental and Social Impacts" of the Isometric Standard. *For each aspect of the assessment, demonstrate how the risks have been assessed and if applicable, what mitigation plan is in place to prevent them. If some aspects are not applicable to your project, justify how you determined it.* *A full Environmental and/or Social Impact Assessment (EIA and/or SIA) is only required if impacts are considered significant and/or if required by the host jurisdiction.*	
	[Optional] I have attached a full EIA document.
	[Optional] I have attached a full SIA document.
X	I acknowledge responsibility for reporting potential environmental and social impacts identified to Isometric and environmental regulators
Environmental Impacts	
Resource efficiency and pollution prevention, including pollutant emissions to air, pollutant discharges to water, noise and vibration, generation of waste and release of hazardous materials, chemical pesticides and fertilizers.	
X	Above risks are applicable to this Project
	Not applicable to this Project
Environmental impacts need to be considered with respect to the "upstream" processes of bio oil feedstock harvest and pyrolysis, as well as the impacts of the injection site. **Biomass Feedstock and Pyrolysis** **Charm Bio-oil** Charm has gone to significant lengths to ensure that both its biomass feedstock sourcing and pyrolysis processes reflect sustainable best practices and are in compliance with all applicable regulations. Before deciding to move forward with a supplier, Charm assesses the counterfactuals for any given biomass source to confirm there is no alternate near-term use case for the material. The forest residues that are used in this Project come from a state-managed and/or state-funded Fuels Reduction and Forest Restoration project. The project area showed signs of high risk factors for a catastrophic wildfire, thus warranting the Colorado State Forest Service a treatment plan that thins the forest to help increase its health and resilience.	

Catastrophic wildfires contribute to a mixture of hazardous air pollutants and release large quantities of carbon dioxide and other greenhouse gasses into the atmosphere. By using the woody biomass (fuels) from these fuels reduction projects, Charm can support and reduce the risk by disposing or reducing the fuels in the forest floor while sequestering the biogenic carbon in this material.

Charm's pyrolysis process is designed to recapture waste heat to dry biomass which minimizes the need for combustion of propane or other fuels. Charm has engaged a qualified emissions testing company to perform stack emissions testing and undergone a rigorous third party analysis of potential emissions from its biomass processing and pyrolysis emissions process. This review has confirmed that Charm's facility emissions are below major air quality permit thresholds.

AECN Bio-oil
The primary environmental impacts upstream of injection are related to the biomass feedstock used for pyrolysis and the process of pyrolysis itself. AECN has gone to significant lengths to ensure that both the biomass feedstock and the pyrolysis process reflect their dedication to sustainability.

All of the wood used as pyrolysis feedstock is harvested from provincially-owned forests in accordance with rights granted under the Quebec Provincial Management Plan (PMP). The PMP is explicitly guided by "sustainable forest management objectives" which dictate the bounds of allowable harvest such that the ecological integrity of provincial forestland is maintained over time.

In addition, the Bureau de Normalisation du Québec, an official certification organization accredited by the Standards Council of Canada, has granted the biomass harvester a certificate of compliance with the SFI Forest Management Standard.

The pyrolysis process itself has also been engineered with sustainability in mind. The facility construction was financed in part by Sustainable Development Technology Canada and Department of Natural Resources Canada. The facility is also certified by SGS Germany as compliant with the EU International Sustainability and Carbon Certification standard and the requirements of Renewable Energy Directive (RED) II.

Kansas Bio-Oil Injection

Vaulted, the injection well operator, has completed all necessary pre-injection studies and analyses to ensure risks to local resources and environment have been appropriately assessed and eliminated or controlled. Charm's contract with Vaulted stipulates that Vaulted has operated and will continue to operate in compliance with all local regulatory requirements.

Please see Vaulted's Environmental Impacts assessment below:
"All necessary pre-injection studies and analyses were conducted before the facility was built, including geologic feasibility studies, local environment and groundwater assessments, and engagement with local community groups and regulators. The Great Plains Facility is already fully permitted and operational.

The Great Plains Facility underwent environmental assessment prior to securing Class V injection permits. The assessment found no material environmental issue with the site.

Regular monitoring is undertaken at the facility, including:
- Quarterly groundwater testing to ensure no groundwater contamination
- Lab analysis on all emplaced material to ensure complies with non-hazardous organic permit
- Monthly reporting on total volume of emplaced material
- Daily readouts of pressures and stability of the subsurface caverns
- Bi-yearly elevation surveys to ensure ground stabilization (and no cavern sinking is occurring)"

Louisiana Bio-Oil Injection
All pre-injection studies and analyses have been conducted as required to secure Class V well permitting from LDENR, ensuring that risks to the local environment and natural resources have been assessed and controlled to the satisfaction of local regulatory authorities. This includes mechanical integrity pressure testing and isolation from USDWs. In addition, the site layout has been designed to ensure that any spilled material drains away from nearby water features. As well as avoiding adverse environmental effects, Charm has a thorough risk management program in place to avoid or mitigate possible future environmental effects that may be encountered during construction and operation.

Assessments required by LDENR permitting did not yield any known or published geological issues such as subsurface seismic activity or faulting in the area which could pose a risk of migration of injectate beyond the approved injection interval.

The Basco 6 Well is located off Louisiana State Highway 115, with an existing access road running from the highway to the well location. This means that equipment needed to convert the Basco 6 Well can travel over existing surface roads, without the need for additional road improvements or construction. Likewise, the existence of a facility road means that Charm will not need to construct additional site access. Bio-oil will be transported via truck accessing the well location via Highway 115 with truck shipments of no greater load weight than permitted by DOT law.

The Basco 6 Well site includes a large, improved rock area and well pad. This pad and site boundaries are of sufficient size to allow much of the new construction (including the aforementioned bio-oil receiving, circulation, and pump systems) to be performed within the existing footprint.The land immediately around the well will be used as part of Charm's operating pad, as is expected with any operation that needs access to a well.

A small portion of the land adjacent to the well (less than .25 acres) was converted back to cropland when the well became inactive, however injection and maintenance operations will require Charm to extend the pad to its original footprint. The existing site pad does reduce the amount of work needed to bring the site back to operating condition.

Charm's planned operations require minimal energy and water resources, creating a small footprint to sustain and scale operations.

A significant advantage of Charm's project design and methodology is it not only minimizes environmental impacts by repurposing existing orphaned oil and gas wells, but ultimately has a positive impact by committing to properly decommissioning and remediating the site upon the project's completion.	
Biodiversity conservation and sustainable management of living natural resources, including terrestrial & marine biodiversity and ecosystems, protecting habitats of rare & endangered species, avoiding conversion of natural forests, grasslands or wetlands, minimizing soil degradation or erosion, minimizing water consumption and stress.	
X	Above risks are applicable to this Project
	Not applicable to this Project

Biomass Sourcing and Pyrolysis – Charm
The forest residues that are used in this Project come from a state managed and/or funded Fuels Reduction and Forest Restoration projects. The treatment plans for those projects require the harvesting of timber to reduce the risk of catastrophic wildfires and improve forest health. The treatment of the forests by these projects can help increase biodiversity for wildlife in the area. By thinning the canopies of the overstocked forest, there is less competition amongst the trees and plants for nutrients and the plant biodiversity in the forest floor can increase. In addition, more sunlight is able to reach the ground and increase forage in the area which can enhance wildlife habitat for species like elk, mule deer, and wild turkey.

Biomass Sourcing and Pyrolysis – AECN
As discussed above, all biomass sourcing occurs under the auspices of the Quebec Provincial Management Plan for forests and is certified by SFI. Both the PMP and the SFI Standard require that the natural resources of forestland are managed sustainably with respect to both plant and animal life.

Bio-Oil Injection – KS
The above discussion of Vaulted's rigorous monitoring of their site extends to these considerations as well. Their ongoing efforts help to ensure against negative impacts on the surrounding ecosystem due to potential risks presented by the injection site. In addition, the Class V injection permit issued by KDHE entails compliance with EPA Underground Injection Control regulations.

Because emplacement is occurring at Vaulted's Great Plains Facility, please reference their account below:

"The Great Plains Facility underwent environmental assessment prior to securing Class V injection permits. The assessment found no material environmental issue with the site."

Bio-Oil Injection – LA
Only minimal costs and no environmental consequences are anticipated to bring the Basco 6 well online for active injection. Charm's planned operations require minimal energy and water resources, creating a small footprint to sustain and scale operations. Additionally, the

well site will not impact any wetland areas. Furthermore, the well facility is located at least 0.5 miles from any apparent occupied residential structures. There are no apparent or documented historically or culturally significant areas found within at least 0.5 miles of the well location. The well location is not subject to frequent flooding and is surrounded to the north, west and south by agricultural land.

Bayou Cocodrie is located to the north at a distance of approximately 0.5 miles away at its nearest proximity to the injection well. The portion of Bayou Cocodrie at this approximate distance from the injection well is listed with the Louisiana Department of Wildlife and Fisheries as a Designated Scenic Waterway. As evidenced by site reconnaissance and available online imagery, there is no apparent location at water or ground level on Bayou Cocodrie where the injection well and well pad location is visible to the naked eye, thus no visual aesthetic impact from the location of the injection well is expected. There are no expected impacts from any potential bio-oil or other surface spillage from operations at the Charm injection well facility on Bayou Cocodrie. Any spilled material will be captured by above ground containment systems and removed from the surface in a timely fashion. Any ambient surface or stormwater generally drains to the south southeast from the Basco 6 Well, away from Bayou Cocodrie. Highway 115 is located between Bayou Cocodrie and the site, providing a further barrier to any potential runoff.

Finally, the Basco 6 Well is not in the vicinity (1 mile) of known critical habitat, as confirmed from the US Fish and Wildlife Service IPaC Information for Planning and Consultation search tool.

Social Impacts

Labor rights and working conditions, including providing safe & healthy working conditions for employees, fair treatment and equal opportunities in your organization; considerations of prevention of forced labor, child labour or trafficked persons protecting workers employed by third parties.

This is required for all Projects.

X	Above risks are applicable to this Project

Charm has a robust Environmental Health and Safety department which ensures a safe and healthy working environment at all Charm facilities including required staff participation in safety training and the provision of task-appropriate PPE. As operations grow and change, additional training and equipment are proactively provided in order to ensure continued high safety performance. A culture of openness and communication around safety and hazard identification is encouraged and reinforced. It is Charm's company policy that any employee or contractor who feels a process is unsafe is encouraged to stop work, communicate the issue to appropriate managers, and not resume work until the issue is resolved. Charm pays living wages for all employees at all facilities.

Third-party partners (AECN and Vaulted) are duly licensed businesses in good standing within their jurisdictions. In both cases, this status means they are governed by laws and regulations which stipulate against practices such as forced labor, child labor, and the employment of trafficked persons.	
Land acquisition and involuntary resettlement in the context of your deployment site selection	
X	Above risks are applicable to this Project
	Not applicable to this Project
All Kansas project sites are owned by entities other than Charm with whom Charm has contracted for specific goods or services. Charm's engagement of these entities for the purpose of this project does not impact land use or acquisition in any material way. That being said, no known involuntary resettlement has occurred due to any third-party commercial activities. Charm leases and operates the Miniforge pyrolysis facility and nearby biomass storage site in Fort Lupton, Co and the Minifinery site in El Dorado, KS where bio oil is pre-processed to ensure suitability for injection. The footprints and usage profiles of these sites are not materially changed due to hosting Charm operations. No known involuntary resettlement has occurred due to the development of these sites. The Basco 6 Well is leased through a private landowner whose family has owned the property for at least three generations.	
Environmental and social justice, Indigenous Peoples, Local Communities, cultural heritage, human rights and gender equality (equal opportunities and pay), as it relates specifically to deployment site selection.	
X	Above risks are applicable to this Project
	Not applicable to this Project
There are no known environmental or social justice risks that result from Charm's site leases/rentals or arising from third-party sites that Charm has contracted with for the purposes of this project. All sites have been long-established as commercial operations and Charm's work does not create a meaningful change in site usage that would impact local communities or natural environments. Charm and all third-party partners operate in good standing regarding regulatory requirements for human rights, worker safety and nondiscrimination.	

Additional risks
In addition to the above areas of environmental and social impacts, I attest that further risks of potential harm arising from the Project have been assessed, and [please select one below option]:

X	A) No such additional risks have been identified
	B) For any such risks identified, appropriate risk mitigations have been put in place and are detailed below.

Please describe any such risks identified, and the corresponding risk mitigations which have been put in place.

Environmental and social risks arising from Charm's bio-oil production have been thoroughly addressed above. From feedstock sourcing to facility operation and interactions with the local community, we take care to embody our company value of "Do No Harm."

Outside of Charm's own bio-oil production, project operations primarily rely on Charm's contracting of products and services with third-party providers. The production of purchased bio-oil, its transportation and sequestration via underground injection in Kansas are all achieved by the continued operations of established businesses whose commercial model is based on the provision of these products and services. As such, Charm's work with these providers does not entail a significant change in their behavior or the assumption of additional risks beyond those associated with their normal operations independent of Charm's intervention. Therefore, we have not identified any additional risks associated with this project.

At the Colorado pyrolysis facility, Kansas pre-processing facility, and injection site in Louisiana, some physical changes to the site have been undertaken, such as pouring concrete pads and adding connections to utilities. However, these developments do not fundamentally change the footprint or nature of activity at those sites from a counterfactual scenario in which the project did not take place, as discussed in the Baseline Scenario section.

For these reasons, there are no additional risks that have been identified as part of this project.

Sustainable Development Impact

Please explain in the below table, where relevant, which Sustainable Development Goals (SDGs) other than SDG13 ("Climate Action") your Project has a positive impact and how this is consistent with the SDG objectives of all jurisdictions in which you operate. If applicable, please include information on any data and parameters monitored before or during the crediting period.

Please see here for a full list and details of all 17 SDGs.

#SDG with name	Description of project impact	Description of alignment with goals of jurisdiction(s), if relevant and applicable	If applicable, unit of measurement (e.g. removal credits, MWh)

8: Decent Work and Economic Growth	Charm's project activities create employment opportunities for living-wage jobs in a work environment that prioritizes safety. See "Labor Rights and Working Conditions"		
9: Industry, Innovation, and Infrastructure	By creating a demand for and sequestering pyrolysis oil, Charm's activities in this project support the development of an industrial infrastructure to remove GHGs that would have contributed to atmospheric warming at scale in the likely counterfactual scenario		
11: Sustainable Cities and Communities	By creating a demand for AECN's pyrolysis of wood residues associated with the production of wood boards for construction and sequestering the carbon associated with production process wastes the project contributes to the sustainability of cities and communities		
12: Responsible Production and Consumption	By creating a demand for AECN's pyrolysis of wood residues associated with the production of wood boards for construction and		

	sequestering the carbon associated with production process wastes the project makes the consumption of these wood products more sustainable		
15: Life on Land	Charm Industrial sources forest residues from a state managed Fuels Reduction and Forest Restoration project. By sourcing material from these types of projects, Charm is enabling this type of project work with the forest service in their conservation efforts with the forests in Colorado. These types of projects and treatment plans are put in place to help sustainably manage the forests. By using the residues from these projects, Charm is helping create more efficiencies in the work the forest service does and mitigating risks involved when these types of treatment plans are not put in place.		
Please explain whether you assessed the SDG impact of your project prior to this qualitative reasoning. If so, please describe if and which standardized assessment tools and methods you relied on.			
There has not been a formal assessment of SDG impacts – The qualitative content above is based on baseline project functions.			

E – Stakeholder Input Process

Stakeholder Input Process Summary
Please provide a description and documentation of how comments by local stakeholders have been invited and compiled, a summary of comments received, and report on how due account was taken of comments received. Refer to Section 3.5 "Stakeholder Input Process" of the Isometric Standard for full requirements.
Charm is committed to developing long lasting relationships and consistent two-way communication with community groups to ensure local economies see the benefits of its rapidly advancing technology and business. Charm's approach to community engagement is to communicate early and often with municipalities, community groups, related businesses and interested citizens in the regions where it operates or hopes to operate. Charm's Community Engagement Manager joined the team in March 2024, and she has been focused on establishing open communication and feedback systems with local groups, developing community and workforce development partnerships, hosting public outreach events, and working towards formalized community community partnerships. In all locations, Charm has done initial outreach to demystify the pyrolysis and injection process and to make Charm accessible to community members in case of any of concerns that may arise. **CO Bio-Oil Production** Charm Industrial has invested extensive time and resources into initial outreach to the community of Fort Lupton in order to best understand local priorities, challenges, and perceptions of carbon removal and Charm specifically. Charm became an official sponsor of the City of Fort Lupton in January 2024 and joined the Chamber of Commerce in June 2024. These sponsorships have allowed us to introduce ourselves to local businesses, nonprofits, academic institutions, and engaged community members, in addition to actively working with the Fort Lupton Planning Commission on permitting requirements. Local permitting requirements include public comment periods and formal public announcements to give communities a chance to provide input or ask questions. From these events and others, Charm keeps record of common questions and concerns, both from the Fort Lupton community as well as questions gathered at external carbon removal community outreach meetings. Examples of Colorado-based meetings include community listening sessions and outreach events hosted by the National Renewable Energy Lab (NREL), the Colorado Energy Office, and more. The most common questions are based around health and safety, such as clarification on the potential impacts on drinking water, any impacts on air quality, confirmation that bio oil is not combustible, and likelihood of and protocol for spills. Once health and safety concerns are addressed, many follow-up questions are about workforce development opportunities, training opportunities, and local economic impact. Due account has been taken of comments received, and has influenced not only the way the company publicly and locally addresses issues of health and safety, but also the type of local partnerships that Charm is dedicated to building in the community for the long-term. The local priorities and challenges that have been conveyed through these events and relationships will continue to guide Charm toward authentic community partnerships and ultimately the creation of an impactful Community Benefit Agreement.

Kansas Injection
Part of the rationale for Charm's decision to emplace bio oil at Vaulted's Great Plains cavern facility is their strong standing in the local community and commitment to social responsibility as cavern operators. Because Charm's work with Vaulted does not require any significant modification of their operations in a manner that would impact local stakeholders, their due diligence also extends to this project, which Charm has contracted with Vaulted in order to undertake. Please see Vaulted's summary of their Stakeholder Input Process below:

"At the Great Plains Facility, multiple sessions were held to solicit feedback from the surrounding community on the site. A site tour was conducted as well as two community meetings held to address concerns and questions. The main voiced question was to inquire about job opportunities at the site. The second question was around maintaining safe drinking water at and around the site. The community was told about the regular monitoring for containment of the formation and the regular groundwater checks.

In general, for future wells, Vaulted sees working with local communities, governments, and other stakeholders as essential to both scaling CDR work and ensuring maximum positive social and environmental impact. Generally, stakeholders include:
- Local, state, and federal regulators (generally, state and local EPA)
- Members of local government
- Nearby residents and landowners (especially within the anticipated radius of injectate migration / influence)
- The waste partners from whom Vaulted offtakes the waste
- Environmental interest groups / NGOs

Vaulted engages with each of these stakeholders across the lifespan of the work of the site - before Vaulted even begins the well permitting process. Before any steel is in the ground, Vaulted submits detailed well plans to the regulators at the start of the permitting process, holds community meetings to answer questions, and works with waste partners to finalize offtake. Vaulted continues to engage each stakeholder throughout the project - as Vaulted secures the permit, builds the well, and operates it on an ongoing basis. Vaulted's sites require regular re-permitting and regular reporting to regulatory and local government agencies, the outputs of which are publicly available. These activities generally entail public engagement via notices, hearing, regular quantification and reporting of net environmental impacts, and public access. The cadence of these activities provides Vaulted with regular input from the public via their elected representatives, responses to public notices, feedback from public presentation, and other vehicles."

Louisiana Injection
In early 2024 Charm started stakeholder engagement in Central Louisiana by arranging meetings with local representatives and Policy Jury members in Evangeline and Rapides Parish, in addition to hosting a presentation for the Evangeline Parish Chamber of Commerce. In summer 2024 Charm officially joined the Evangeline Parish Chamber of Commerce and has become a regular attendee for local meetings and workshops. Charm's first public outreach meeting is planned for spring 2025, however over the past year Charm has engaged with conservation and forestry institutions such as the Louisiana Fish &

Wildlife Foundation, academic institutions such as South Louisiana Community College (SOLACC), LSU's Department of Petroleum Engineering, regional economic development groups such as Louisiana Central and GNO, Inc, as well as workforce development and transition groups such as Acadiana Workforce Solutions and True Transitions. Many of the learnings collected from these conversations include:

- Broad concern about high numbers of young people leaving home and not returning because there are few local opportunities for gainful employment
- Citizens want safety regulations in place, especially around injection (ex. Ensure protection of aquifers and soil)
- Transportation is often a challenge when maintaining a consistent workforce in rural areas (workers sometimes don't have a car or a license)
- The co-benefits of Charm's operations are very appealing to a broad variety of stakeholders who are otherwise neutral about climate initiatives (farmers, landowners, workforce development institutions, economic development groups, and beyond)

Charm's injection permitting process includes a public comment period, during which any local citizen can submit questions or concerns which must be addressed by LDENR. Charm has requested a public outreach meeting as part of this process in order to address any public concerns in a timely manner. In preparation for this and other public outreach events, common questions and concerns are collected in Charm's Louisiana Stakeholder Mapping document. This document captures questions and readily shareable responses, especially those around health and safety. Charm continues to present to local municipal and community entities in Evangeline Parish and beyond.

Grievance Mechanism

Please outline the mechanism for stakeholders to voice, process and resolve grievances.

Charm Pyrolysis
For the Fort Lupton, CO facility, Charm's basic contact information is available on the Google listing as well as the Fort Lupton Chamber of Commerce website. This phone number goes to our Community Engagement Manager, who can relay the message to the appropriate Charm Industrial employee. Charm has made itself readily available to the local community as a member of the local Chamber of Commerce and as a sponsor of municipal events. Public comment periods are built into all Fort Lupton City Commission permitting processes, which give the public an opportunity to ask questions and voice concerns. These meetings are open to the public and can be attended in person or virtually.

Vaulted Injection
Because Charm's activities do not constitute a material change to Vaulted's operations, Vaulted's established grievance mechanism also applies to this project. Please see below for Vaulted's summary of their grievance mechanism:

“In the Great Plains Facility, Vaulted regularly interfaces with the Kansas Department of Health and Environment (KDHE), particularly their Underground Injection Control Division.
Vaulted submits monthly and quarterly reports to KDHE showing that the site is running safely, including data on groundwater quality, emplaced material spec, volume of injected material, and pressures and stability in the subsurface caverns. Vaulted has a strong ongoing relationship with KDHE, with regular calls and site visits to ensure they’re comfortable with the facility’s operations. Vaulted has also engaged with federal, local and county representatives, as well as residents and landowners proximate to the site.
Because the site is in a small community rural setting, Vaulted made neighbourhood outreach a top priority. Vaulted engaged the community before filing for permits at the project definition stage, throughout the permitting process, and once the permit was issued before commencing operations. These touch points included when the landowner originally permitted the site for waste emplacement, as well as when Vaulted filed for an organic waste permit in 2021 year through Advantek, Vaulted’s incubating company. In both cases, Vaulted/Advantek used county records to identify interested parties, sent letters to the nearby residents and landowners, hosted open houses (information sessions and site tours). On each occasion, Vaulted had local regulators from KDHE and local government officials present. Additionally, Vaulted posted the facility permits into the federal register and other publications as applicable to provide adequate public notice and opportunities for feedback, engagement, or protest.

For any issues or questions that arise community members and other stakeholders can reach out to the facility via phone at 620-662-6367.”

Louisiana Injection
Charm Industrial has publicly available phone numbers on our Google listing. Both of these phone lines connect directly to Charm’s Community Engagement Manager who can either address concerns directly or connect the caller with the appropriate Charm staff members. Charm’s contact information is also available on the Evangeline Chamber of Commerce Member Portal (public website) as well as the Fort Lupton Chamber of Commerce website.

Section F – Project Changes

F.1 Summary of changes

Please provide details of the proposed or actual changes to the implementation, operation or monitoring of the project activity description submitted to Isometric.
Project Proponents must determine whether the proposed or actual changes are temporary deviations or permanent changes

If there are temporary changes or deviations which are only relevant to a specific Reporting Period, please note this instead in Section C in the corresponding GHG Statement Report.

(Kindly note that this section may not be applicable during the initial validation. If so, kindly leave the section blank and mention "Section is intentionally left blank.")

9/9/25
During the past several months, Charm has worked together with Isometric to identify and implement some MRV changes in order to better fit evolving on-the-ground operations. The changes are as follows:

1. Charm pyrolysis currently produces 3 distinct products: Bio char and two bio-oil fractions “oily” and “aqueous”. Initially, it was not clear whether or not we would be able to successfully modify pre-processing operations to successfully render the aqueous oil non-hazardous. For that reason, we did not allocate production emissions to the aqueous oil, as the risk that it would not end up being injectable was too significant. By mid-year we had successfully proved out this process. As a result, starting with Production Period 7, which covers June production, we began to allocate emissions to all 3 products. Aqueous oil produced before June 2025 has no production emissions associated with it as all emissions have already been allocated to biochar and “oily” bio-oil
2. AECN bio-oil is weighed on a calibrated scale at the origination point and prior to injection. As the material is transferred between vessels during processing, volume and density are used to estimate the mass as it is not practical to weigh the material *in situ*. This can create upstream discrepancies as there are often small deltas between the product of volume and density and the mass measured on a scale. While this can be reconciled on a small scale, scaling up removals also scales up the opportunity for error in these manual adjustments. To avoid this, upstream emissions are allocated to injection batches by subtracting the Mass of Bio Oil Injected from the estimated Input Mass. Because the Input Mass is an estimate, that loss estimate is increased by 25% in order to ensure removals are accounted for conservatively, as this increases the production and transport emissions allocated to a removal, while holding the Mass of Bio Oil Injected constant. The emissions are allocated against the oldest BOL first, in order of delivery to the KS pre-processing site

3. KS permits require salt to be added to bio-oil at a level that significantly super-saturates the material. As a result, large deposits of salt are left over in the blending tank when the injectate is removed. Site engineering resources are able to volumetrically estimate the mass of salt in the tank by measuring deposit depth. In addition, multiple rounds of samples have been taken which show that the residue ranges from roughly 75-85% salt, with the remainder composed of entrained bio-oil. From the April/May Reporting Period on, 70% (used as a conservative estimate) of the estimated mass of the remaining residue may be counted towards the total salt added to the *subsequent* injection batch, provided the tank is not cleaned in between batches.
4. Over the course of processing and injecting bio-oil, Charm operators took full and empty weights of many IBC totes used to store finished bio-oil. This practice yielded a large dataset, which we used to create a conservative standard loss estimate. This was computed by taking the full and empty weights of 59 bio-oil totes and taking the average. That average was then increased by 25% to ensure that the estimate would be conservative. Rather than take the empty weight of each individual tote, this standard loss (which came out to 5.33kg) is assumed for each tote of "oily fraction" Charm bio-oil used. There will be an annual revision of this estimate to ensure it stays current with the physical character of Charm's bio-oil.

7/18/25

1. Charm Industrial has recently closed the public comment period for its permit to operate the Basco 6 injection well at an abandoned oil and gas reservoir in Evangeline Parish, LA. In anticipation of the site entering into operation in August 2025, we have expanded the PDD to include a new injection facility
2. This expansion required adding to the following sections in order to include the full details for GHG accounting associated with Charm pyrolysis as detailed in the Bio-Oil Geological Storage module as well as associated modules covering transport, energy use, etc.:
    1. Title
    2. Description
    3. Project Location(s)
    4. Technical Description of Project Activity
    5. Estimated Carbon Removal Capacity
    6. Project Boundary
    7. Baseline Scenario
    8. Durability Assessment
    9. Monitoring Plan
    10. Data Collection and Storage
    11. Uncertainty Assessment
    12. Storage Well Overview
    13. Project Closure
    14. Environmental Impacts

15. Biodiversity Conservation and Sustainable Management of Living Natural Resources
16. Land Acquisition and Involuntary Resettlement
17. Environmental and Social Justice, Indigenous Peoples, Local Communities, Cultural Heritage, Human Rights and Gender Equality (equal opportunities and pay) as it relates specifically to deployment site selection
18. Additional Risks
19. Stakeholder Input Process Summary
20. Grievance Mechanism
21. Appendix 6 and 7
22. Supplemental doc 3

3. Appendix 3 has been added to account for a new feedstock being processed by Charm Industrial
4. Minor edits have been made to update existing PDD content to reflect Charm operations

4/8/25

1. Charm Industrial will begin using Renewable Energy Credits (RECs) in accordance with the calculation approach outlined in the Energy Use Accounting module Section 5 and the eligibility criteria outlined in Section 6. The electricity accounting details provided in Appendix 4 have been updated to reflect this change.

2/19/25

1. Charm Industrial will begin using Book-and-Claim Units (BCUs) from Renewable Diesel in our Net CDR calculations, in accordance with the eligibility criteria outlined in the Transportation Emissions Accounting module v1.1. Per Section 4.2.2, Charm has elected to use a bespoke calculation approach which has been reviewed and approved by Isometric. Details on the calculation approach are included in Appendix 4 below.
2. Charm has begun injecting bio-oil into a new cavern at Vaulted, AR-9, following the closure and sealing of SW-12, which had been used previously. This is still part of Vaulted's facility, utilizing their existing equipment and is part of the same cavern system documented in this PDD. However, Charm has conducted diligence and documentation of the cavern's characteristics in accordance with the requirements of the Biomass or Bio-Oil Storage in Salt Caverns module and submitted these documents to Isometric and the VVB.
3. Site operations began to generate bio-oil process wastes in H2 of 2024. In order to account for emissions resulting from disposal processes, Charm is using an emissions calculation approach that accounts for both the transport of the wastes to the disposal site, and the emissions impact of the disposal itself.
    a. Calculation of transport emissions for waste disposal are conducted using the same emission factors and methods as all other transport emissions for the project

b. The emission impacts of the physical disposal are estimated using emission factors from reputable, peer-reviewed sources such as Open LCA, Ecoinvent, etc.
    i. Emission factors are used as-is where feasible. To account for the disposal of less-studied material such as bio-oil, a reasonable estimate may be made using proxy values. In this case, references to baseline values and sources, as well as any additional assumptions and justifications will be documented by the project and are subject to review and approval by Isometric
c. Details regarding the data sources and calculation methods described have been added to Appendix 3

9/29/24

The primary changes made to this PDD are an expansion to include Charm's own pyrolysis as a bio-oil source alongside offtakes from AECN.

This expansion required adding to the following sections in order to include the full details for GHG accounting associated with Charm pyrolysis as detailed in the Bio-Oil Geological Storage module as well as associated modules covering transport, energy use, etc.:

23. Project Location(s)
24. Technical Description of Project Activity
25. Project Boundary
26. Baseline Scenario
27. Leakage Assessment
28. Demonstration of Additionality: Financial Additionailty
29. Monitoring Plan
30. Data Collection and Storage
31. Uncertainty Assessment
32. Environmental Impacts
33. Biodiversity Conservation and Sustainable Management of Living Natural Resources
34. Land Acquisition and Involuntary Resettlement
35. Environmental and Social Justice, Indigenous Peoples, Local Communities, Cultural Heritage, Human Rights and Gender Equality (equal opportunities and pay) as it relates specifically to deployment site selection
36. Additional Risks
37. Sustainable Development Impact
38. Stakeholder Input Process Summary
39. Grievance Mechanism
40. Biomass Feedstock Appendix
    a. New appendix (Appendix 2) added
41. GHG Statement Methodology – Data
42. GHG Statement Methodology – Calculations

F.2 Impact of changes
Please provide details on the impact of the proposed or actual changes (temporary/permanent) on the following sections of the project: *a) Additionality* *b) Applicability of Protocol/ Module* *c) Quantification of the carbon removals* *d) Any other (stakeholder consultation, safeguard assessment etc.)* *(Kindly note that this section may not be applicable during the initial validation. If so, kindly leave the section blank and mention "Section is intentionally left blank.")*
9/10/25 **a) Additionality** i) There is no significant change to project additionality as a result of changes documented at this time **b) Applicability of Protocol/Module** i) There is no significant change to the applicability of any protocol or module to the project as a result of changes documented at this time **c) Quantification of Carbon Removals** i) Changes consist of minor changes to better accommodate evolving operations. Any impact on the quantification of carbon removals will be *de minimis* **7/18/25** **d) Additionality** i) There is no significant change to project additionality as a result of changes documented at this time **e) Applicability of Protocol/Module** i) There is no significant change to the applicability of any protocol or module to the project as a result of changes documented at this time **f) Quantification of Carbon Removals** i) While largely unchanged, specific calculation modifications made to reflect Louisiana well operations are summarized in the Comprehensive Emissions Monitoring and Accounting Plan included as a PDD supplement **4/8/25** **g) Additionality** i) There is no significant change to project additionality as a result of changes documented at this time **h) Applicability of Protocol/Module** i) There is no significant change to the applicability of any protocol or module to the project as a result of changes documented at this time **i) Quantification of Carbon Removals**

i) Net CDR calculations will only include the grid emissions that are not matched by an equivalent volume of verifiable RECs

2/19/25

j) Additionality

i) There is no significant change to project additionality as a result of changes documented at this time

k) Applicability of Protocol/Module

i) There is no significant change to the applicability of any protocol or module to the project as a result of changes documented at this time

l) Quantification of Carbon Removals

i) Net CDR calculations will only include diesel emissions that are not matched by emissions avoided by the verifiable substitution of renewable diesel for fossil diesel outside of an LCFS jurisdiction

9/29/24

a) Additionality

i) There is no significant change in project additionality as a result of using Charm-produced oil. As noted in the Financial Additionality section, biochar is a co-product of bio-oil production via pyrolysis. Charm may sell this biochar to an offtaker or use it for additional credit generation, but even in that case the revenue from these activities would not approach netting out the cost of bio-oil production

b) Applicability of Protocol/Module

i) Regardless of the supplier, the project activity is still the sequestration of bio-oil through geological sequestration, so the Bio-Oil Geological Storage module is still applicable

c) Quantification of Carbon Removals

i) Charm's control of its own bio-oil production will allow for a more granular and frequent quantification of the emissions impact of its production operations. That being said, the methodology for quantification of carbon removals is not fundamentally changed as a result of this PDD expansion

F.3 History of changes

Please provide the summary of the changes in the table
(Kindly note that this section may not be applicable during the initial validation. If so, kindly leave the section blank and mention "Section is intentionally left blank.")

A change log has been provided as a separate document alongside this PDD

Appendix 1: AECN Biomass Feedstock Information

This Appendix must be completed for each feedstock type and feedstock provider used in this Project.

Feedstock Summary
Please describe your Feedstocks used.
The biomass feedstock used for this Project is sawdust and shavings from a lumber mill adjacent to the bio-oil provider, AECN. Groupe Remabec, which is a co-owner of AECN, harvests trees from publicly-owned forests in accordance with rights granted by the Province of Quebec under the Provincial Management Plan (provided as supporting documentation for this Project). The harvest activity is also SFI-certified through the Bureau de Normalisation du Québec, a copy of the certificate is also provided as additional documentation. The sawdust and shavings are a byproduct of the Arbec sawmill's production of wood boards, which are the primary revenue source for the mill. A small portion (10%) of the feedstock consists of chipped unmerchantable roundwood that had been previously harvested and transported to AECN.

Feedstock Hazardous status
Please describe how you are demonstrating that the feedstock you are using is not hazardous. This may either be done by providing evidence from tests of the feedstock or by providing evidence that the relevant injection permits only allow non-hazardous materials.
The feedstock is non-hazardous in itself. The resulting pyrolysis oil is pre-treated by Charm and tested to assure it complies with UIC regulations prior to its emplacement at Vaulted.

Feedstock Provider Organizations
Please provide a complete list of organizations involved in providing the feedstock, clarifying the organization's role and providing contact information for each. (*Please duplicate the below rows for each additional organization you wish to add)*
Organization 1
Company registration number (*Unique business identification number in your country of registration*): 1168427111
Organization Name: AE Cote-Nord Canada Bioenergy Inc.
Organization Address: 210-8000 Boulevard Langelier Sant-Leonard, QC H1P3K2 Canada
Contact Person:

Contact Email Address: pct.rh@arbec.co
[Optional] Contact Phone Number:
Organization role in project: Bio-Oil Vendor

Potential Market Leakage Impacts Eligibility	
Please select which of the Potential Market Leakage Impact Eligibility Criteria you're using to demonstrate eligibility for this feedstock.	
	EC1: Project Proponent does not pay for the feedstock
	EC2: Project Proponent receives a payment for the feedstock
	EC3: Project Proponent pays for recovery & replacement activities only
	EC4: Project Proponent pays to a 3rd party, not entity producing feedstock
X	EC5: Publicly managed forest management activity
	EC6: Certified forest management activity in increasing carbon stock areas
	EC7: Certified forest management activity in exceptional circumstances
	EC8: Sustainably sourced agricultural crop residue
	EC9: Surplus residue with no demonstrated growth of supply
Demonstrate how your feedstock meets it by providing the required documentation or analysis.	
Groupe Remabec, which is a parent company of AECN, harvests trees from publicly-owned forests in accordance with rights granted by the Province of Quebec under the Provincial Management Plan provided as supporting documentation for this Project. The harvest activity is also SFI-certified through the Bureau de Normalisation du Québec, a copy of the certificate is provided as additional documentation.	

Counterfactual Storage Eligibility
Please describe and attach any relevant evidence to demonstrate that your feedstock would have emitted the biogenic CO_2 to the atmosphere sooner than the required threshold period. If only a portion of your feedstock would have emitted the CO_2 after the threshold

period, attach relevant evidence and confirm that you have incorporated the relevant calculation into your LCA.

The Arbec sawmill, which provides AECN the sawdust and shavings used in pyrolysis, has a pile of sawmill residues that has been on site for years. The sawmill accumulated the pile of residues as a result of the shutdown of local businesses who had purchased it in the past. AECN was developed as a solution to this issue of residue accumulation. The historic fate of these residues absent AECN or other offtakers has been accumulation in this large pile.

Pier and Kelly's study of waste sawdust in the Tennessee Valley (Pier and Kelly, 1997), included as documentation with this PDD, shows that anaerobic conditions in the center of such piles lead to significant methane production.

In order to compute a conservative estimation of the counterfactual storage of piled sawdust, we assume that 1% of the biogenic carbon lost to decomposition is emitted as methane. In terms of the overall decomposition rate, studies of woody debris decay vary widely. However, Fasth et al. (2011) found an average of a 24% annual decay rate for smaller-sized particles in a study of 11 different tree species. This is confirmed by Blasdell's research on the decay rates of forest litter — This found a median 50% residence time of 2.48 years, from which a comparable decay rate can be inferred.

Using this as a benchmark, the total CO2e released by decomposing wood within 15 years, from emission of CO2 and CH4, would exceed the total stored carbon of the wood. Additionally, at a 24% decay rate, after 50 years the piled woody material would have effectively decayed entirely. For this reason, we are assigning the sawmill residues used by AECN as a pyrolysis feedstock a counterfactual storage discount of 0. This calculation is in accordance with EC13 of the Biomass Feedstock Accounting module.

Dedicated Energy Feedstock Eligibility

This is only applicable to non-forestry feedstocks.

Please describe your analysis in how you determined that your feedstock isn't grown for the purposes of energy production.

N/A

Counterfactual Fate of Feedstock

Please describe and attach any relevant evidence to demonstrate what the most likely counterfactual scenario would be for your feedstock, using guidance outlined in Section 3.2.1 of the Biomass Feedstock Module.

The bio-oil Charm sources from AECN is produced using sawdust and shavings generated by the Arbec lumber mill adjacent to AECN's pyrolysis facility. By purchasing the oil, Charm creates demand that AECN meets by offtaking additional residues from the mill.

AECN's development was in large part triggered by the closure of other businesses in the region who had historically purchased these residues as a production input. Due to the remote location of the mill, the development of economically viable offtake relationships with businesses outside the region is a challenge. This is evidenced by Arbec and Groupe Remabec's decision to invest over $100 million CAD to start AECN and build out an onsite pyrolysis facility, as well as the accumulation of a large pile of sawdust as sawmill operations continued without a residue offtaker for a significant period of time. The AECN facility was the solution to this problem.

Because its primary revenue stream is the production of wood boards for construction, the mill would have reason to continue to operate and create the residues currently used as pyrolysis feedstock by AECN if the pyrolysis facility were absent. Given the demonstrated difficulty of securing alternative offtake relationships, the historical fate of the feedstock in this scenario has been accumulation and piling on site.

Replacement Emissions	
Please select which method of replacement emissions accounting you've selected given the nature of your feedstock	
	Accounting for the feedstock replacement emissions in the LCA
X	Not accounting for replacement emissions due to exemption C1 *(Feedstock has no counterfactual use)*
	Not accounting for replacement emissions due to exemption C2 *(Feedstock counterfactual use is most likely replaced with a feedstock with no counterfactual use)*
	Not accounting for replacement emissions due to exemption C3 *(Feedstock has no counterfactual use due to surplus)*
If you selected an exemption above, please demonstrate how your feedstock meets it by providing the required documentation or analysis.	
See "Counterfactual Fate of Feedstock"	

Appendix 2: Charm Biomass Feedstock Information – MacArthur Gulch

This Appendix must be completed for each feedstock type and feedstock provider used in this Project.

Feedstock Summary
Please describe your Feedstocks used.
The biomass feedstock used for this Project is non-sawmill quality Ponderosa Pines and Douglas Firs from a wildfire mitigation project managed by the Colorado State Forest Service (CSFS) called the McArthur Gulch Fuels Reduction and Forest Restoration project. Due to the risk of catastrophic wildfires in certain areas of Colorado and concerns of overstocked forests showing signs and symptoms of insect infestation and disease, collaborative partnerships across the state have been formed to help focus resources to identify high-priority critical landscapes and create forest treatment plans to ensure resiliency within these landscapes. One of these collaboratives, The Rocky Mountain Restoration Initiative (RMRI), has identified the Upper South Platte, the location of this project, a high-priority critical area. Altitude Forestry is the sole contractor hired by the CSFS to conduct the treatment for this project, and has supplied the feedstock used in Charm's operations. As part of the CSFS contract agreements for this project, the disposal of the forest residues are the responsibility of the contractor. As the treatment plan for this project is managed and organized by the CSFS, a copy of the state forest project RFP and contractor bid is provided as supporting documentation.

Feedstock Hazardous status
Please describe how you are demonstrating that the feedstock you are using is not hazardous. This may either be done by providing evidence from tests of the feedstock or by providing evidence that the relevant injection permits only allow non-hazardous materials.
The feedstock is non-hazardous in itself. The resulting pyrolysis oil is tested to assure it complies with UIC regulations prior to its emplacement.

Feedstock Provider Organizations
Please provide a complete list of organizations involved in providing the feedstock, clarifying the organization's role and providing contact information for each. (*Please duplicate the below rows for each additional organization you wish to add)*
Organization 1
Company registration number (*Unique business identification number in your country of registration*):

Organization Name: Altitude Forestry
Organization Address: 13521 Spruce Creek Circle Larkspur, CO 80118 US
Contact Person: Tyler Matthews
Contact Email Address: tmatthews@altitude.com
[Optional] Contact Phone Number:
Organization role in project: Biomass Supplier

Potential Market Leakage Impacts Eligibility	
Please select which of the Potential Market Leakage Impact Eligibility Criteria you're using to demonstrate eligibility for this feedstock.	
	EC1: Project Proponent does not pay for the feedstock
	EC2: Project Proponent receives a payment for the feedstock
	EC3: Project Proponent pays for recovery & replacement activities only
	EC4: Project Proponent pays to a 3rd party, not entity producing feedstock
X	EC5: Feedstock is a non-marketable forest residue
	EC6: Feedstock is an agricultural crop residue or animal waste
	EC7: Feedstock is a biogenic product that is a non-marketable waste product
X	EC8: Publicly managed forest management activity
	EC9: Certified forest management activity in increasing carbon stock areas
	EC10: Certified forest management activity in exceptional circumstances
	EC11: Sustainably sourced agricultural crop residue
	EC12: Sustainably sourced animal waste
Demonstrate how your feedstock meets it by providing the required documentation or analysis.	

The supplier, Altitude Forestry, was contracted to treat the area and harvest the wood for the McArthur Gulch project, a wildfire mitigation project managed by the Colorado State Forest Service (CSFS). An official project description from the CSFS, and the bid provided by the contractor to the State Forest Service are provided as supporting documentation for this Project.

Counterfactual Storage Eligibility

Please describe and attach any relevant evidence to demonstrate that your feedstock would have emitted the biogenic CO_2 to the atmosphere sooner than the required threshold period. If only a portion of your feedstock would have emitted the CO_2 after the threshold period, attach relevant evidence and confirm that you have incorporated the relevant calculation into your LCA.

Charm procured full logs from Altitude Forestry, a logging operations contractor, that was hired by the Colorado State Forest Service (CSFS) to perform the forest treatment plan for the McArthur Gulch project set out by the CSFS. The project was managed and considered high-priority by the CSFS for fuels reduction and forest restoration, due to the numerous beneficial outcomes from the treatment including a risk reduction of catastrophic wildfires, a protection of watershed health, and an enhancement of wildlife habitats for animals.

As part of the contract agreement between Altitude and the CSFS, Altitude Forestry is in charge of the disposal and removal of the biomass from the project site post-treatment. The logs were harvested in the project area as part of the forest treatment plan, then stacked and brought to a decking site in the project area post-harvest.

According to Altitude Forestry and the CSFS, there is limited to no utilization for this type of feedstock in Colorado due to the quality of the wood and the species available. The wood is either considered crooked and does not meet sawmill standards or is not economically viable to transport due to the lack of sawmills in the area and the high cost of transportation. Supporting documentation was provided from Altitude Forestry and the Colorado State Forest Service highlighting that the logs sourced from this project are considered non-merchantable biomass with limited disposal options in a fire-prone ecosystem. With this lack of utilization, the biomass from this project would most likely sit in the decking site of the area.

According to the Colorado State Wildfire Risk map, this project area is considered to have a high level of wildfire risk, and a very high level for Wildland Urban Interface (WUI) risk. According to the U.S. Fire Administration, WUI is a term commonly known in areas that experience wildfires. It is a zone between wilderness land and human activity that is considered to have greater risk of catastrophic wildfires.

Given the above, the biomass clearly falls under EC14 – Feedstocks that are sourced through government managed or funded wildfire mitigation or restoration activities. For this reason, we are assigning the feedstock from this wildfire mitigation project a counterfactual storage discount of 0.

Dedicated Energy Feedstock Eligibility
This is only applicable to non-forestry feedstocks. *Please describe your analysis in how you determined that your feedstock isn't grown for the purposes of energy production.*
N/A – All feedstocks described in this appendix are forestry feedstocks

Counterfactual Fate of Feedstock
Please describe and attach any relevant evidence to demonstrate what the most likely counterfactual scenario would be for your feedstock, using guidance outlined in Section 3.2.1 of the Biomass Feedstock Module.

The biomass Charm sources are forest residues (green logs) harvested by a contractor, Altitude Forestry, from a government regulated and funded forest management project. The project, known as the McArthur Gulch Forest Restoration and Fuels Reduction project, was developed by the Colorado State Forest Service (CSFS) with one of the desired primary outcomes being to reduce the risk of catastrophic wildfires and facilitate fire suppression in the project area.

According to Altitude Forestry and the CSFS, there is limited to no utilization for this type of feedstock in Colorado due to the quality of the wood and the species available. The wood is either considered crooked and does not meet sawmill standards or does not become economically viable to transport due to the lack of sawmills in the area and the high cost of transportation. Supporting documentation was provided from Altitude Forestry and the Colorado State Forest Service highlighting that the logs sourced from this project are considered non-merchantable biomass with limited disposal options in a fire-prone ecosystem. With this lack of utilization, the biomass from this project would most likely sit in the decking site of the area. For this reason, the feedstock falls under Condition C1 in Section 3.2.2 of the Biomass Feedstock Accounting Module and can be considered to have 0 replacement emissions.

Given that the feedstock source both presents little to no risk of market leakage (EC 5 and 8) and has no replacement counterfactual (C1), counterfactual emissions in general are considered to be 0.

Replacement Emissions	
Please select which method of replacement emissions accounting you've selected given the nature of your feedstock	
	Accounting for the feedstock replacement emissions in the LCA

X	Not accounting for replacement emissions due to exemption C1 *(Feedstock has no counterfactual use)*
	Not accounting for replacement emissions due to exemption C2 *(Feedstock counterfactual use is most likely replaced with a feedstock with no counterfactual use)*
	Not accounting for replacement emissions due to exemption C3 *(Feedstock has no counterfactual use due to surplus)*
If you selected an exemption above, please demonstrate how your feedstock meets it by providing the required documentation or analysis.	
See “Counterfactual Fate of Feedstock”	

Appendix 3: Charm Biomass Feedstock Information – Arkansas Mountain

This Appendix must be completed for each feedstock type and feedstock provider used in this Project.

Feedstock Summary
Please describe your Feedstocks used.
The biomass feedstock used for this Project is non-sawmill quality Ponderosa Pines and Douglas Firs from a wildfire mitigation project funded by the Colorado State Forest Service (CSFS) called the Arkansas Mountain Fuels Reduction project. Due to the risk of catastrophic wildfires in certain areas of Colorado and concerns of overstocked forests showing signs and symptoms of insect infestation and disease, collaborative partnerships across the state have been formed to help focus resources to identify high-priority critical landscapes and create forest treatment plans to ensure resiliency within these landscapes. One of these collaboratives, The Boulder Watershed Collective (BWC), has identified Arkansas Mountain, the location of this project, a high-priority critical area. Altitude Forestry is hired by the BWC using fuels reduction grant funds from the CSFS to conduct the treatment for this project, and has supplied the feedstock used in Charm's operations. As part of the BWC contract agreements for this project, the disposal of the forest residues are the responsibility of the contractor. As the treatment plan for this project is managed and organized by the CSFS, a copy of the state forest project RFP and contractor bid is provided as supporting documentation.

Feedstock Hazardous status
Please describe how you are demonstrating that the feedstock you are using is not hazardous. This may either be done by providing evidence from tests of the feedstock or by providing evidence that the relevant injection permits only allow non-hazardous materials.
The feedstock is non-hazardous in itself. The resulting pyrolysis oil is tested to assure it complies with UIC regulations prior to its emplacement.

Feedstock Provider Organizations
Please provide a complete list of organizations involved in providing the feedstock, clarifying the organization's role and providing contact information for each. (*Please duplicate the below rows for each additional organization you wish to add)*
Organization 1
Company registration number (*Unique business identification number in your country of registration*):

Organization Name: Altitude Forestry
Organization Address: 13521 Spruce Creek Circle Larkspur, CO 80118 US
Contact Person: Tyler Matthews
Contact Email Address: tmatthews@altitude.com
[Optional] Contact Phone Number:
Organization role in project: Biomass Supplier

Potential Market Leakage Impacts Eligibility	
Please select which of the Potential Market Leakage Impact Eligibility Criteria you're using to demonstrate eligibility for this feedstock.	
	EC1: Project Proponent does not pay for the feedstock
	EC2: Project Proponent receives a payment for the feedstock
	EC3: Project Proponent pays for recovery & replacement activities only
	EC4: Project Proponent pays to a 3rd party, not entity producing feedstock
X	EC5: Feedstock is a non-marketable forest residue
	EC6: Feedstock is an agricultural crop residue or animal waste
	EC7: Feedstock is a biogenic product that is a non-marketable waste product
X	EC8: Publicly managed forest management activity
	EC9: Certified forest management activity in increasing carbon stock areas
	EC10: Certified forest management activity in exceptional circumstances
	EC11: Sustainably sourced agricultural crop residue
	EC12: Sustainably sourced animal waste
Demonstrate how your feedstock meets it by providing the required documentation or analysis.	

The supplier, Altitude Forestry, was contracted to treat the area and harvest the wood for the Arkansas Mountain Fuels Reduction project, a wildfire mitigation project managed by the Boulder Watershed Collective (BWC) and funded by the Colorado State Forest Service (CSFS). An official project description from the BWC, and the bid provided by the contractor to the State Forest Service are provided as supporting documentation for this Project.

Counterfactual Storage Eligibility

Please describe and attach any relevant evidence to demonstrate that your feedstock would have emitted the biogenic CO_2 to the atmosphere sooner than the required threshold period. If only a portion of your feedstock would have emitted the CO_2 after the threshold period, attach relevant evidence and confirm that you have incorporated the relevant calculation into your LCA.

Charm procured full logs from Altitude Forestry, a logging operations contractor, that was hired by the Boulder Watershed Collective (BWC) to perform the forest treatment plan for the Arkansas Mountain Fuels Reduction project funded by the CSFS. The project was managed and considered high-priority by the BWC and CSFS for fuels reduction and forest restoration, due to the numerous beneficial outcomes from the treatment including a risk reduction of catastrophic wildfires, an improvement of forest health, a promotion of healthy regeneration of the understory, and an enhancement of wildlife habitats and species diversity.

As part of the contract agreement between Altitude and the BWC, Altitude Forestry is in charge of the disposal and removal of the biomass from the project site post-treatment. The logs were harvested in the project area as part of the forest treatment plan, then stacked and brought to a decking site in the project area post-harvest.

According to Altitude Forestry and the BWC, there is limited to no utilization for this type of feedstock in Colorado due to the quality of the wood and the species available. The wood is either considered crooked and does not meet sawmill standards or is not economically viable to transport due to the lack of sawmills in the area and the high cost of transportation. Supporting documentation was provided from Altitude Forestry and the Colorado State Forest Service highlighting that the logs sourced from this project are considered non-merchantable biomass with limited disposal options in a fire-prone ecosystem. With this lack of utilization, the biomass from this project would most likely sit in the decking site of the area.

According to the Colorado State Wildfire Risk map, this project area is considered to have a high level of wildfire risk, and a very high level for Wildland Urban Interface (WUI) risk. According to the U.S. Fire Administration, WUI is a term commonly known in areas that experience wildfires. It is a zone between wilderness land and human activity that is considered to have greater risk of catastrophic wildfires.

Given the above, the biomass clearly falls under EC14 – Feedstocks that are sourced through government managed or funded wildfire mitigation or restoration activities. For

this reason, we are assigning the feedstock from this wildfire mitigation project a counterfactual storage discount of 0.

Dedicated Energy Feedstock Eligibility

This is only applicable to non-forestry feedstocks.

Please describe your analysis in how you determined that your feedstock isn't grown for the purposes of energy production.

N/A – All feedstocks described in this appendix are forestry feedstocks

Counterfactual Fate of Feedstock

Please describe and attach any relevant evidence to demonstrate what the most likely counterfactual scenario would be for your feedstock, using guidance outlined in Section 3.2.1 of the Biomass Feedstock Module.

The biomass Charm sources are forest residues (green logs) harvested by a contractor, Altitude Forestry, from a government-funded forest management project. The project, known as the Arkansas Mountain Fuels Reduction project, was developed by the Boulder Watershed Collective (BWC) and funded by the Colorado State Forest Service (CSFS) with one of the desired primary outcomes being to reduce the risk of catastrophic wildfires and facilitate fire suppression in the project area.

According to Altitude Forestry and the BWC, there is limited to no utilization for this type of feedstock in Colorado due to the quality of the wood and the species available. The wood is either considered crooked and does not meet sawmill standards or does not become economically viable to transport due to the lack of sawmills in the area and the high cost of transportation. Supporting documentation was provided from Altitude Forestry and the BWC highlighting that the logs sourced from this project are considered non-merchantable biomass with limited disposal options in a fire-prone ecosystem. With this lack of utilization, the biomass from this project would most likely sit in the decking site of the area. For this reason, the feedstock falls under Condition C1 in Section 3.2.2 of the Biomass Feedstock Accounting Module and can be considered to have 0 replacement emissions.

Given that the feedstock source both presents little to no risk of market leakage (EC 5 and 8) and has no replacement counterfactual (C1), counterfactual emissions in general are considered to be 0.

Replacement Emissions

Please select which method of replacement emissions accounting you've selected given the nature of your feedstock	
	Accounting for the feedstock replacement emissions in the LCA
X	Not accounting for replacement emissions due to exemption C1 *(Feedstock has no counterfactual use)*
	Not accounting for replacement emissions due to exemption C2 *(Feedstock counterfactual use is most likely replaced with a feedstock with no counterfactual use)*
	Not accounting for replacement emissions due to exemption C3 *(Feedstock has no counterfactual use due to surplus)*
If you selected an exemption above, please demonstrate how your feedstock meets it by providing the required documentation or analysis.	
See "Counterfactual Fate of Feedstock"	

Appendix 4: Charm Biomass Feedstock Information – Arroyo Chico

This Appendix must be completed for each feedstock type and feedstock provider used in this Project.

Feedstock Summary
Please describe your Feedstocks used.
The biomass feedstock used for this Project is non-sawmill quality Ponderosa Pinesfrom a wildfire mitigation project funded by the Colorado State Forest Service (CSFS) called the Four Mile WUI Priority Project/Arroyo Chico II Fuels Reduction project. Due to the risk of catastrophic wildfires in certain areas of Colorado and concerns of overstocked forests showing signs and symptoms of insect infestation and disease, collaborative partnerships across the state have been formed to help focus resources to identify high-priority critical landscapes and create forest treatment plans to ensure resiliency within these landscapes. One of these collaboratives is a fire department called the Four Mile Fire Protection District (FMFPD), has identified Arroyo Chico, the location of this project, a high-priority critical area. Altitude Forestry is hired by the FMFPD using fuels reduction grant funds from the CSFS to conduct the treatment for this project, and has supplied the feedstock used in Charm's operations. As part of the FMFPD contract agreements for this project, the disposal of the forest residues are the responsibility of the contractor. The treatment plan for this project is managed and organized by FMFPD, but funded by and coordinated with the CSFS, The forest project RFP, contractor bid and CSFS grant award letters are provided as supporting documentation.

Feedstock Hazardous status
Please describe how you are demonstrating that the feedstock you are using is not hazardous. This may either be done by providing evidence from tests of the feedstock or by providing evidence that the relevant injection permits only allow non-hazardous materials.
The feedstock is non-hazardous in itself. The resulting pyrolysis oil is tested to assure it complies with UIC regulations prior to its emplacement.

Feedstock Provider Organizations
Please provide a complete list of organizations involved in providing the feedstock, clarifying the organization's role and providing contact information for each. (*Please duplicate the below rows for each additional organization you wish to add)*
Organization 1
Company registration number (*Unique business identification number in your country of registration*):

Organization Name: Altitude Forestry
Organization Address: 13521 Spruce Creek Circle Larkspur, CO 80118 US
Contact Person: Tyler Matthews
Contact Email Address: tmatthews@altitude.com
[Optional] Contact Phone Number:
Organization role in project: Biomass Supplier

Potential Market Leakage Impacts Eligibility	
Please select which of the Potential Market Leakage Impact Eligibility Criteria you're using to demonstrate eligibility for this feedstock.	
	EC1: Project Proponent does not pay for the feedstock
	EC2: Project Proponent receives a payment for the feedstock
	EC3: Project Proponent pays for recovery & replacement activities only
	EC4: Project Proponent pays to a 3rd party, not entity producing feedstock
X	EC5: Feedstock is a non-marketable forest residue
	EC6: Feedstock is an agricultural crop residue or animal waste
	EC7: Feedstock is a biogenic product that is a non-marketable waste product
X	EC8: Publicly managed forest management activity
	EC9: Certified forest management activity in increasing carbon stock areas
	EC10: Certified forest management activity in exceptional circumstances
	EC11: Sustainably sourced agricultural crop residue
	EC12: Sustainably sourced animal waste
Demonstrate how your feedstock meets it by providing the required documentation or analysis.	

The supplier, Altitude Forestry, was contracted to treat the area and harvest the wood for the Arroyo Chico II Fuels Reduction Project, a wildfire mitigation project managed by the Four Mile Fire Protection District (FMFPD) and funded by the Colorado State Forest Service (CSFS). An official project description from the BWC, and the bid provided by the contractor to the State Forest Service are provided as supporting documentation for this Project.

Counterfactual Storage Eligibility

Please describe and attach any relevant evidence to demonstrate that your feedstock would have emitted the biogenic CO_2 to the atmosphere sooner than the required threshold period. If only a portion of your feedstock would have emitted the CO_2 after the threshold period, attach relevant evidence and confirm that you have incorporated the relevant calculation into your LCA.

Charm procured full logs from Altitude Forestry, a logging operations contractor, that was hired by the Four Mile Fire Protection District (FMFPD) to perform the forest treatment plan for the Arroyo Chico II Fuels Reduction Project funded by the CSFS. The project was managed and considered high-priority by the FMFPD and CSFS for fuels reduction and forest restoration, due to the numerous beneficial outcomes from the treatment including a risk reduction of catastrophic wildfires, an improvement of forest health, a promotion of healthy regeneration of the understory, and an enhancement of wildlife habitats and species diversity.

As part of the contract agreement between Altitude and the FMFPD, Altitude Forestry is in charge of the disposal and removal of the biomass from the project site post-treatment. The logs were harvested in the project area as part of the forest treatment plan, then stacked and brought to a decking site in the project area post-harvest.

According to Altitude Forestry and the FMFPD, there is limited to no utilization for this type of feedstock in Colorado due to the quality of the wood and the species available. The wood is either considered crooked and does not meet sawmill standards or is not economically viable to transport due to the lack of sawmills in the area and the high cost of transportation. Supporting documentation was provided from Altitude Forestry and the FMFPD highlighting that the logs sourced from this project are considered non-merchantable biomass with limited disposal options in a fire-prone ecosystem. With this lack of utilization, the biomass from this project would most likely sit in the decking site of the area.

According to the Colorado State Wildfire Risk map, this project area is considered to have a high level of wildfire risk, and a very high level for Wildland Urban Interface (WUI) risk. According to the U.S. Fire Administration, WUI is a term commonly known in areas that experience wildfires. It is a zone between wilderness land and human activity that is considered to have greater risk of catastrophic wildfires.

Given the above, the biomass clearly falls under EC14 – Feedstocks that are sourced through government managed or funded wildfire mitigation or restoration activities. For

this reason, we are assigning the feedstock from this wildfire mitigation project a counterfactual storage discount of 0.

Dedicated Energy Feedstock Eligibility

This is only applicable to non-forestry feedstocks.

Please describe your analysis in how you determined that your feedstock isn't grown for the purposes of energy production.

N/A – All feedstocks described in this appendix are forestry feedstocks

Counterfactual Fate of Feedstock

Please describe and attach any relevant evidence to demonstrate what the most likely counterfactual scenario would be for your feedstock, using guidance outlined in Section 3.2.1 of the Biomass Feedstock Module.

The biomass Charm sources are forest residues (green logs) harvested by a contractor, Altitude Forestry, from a government-funded forest management project. The project, known as the Arroyo Chico II Fuels Reduction Project, was developed by the FMFPD and funded by the CSFS with one of the desired primary outcomes being to reduce the risk of catastrophic wildfires and facilitate fire suppression in the project area.

According to Altitude Forestry and the FMFPD, there is limited to no utilization for this type of feedstock in Colorado due to the quality of the wood and the species available. The wood is either considered crooked and does not meet sawmill standards or does not become economically viable to transport due to the lack of sawmills in the area and the high cost of transportation. Supporting documentation was provided from Altitude Forestry and the FMFPD highlighting that the logs sourced from this project are considered non-merchantable biomass with limited disposal options in a fire-prone ecosystem. With this lack of utilization, the biomass from this project would most likely sit in the decking site of the area. For this reason, the feedstock falls under Condition C1 in Section 3.2.2 of the Biomass Feedstock Accounting Module and can be considered to have 0 replacement emissions.

Given that the feedstock source both presents little to no risk of market leakage (EC 5 and 8) and has no replacement counterfactual (C1), counterfactual emissions in general are considered to be 0.

Replacement Emissions

Please select which method of replacement emissions accounting you've selected given the nature of your feedstock

	Accounting for the feedstock replacement emissions in the LCA
X	Not accounting for replacement emissions due to exemption C1 *(Feedstock has no counterfactual use)*
	Not accounting for replacement emissions due to exemption C2 *(Feedstock counterfactual use is most likely replaced with a feedstock with no counterfactual use)*
	Not accounting for replacement emissions due to exemption C3 *(Feedstock has no counterfactual use due to surplus)*
If you selected an exemption above, please demonstrate how your feedstock meets it by providing the required documentation or analysis.	
See “Counterfactual Fate of Feedstock”	

Appendix 5: GHG Statement Report

Information requirements for verification are as follows:

- GHG Statement Report (this Appendix) containing information relating to the GHG Statement
- GHG Statement calculations (provided separately)
- Supporting information including copies of raw data used

An appropriate level of information should be provided with a clearly referenced and transparent audit trail of decision making, assumptions, explanations, such that a verifier can trace all inputs, outputs and decision making. All data sources and assumptions must be clearly referenced, transparent and traceable.

After this Appendix has been completed for the first time, it may be acceptable to refer back to relevant sections for subsequent Reporting Periods, for example if there have been no changes to the GHG Assessment scope, data collection methodology, calculation methodology or assumptions.

There is no word limit for answers in sections below.

General Information	
Name of practitioner who prepared the GHG Statement, and relevant competencies. This should include an overview of relevant qualifications and experience undertaking GHG assessments.	Max Lavine, Charm Industrial Operations
Date of report	2024/03/01 *[Updated 2/2025]*
Please specify the Reporting Period this GHG Statement has been prepared for (the Reporting Period describes the time period over which the carbon removal activity assessed in the GHG Statement occurred)	At the time of this report, the initial injections associated with this project have not begun. It is anticipated that injections will occur in between now and the completion of project validation.
Relevant project information if not included in PDD	All data related to the calculation of removals will be transmitted from Charm's Ledger software to Isometric directly via API. What follows is an account of the data and calculations that will occur in Charm's Ledger software.
Details and links to any supporting documentation	Site-specific supplemental documents have been submitted alongside the main PDD. The documents shows the calculations for "Sample Removals" in order to demonstrate calculation methodology and an estimation of the impact

	of the different emissions within the project boundary on overall removals.
Please confirm that the GHG Statement has been undertaken in accordance with the information provided in the project boundary section of this PDD. Provide any additional information relating to the GHG Statement project boundary here.	The GHG Statement has been undertaken in accordance with the information provided in the project boundary section of this PDD. All data collected relates to listed GHG Emissions Sources/Fluxes as identified in the main body of the document. There are a few exclusions from the Project Boundary that merit further explanation: 1. **Salt used in bio-oil blending** a. The salt used in bio oil blending is off-spec salt from a nearby Morton's production facility. Because the salt is a waste product from an independently-occurring production process, we have not included the associated embodied emissions within the project boundary. 2. **Injection Fuel Use** a. Vaulted has submitted their own PDD which has been validated by Isometric and a third-party verifier. It does not report any fuel use, therefore fuel use is not accounted for here. If that were to change, it would be reported in Vaulted's monthly energy consumption report and accounted for in a manner analogous to that documented in Fuel Use for Injectate Pre-Processing sections below

Note: GHG data and calculation information for Charm's Basco 6 injection site are detailed in the Combined Emissions Monitoring and Accounting Plan in Supplemental Doc 3

GHG Statement Methodology - Data
Provide information on the GHG Statement approach and methodology in relation to data collection. *This should include the following information as a minimum:* ● *Data collection procedures - how data was requested and gathered, who supplied each data point, the format data was received in, and the date data points were received*

- *How data quality was assessed and how the data hierarchy was followed in terms of using measured, calculated and estimated data. Evidence and justification should be provided in any cases where measured data was not used*
- *Details of data validation processes followed, including treatment of missing data*

Evidence of all raw data that informed the assessment (including data that informed any assumptions) must be appended to the PDD, or linked within the GHG Statement.

Operational data and supporting documents will be transferred to Isometric via API. What is below is a summary of each data source and account of collection methods.

1. Fuel Use from Feedstock Transport – AECN
    a. Data Source: Vendor LCA
        i. The oil vendor (AECN) completed an LCA including the fuel used in feedstock transport using GHGenius Software
    b. Data Quality Assessment
        i. LCA is based on directly-measured data and calculations based on emissions factors
            1. Data for this LCA was collected in Spring/Summer of 2023. The expectation is that the vendor will update the LCA annually, or in light of major operational changes, whichever comes first
        ii. GHGenius software is a long-standing and well reputed LCA software whose data is primarily sourced from the Canadian government
        iii. Values are included in the bio oil Process Emissions and have been modified to ensure a conservative estimate of removals as detailed in the Uncertainty Analysis
2. Fuel Use from Feedstock Transport – Charm
    a. Data Source
        i. Feedstock is handled using a self-loading truck to facilitate removal from collection area – Fuel used in handling accounted for by the total weight of biomass transported and a standard emission factor for lumber handling published by Auburn University Forestry and Wildlife Sciences. Emissions are quantified using the GLEC 3.0 emissions factor for diesel combustion
        ii. Transportation emissions are calculated using the distance-based method outlined in the Transportation Emissions Accounting module
            1. Cargo is weighed using a calibrated truck scale when loaded by the vendor and documented with a Bill of Lading, scale ticket, and/or invoice document
            2. The vehicle routing is assessed using Google Maps, using the longest given route in order to ensure a conservative estimate of removals
            3. Vehicle Type is always a Heavy Duty Vehicle (flatbed truck) due to the quantity and nature of cargo
            4. This information allows for the calculation of WTW emissions from fuel use using GLEC 3.0 emission factors
            5. Where consumables are shared between Production and other functions outside of the system boundary, usage will be allocated

appropriately as evidenced by pyrolyzer up-time and/or historical usage rates

b. Data Quality Assessment
    i. Fuel use and weight of delivered biomass are measured directly by truck fuel meters and calibrated scales respectively
    ii. Vehicle Type is always a Heavy Duty Vehicle (specialty truck) due to the quantity and nature of cargo
    iii. Google Maps routing is a reliable source of mileage information and is controlled for uncertainty as described in the Uncertainty Analysis section
    iv. Scales used are calibrated and the result is adjusted for the maximum allowable error for a calibrated legal-for-trade truck scale to ensure a conservative estimate of removals as detailed in the Uncertainty Analysis
    v. GLEC and Auburn University qualify as a "reputable source" for emissions factors given the Isometric Protocol's definition

3. Embodied Emissions from Feedstock Transport – AECN
    a. See "Embodied Emissions from Equipment and Facilities for Biomass Pyrolysis – AECN" below
4. Embodied Emissions from Feedstock Transport – Charm
    a. Data Sources: Vendor LCA, GREET, USEPA
        i. For all truck transport, the Vehicle Embodied Emissions are calculated using the GREET 2023 emissions factors for Medium-and-Heavy-Duty (MHDV) Trucks and Trailers. The total value for the embodied emissions is divided by the expected useful life for a Heavy Duty compression ignition (diesel) engine established by US EPA Office of Transportation and Air Quality in 2016 for engines manufactured after 2004 at 435,000 miles. This yields a quantity of embodied carbon per mile traveled.
            1. The trucks being used in this project are provided by a third-party vendor. Therefore, only the mileage driven to transport biomass used for Charm bio oil is considered.
            2. The emissions factor is modified to include the average "deadhead" or unloaded journey distance for a vehicle of this type according to ATRI analysis
                a. Where consumables are shared between Production and other functions outside of the system boundary, usage will be allocated appropriately as evidenced by pyrolyzer up-time and/or historical usage rates
    b. Data Quality Assessment
        i. GREET, ATRI, and the US EPA qualify as "reputable sources" given the Isometric Protocol's definition of "A source that would be widely considered trustworthy based on the process undertaken (e.g., peer review) or origin of the information (e.g., government body)."
        ii. Google Maps routing is a reliable source of mileage information and is controlled for uncertainty as described in the Uncertainty Analysis section
5. Fuel Use from Biomass Pyrolysis – AECN
    a. Data Source: Vendor LCA

        i. The oil vendor (AECN) completed an LCA including the fuel used in feedstock transport using GHGenius Software
    b. Data Quality Assessment
        i. LCA is based on directly-measured data and calculations based on emissions factors
            1. Data for this LCA was collected in Spring/Summer of 2023. The expectation is that the vendor will update the LCA annually, or in light of major operational changes, whichever comes first
        ii. GHGenius software is a long-standing and well reputed LCA software whose data is primarily sourced from the Canadian government
        iii. Values are included in the bio oil Process Emissions and have been modified to ensure a conservative estimate of removals as detailed in the Uncertainty Analysis
6. Fuel Use from Biomass Pyrolysis – Charm
    a. Data Source
        i. Fuel (diesel, propane, gasoline) is purchased on an ongoing basis – Invoices for fuel purchases are used to measure fuel usage on site
            1. Where consumables are shared between Production and other functions outside of the system boundary, usage will be allocated appropriately as evidenced by pyrolyzer up-time and/or historical usage rates
        ii. Fuel quantity is multiplied by the appropriate emissions factor for combustion of that fuel published in the most recent GLEC framework (v3.0)
    b. Data Quality Assessment
        i. Invoices provide an accurate representation of fuel purchases for the site. The result is adjusted for the maximum allowable error for a calibrated legal-for-trade fuel meter to ensure a conservative estimate of removals as detailed in the Uncertainty Analysis
        ii. As a source for emissions factors, GLEC is considered a “reputable source” under the Isometric Protocol’s definition
7. Electricity Use from Biomass Pyrolysis – AECN
    a. Data Source: Vendor LCA
        i. The oil vendor (AECN) completed an LCA including the fuel used in feedstock transport using GHGenius Software
    b. Data Quality Assessment
        i. LCA is based on directly-measured data and calculations based on emissions factors
            1. Data for this LCA was collected in Spring/Summer of 2023. The expectation is that the vendor will update the LCA annually, or in light of major operational changes, whichever comes first
        ii. GHGenius software is a long-standing and well reputed LCA software whose data is primarily sourced from the Canadian government
        iii. Values are included in the bio oil Process Emissions and have been modified to ensure a conservative estimate of removals as detailed in the Uncertainty Analysis
8. Electricity Use from Biomass Pyrolysis – Charm
    a. Data Source
        i. Total facility electricity consumption is measured by a utility-provided

electricity meter

1. Where consumables are shared between Production and other functions outside of the system boundary, usage will be allocated appropriately as evidenced by pyrolyzer up-time and/or historical usage rates

ii. The emissions impact of electricity generation will be determined using the US EPA eGrid emissions factor for the RMPA subregion, where the pre-processing site is located and adjusted to show the residual grid mix by Green-e. To account for upstream emissions associated with generation infrastructure, a weighted average is calculated using the generation technology mix reported by EPA and the associated infrastructure emissions factors as determined by GREET.

iii. Where RECs are used, the emissions from generator infrastructure are sourced from GREET and the grid-region derating factor is sourced from US EPA eGrid

b. Data Quality Assessment

i. Electricity use is measured using a calibrated instrument from the local utility provider calibrated and purpose-built for measuring power usage over time. Given that the readings are the basis for billing for power usage, the utility has a direct incentive to ensure their accuracy

ii. Subregional grid emissions factors and generation technology mix are provided by the US EPA and grid infrastructure emissions factors from GREET, which are reputable sources under the Isometric Protocol's definition

9. Stack Emissions from Pyrolysis Biomass Dryer – AECN

a. Data Source

i. The vendor has not analyzed stack emissions from their pyrolysis process. In absence of direct data, we were provided a modeling of process emissions from the technology in use (Ensyn RTP) processing woody biomass prepared for Natural Resources Canada detailing the addition of pyrolysis oil pathways to GHGenius software.

b. Data Quality Assessment

i. GHGenius software is a long-standing and well reputed LCA software whose data is primarily sourced from the Canadian government

ii. Values are included in the bio oil Process Emissions and have been modified to ensure a conservative estimate of removals as detailed in the Uncertainty Analysis

10. Stack Emissions from Pyrolysis – Charm

a. Data Source

i. Charm has engaged a qualified emissions testing company approved by a state regulatory authority to perform compliance emissions testing for its pyrolysis process. This testing characterizes the GHG composition and flow rate for pyrolysis tailgas emissions.

ii. This data is used by Charm's engineering team to calculate a quantity of tailags emitted per kg of biomass processed. This quantity is then scaled based on the biomass throughput recorded for production

b. Data Quality Assessment

i. The testing company's approval by an appropriate oversight authority to perform compliance emissions testing qualifies it as an acceptable

source of such data per the Isometric Bio-Oil Geological Storage Protocol Section 7.4.3.5

ii. Charm's engineering team has unique process-specific knowledge required for a reasonable estimate of the relationship of the tailgas data provided by the emissions testing company to the emissions rate of a pyrolyzer as tested

iii. Operators are best positioned to log the quantity of biomass used during Production Periods

11. Embodied Emissions from Equipment and Facilities for Biomass Pyrolysis – AECN
    a. Data Sources
        i. Vendor has provided a total cost of facility build-out divided into construction, equipment, and installation costs
        ii. Emissions impact is calculated using NAICS Supply Chain Emissions Factors (cost-based) for Industrial Building Construction, Industrial Machinery Manufacturing, and Installation
    b. Data Quality Assessment
        i. The Vendor is the best available source for cost information on the build of their facility. The figure provided exceeds what has been described in the public record, ensuring a conservative estimate of impacts using a cost-based model
        ii. NAICS Supply Chain Greenhouse Gas Emissions Factors are provided by the US EPA which qualifies as a "reputable source" given the Isometric Protocol's definition
12. Embodied Emissions from Equipment and Facilities for Biomass Pyrolysis – Charm
    a. Data Sources
        i. Multiple data sources may be used for LCA depending on the object of analysis and data availability
            1. For equipment, specific model information is often enough to find comparable entries in a reputable database
                a. Smaller, regularly-exchanged components (e.g. nuts and bolts) are estimated on a per-machine, per-month basis.
            2. For facilities improvements, the quantity, type and estimated mass of materials used is often available and sufficient for LCA; for certain types of simple equipment that doesn't have a clear analogue in an LCA database this may be used as well
            3. When the above is not available and/or in the case of complex equipment that does not have a clear analogue in an LCA database, cost may be used to estimate the embodied emissions as determined through either invoices or typical values on online marketplaces (if rented)
        ii. Embodied emissions are quantified based on the data above using emissions factors from reputable sources including but not limited to the GREET database, the EcoInvent database, NAICS supply Chain Emission Factors, USLCI database, ICE database, and/or the Athena Impact Estimator
        iii. Pyrolysis facility embodied emissions are amortized based on total biomass throughput
    b. Data Quality Assessment
        i. The LCA database sources listed above qualify as "reputable sources"

given the Isometric Protocol's definition

ii. Detailed model information can be effectively used to correlate a piece of pieces of equipment with comparable data in an LCA database

iii. The site operations team is in the best position to evaluate the usage and cost of smaller regularly-exchanged components

iv. The contractor and/or site manager's assessment constitutes reasonable expert judgment regarding material mass estimates for facility improvements

v. Online marketplaces provide a relatively reliable source of pricing data due to the high transaction volume and number of sellers on prominent sites for buying and selling equipment

vi. Biomass throughput is the clearest metric to quantify equipment utilization – Output is problematic because pyrolyzers produce 5 distinct products at rates that vary depending on operating conditions

13. Fuel Use from Injectate Transport to Pre-Processing
    a. Trucking
        i. Data Sources
            1. Cargo is weighed using a calibrated truck scale when loaded by the vendor and documented with a Bill of Lading, scale ticket, and invoice document
            2. The vehicle routing is assessed using Google Maps, using the longest given route in order to ensure a conservative estimate of removals
            3. Vehicle Type is always a Heavy Duty Vehicle (tanker truck) due to the quantity and nature of cargo
            4. This information allows for the calculation of WTW emissions from fuel use using GLEC 3.0 emission factors
        ii. Data Quality Assessment
            1. Vehicle Type is always a Heavy Duty Vehicle (tanker truck) due to the quantity and nature of cargo
            2. Google Maps routing is a reliable source of mileage information and is controlled for uncertainty as described in the Uncertainty Analysis section
            3. Vendor scales are calibrated and the result is adjusted for the maximum allowable error for a calibrated legal-for-trade truck scale to ensure a conservative estimate of removals as detailed in the Uncertainty Analysis
            4. GLEC qualifies as a "reputable source" for emissions factors given the Isometric Protocol's definition
    b. Rail
        i. Data Sources
            1. Cargo is weighed using a calibrated truck scale when loaded by the vendor and documented with a Bill of Lading, scale ticket, and invoice document
            2. The vehicle routing is provided by the rail transport provider
            3. Vehicle Type is always a DOT-111 tanker truck, as this is the only type in Charm's rail fleet.
            4. This information allows for the calculation of WTW emissions

from fuel use using GLEC 3.0 emission factors

ii. Data Quality Assessment

1. Vehicle Type is always a DOT-111 tanker truck, as this is the only type in Charm's rail fleet.
2. The rail transport provider is a reliable source of mileage information
    a. The variable of mileage has a low uncertainty due to the number of alternative routes for rail transport being very limited relative to highway travel
3. Vendor scales are calibrated and the result is adjusted for the maximum allowable error for a calibrated legal-for-trade truck scale to ensure a conservative estimate of removals as detailed in the Uncertainty Analysis
4. GLEC qualifies as a "reputable source" for emissions factors given the Isometric Protocol's definition

14. Vehicle Embodied Emissions from Injectate Transport to Pre-Processing

a. Truck

i. Data Sources

1. For all truck transport, the Vehicle Embodied Emissions are calculated using the GREET 2023 emissions factors for Medium-and-Heavy-Duty (MHDV) Trucks and Trailers. The total value for the embodied emissions is divided by the expected useful life in miles for a Heavy Duty compression ignition (diesel) engine established by US EPA Office of Transportation and Air Quality in 2016 for engines manufactured after 2004. This yields a quantity of embodied carbon per mile traveled.
    a. The trucks being used in this project are not used by Charm, but by a third-party transport provider contracted by Charm. Therefore, a method in which only the mileage driven while under contract is considered is necessary.
    b. The emissions factor is modified to include the average "deadhead" or unloaded journey distance for a vehicle of this type according to American Transportation Research Institute analysis

ii. Data Quality Assessment

1. GREET, ATRI, and the US EPA qualify as "reputable sources" given the Isometric Protocol's definition
2. The variable of mileage has a low uncertainty, especially given that the routing between points is generally consistent over time and Google Maps route selection controls for uncertainty by selecting the longest route when multiple are presented to ensure a conservative estimate of removals

b. Rail

i. Data Sources

1. For all rail transport, the Vehicle Embodied Emissions are calculated using the DOT average purchase cost of a DOT-111 railcar multiplied by the NAICS Supply Chain Emissions

Factors (cost-based) for Railroad Rolling Stock. The total value for the embodied emissions is divided by the expected useful life of DOT-111 rail cars in miles as assessed by Cambridge Systematics. This yields a quantity of embodied carbon per mile traveled

a. The railcars being used in this project are leased to Charm but only used by Charm for a portion of their useful life. Therefore, a method in which only the mileage traveled while under lease is necessary.
b. Vehicle routing is provided by the transport provider
c. Journey distance is doubled to account for an unloaded round-trip because cars are leased to Charm specifically

2. Data Quality Assessment
   a. DOT, NAICS, and Cambridge Systematics qualify as "Reputable sources" given the Isometric protocol's definition
   b. The rail transport provider is a reliable source of mileage information
      i. The variable of mileage has a low uncertainty due to the number of alternative routes for rail transport being very limited relative to highway travel

15. Electricity Use from Injectate Pre-Processing
    a. Data Sources
       i. The emissions impact of electricity generation will be determined using the US EPA eGrid emissions factor for the SPNO subregion, where the pre-processing site is located and adjusted to show residual grid mix by Green-e. To account for upstream emissions associated with generation infrastructure, a weighted average is calculated using the generation technology mix reported by EPA and the associated infrastructure emissions factors as determined by GREET. See KS Grid Emissions sheet in the supplemental doc included with this PDD.
       ii. Where RECs are used, the emissions from generator infrastructure are sourced from GREET and the grid-region derating factor is sourced from US EPA eGrid
    b. Data Quality Assessment
       i. Electricity use is measured using a calibrated instrument from the local utility provider calibrated and purpose-built for measuring power usage over time. Given that the readings are the basis for billing for power usage, the utility has a direct incentive to ensure their accuracy
       ii. Subregional grid emissions factors and generation technology mix are provided by the US EPA and grid infrastructure emissions factors from GREET, which are reputable sources under the Isometric Protocol's definition

16. Fuel Use from Injectate Pre-Processing
    a. Data Sources
       i. Fuel for site operations is purchased on an ongoing basis – Invoices for fuel purchases are used to measure fuel usage on-site
       ii. Fuel quantity is multiplied by the emissions factor for diesel fuel

combustion published in the most recent GLEC Framework (V3.0)

b. Data Quality Assessment
    i. Invoices provide an accurate representation of diesel purchases for the site, which constitute the sole source of fuel used. The result is adjusted for the maximum allowable error for a calibrated legal-for-trade fuel meter to ensure a conservative estimate of removals as detailed in the Uncertainty Analysis
    ii. As a source for emissions factors, GLEC is considered a "reputable source" under the Isometric Protocol's definition

17. Embodied Emissions from Equipment for Injectate Pre-Processing
    a. Data Sources
        i. Multiple data sources may be used for LCA depending on the object of analysis and data availability
            1. For equipment, specific model information is often enough to find comparable entries in a reputable database
            2. For facilities improvements, the quantity, type and estimated mass of materials used is often available and sufficient for LCA; for certain types of simple equipment that doesn't have a clear analogue in an LCA database this may be used as well
            3. When the above is not available and/or in the case of complex equipment that does not have a clear analogue in an LCA database, cost may be used to estimate the embodied emissions as determined through either invoices or typical values on online marketplaces (if rented)
        ii. Embodied emissions are quantified based on the data above using emissions factors from reputable sources including but not limited to the GREET database, the EcoInvent database, NAICS supply Chain Emission Factors, USLCI database, ICE database, and/or the Athena Impact Estimator
        iii. Expected service life is estimated by internal engineering resources and/or relevant literature
    b. Data Quality Assessment
        i. The LCA database sources listed above qualify as "reputable sources" given the Isometric Protocol's definition
        ii. Materials mass can be effectively used to complete an LCA for facilities improvements or simple equipment
        iii. The contractor and/or site manager's assessment constitutes reasonable expert judgment regarding material mass estimates
        iv. Online marketplaces provide a relatively reliable source of pricing data due to the high transaction volume and number of sellers on prominent sites for buying and selling equipment
        v. Relevant literature and the estimation of internal engineering resources provide a reasonable expectation regarding the overall service life of a given piece of equipment

18. Embodied Emissions from Consumables for Injectate Pre-Processing
    a. Data Sources
        i. Liquid Caustic Soda (50%) is added to bio oil in amounts measured by on-site staff using a flow meter

ii. GREET emissions factors are used to calculate the emissions impact of LCS used
iii. Salt is used as well, but it is a waste byproduct from a nearby Morton salt factory, so embodied emissions are not considered

b. Data Quality Assessment
i. The onsite flow meter provides a reliable measure of the quantity of LCS used. The result is adjusted for the uncertainty for a typical flow meter to ensure a conservative estimate of removals as detailed in the Uncertainty Analysis
ii. As a source for emissions factors, GREET is considered a "reputable source" under the Isometric Protocol's definition

19. Fuel Use from Delivery of Consumables to Injectate Pre-Processing
a. Data Sources
i. Delivery invoices establish the mass of cargo transported to the Pre-Processing site and location of product origin, which is used to calculate the distance traveled
ii. The vehicle routing is assessed using Google Maps, using the longest given route in order to ensure a conservative estimate of removals
iii. Vehicle Type is always a Heavy Duty Vehicle (flatbed truck) due to the quantity and nature of cargo
iv. This information allows for the calculation of WTW emissions from fuel use using GLEC 3.0 emission factors

b. Data Quality Assessment
i. Because delivery invoices are the basis for billing for goods delivered, vendors have a direct incentive to ensure their accuracy
ii. Vehicle Type is always a Heavy Duty Vehicle (flatbed truck) due to the quantity and nature of cargo
iii. Google Maps routing is a reliable source of mileage information and is controlled for uncertainty as described above
iv. GLEC qualifies as a "reputable source" for emissions factors given the Isometric Protocol's definition

20. Embodied Emissions from Delivery of Consumables to Injectate Pre-Processing
a. Data Sources
i. For all truck transport, the Vehicle Embodied Emissions are calculated using the GREET 2023 emissions factors for Medium-and-Heavy-Duty (MHDV) Trucks and Trailers. The total value for the embodied emissions is divided by the expected useful life for a Heavy Duty compression ignition (diesel) engine established by US EPA Office of Transportation and Air Quality in 2016 for engines manufactured after 2004 at 435,000 miles. This yields a quantity of embodied carbon per mile traveled.
1. The trucks being used in this project are not used by Charm, but by a third-party transport provider contracted by Charm. Therefore, only the mileage driven while under contract is considered.
2. The emissions factor is modified to include the average "deadhead" or unloaded journey distance for a vehicle of this type according to American Transportation Research Institute analysis

b. Data Quality Assessment
    i. GREET, ATRI, and the US EPA qualify as “reputable sources” given the Isometric Protocol’s definition
    ii. The variable of mileage has a low uncertainty, especially given that the routing between points is generally consistent over time and Google Maps route selection controls for uncertainty by selecting the longest route when multiple are presented to ensure a conservative estimate of removals

21. Fuel Use from Processed Injectate Transport to Injection Site
    a. Data Sources
        i. Cargo is weighed using a calibrated truck scale at the injection site and documented with a weigh scale ticket
        ii. The vehicle routing is assessed using Google Maps, using the longest given route in order to ensure a conservative estimate of removals
        iii. Vehicle Type is always a Heavy Duty Vehicle (tanker truck) due to the quantity and nature of cargo
        iv. This information allows for the calculation of WTW emissions from fuel use using GLEC 3.0 emission factors
    b. Data Quality Assessment
        i. Vehicle Type is always a Heavy Duty Vehicle (tanker truck) due to the quantity and nature of cargo
        ii. Google Maps routing is a reliable source of mileage information and is controlled for uncertainty as described above
        iii. Injection site scales are calibrated and the result is adjusted for the maximum allowable error for a calibrated legal-for-trade truck scale to ensure a conservative estimate of removals as detailed in the Uncertainty Analysis
        iv. GLEC qualifies as a “reputable source” for emissions factors given the Isometric Protocol’s definition

22. Vehicle Embodied Emissions for Processed Injectate Transport to Injection Site
    a. Data Sources
        i. For all truck transport, the Vehicle Embodied Emissions are calculated using the GREET 2023 emissions factors for Medium-and-Heavy-Duty (MHDV) Trucks and Trailers. The total value for the embodied emissions is divided by the expected useful life for a Heavy Duty compression ignition (diesel) engine established by US EPA Office of Transportation and Air Quality in 2016 for engines manufactured after 2004 at 435,000 miles. This yields a quantity of embodied carbon per mile traveled.
            1. The trucks being used in this project are not used by Charm, but by a third-party transport provider contracted by Charm. Therefore, only the mileage driven while under contract is considered.
            2. The emissions factor is modified to include the average “deadhead” or unloaded journey distance for a vehicle of this type according to American Transportation Research Institute analysis
    b. Data Quality Assessment
        i. GREET, ATRI, and the US EPA qualify as “reputable sources” given

the Isometric Protocol's definition

ii. The variable of mileage has a low uncertainty, especially given that the routing between points is generally consistent over time and Google Maps route selection controls for uncertainty by selecting the longest route when multiple are presented to ensure a conservative estimate of removals

23. Final Injectate Mass
    a. Data Sources
        i. Delivered injectate will be weighed using the Well Operator's on-site truck scale by taking the weight of the delivery vehicle prior to and following offloading the injectate as specified in the written agreement between Charm and the Well Operator contracting emplacement services. Weigh scale tickets are produced for each load to document the mass injected
    b. Data Quality Assessment
        i. Well Operator scales are calibrated and the result is adjusted for the maximum allowable error for a calibrated legal-for-trade truck scale to ensure a conservative estimate of removals as detailed in the Uncertainty Analysis
24. Electricity Use from Injection
    a. Data Sources
        i. Agreement between the Charm and Well Operator specifies that the Well Operator will provide regular reports of the energy consumption on site, including electricity as metered by the local utility, as a reflection of the total hours of operations during services provided to Charm
            1. The emissions impact of electricity generation will be determined using the US EPA eGrid emissions factor for the SPNO subregion, where the pre-processing site is locatedand adjusted to show residual grid mix by Green-e. To account for upstream emissions associated with generation infrastructure, a weighted average is calculated using the generation technology mix reported by EPA and the associated infrastructure emissions factors as determined by GREET. This is covered in the supplemental doc included with this PDD.
    b. Data Quality Assessment
        i. Electricity use is measured using a calibrated instrument from the local utility provider calibrated and purpose-built for measuring power usage over time. Given that the readings are the basis for billing for power usage, the utility has a direct incentive to ensure their accuracy
        ii. Subregional grid emissions factors and generation technology mix are provided by the US EPA and grid infrastructure emissions factors, which are reputable sources under the Isometric Protocol's definition
25. Embodied Emissions from Injection Equipment
    a. Data Sources
        i. In the GHG Statement provided as part of their own PDD, Vaulted calculated that their onsite equipment accounts for 0.0000351566 tons of steel and 0.000077 m3 of concrete per ton of waste injected, given their conservative estimate of a 2 million metric ton overall well capacity.

ii. The embodied emissions impact of these material quantities per ton of injectant are calculated using the GREET emissions factors for these materials

b. Data Quality Assessment

i. Vaulted's calculation of on-site materials was validated and verified under the Isometric protocol and is therefore considered a reliable source of data

ii. Both GREET qualifies as a "reputable source" for emissions factors given the Isometric Protocol's definition

26. Carbon Content of Injected Bio Oil

a. Data Sources

i. Mass of Injectate will be determined by using an on-site truck scale that will be used to weigh the delivery truck prior to and after injection. A weigh station ticket will be produced to verify mass injected

ii. Mass of Injected Bio Oil will be determined by subtracting the mass of salt and LCS used for a particular batch of injectate from the Mass of Injectate

iii. Carbon content of the oil will be measured using Isometric Protocol Method A – Each injection batch will be sampled prior to the addition of salt and LCS and one sample will be sent to an analytic lab to measure the carbon content of that batch.

b. Data Quality Assessment

i. The scale used at Vaulted's on-site weigh station is properly calibrated and maintained by Vaulted and the result is adjusted for the maximum allowable error for a calibrated legal-for-trade truck scale to ensure a conservative estimate of removals as detailed in the Uncertainty Analysis

ii. All additions to the injectate are monitored using an on-site scale (solid additions) and flow meter (liquid additions) for each batch

iii. Carbon content is measured by analytic testing at an ISO-certified lab and based directly on the batch being injected. The result is adjusted for the maximum allowable error for lab analysis to ensure a conservative estimate of removals as detailed in the Uncertainty Analysis

27. Emissions from Sample Transport to Lab

a. Data Sources

i. The number of tests, samples, and sample weights are internally standardized for each injection batch and calculated as reported by the Charm Chemistry and Research teams

ii. Distance between the sampling site and labs is determined according to Google Maps routing. Road distances are used despite the transport assumption being air freight. This is due to the fact that road distances will reliably be longer than the more direct route allowed by air travel, ensuring a conservative estimate of removals

iii. This information allows Charm to calculate the Well to Wheel (WTW) emissions associated with air transport using emissions factors from the GLEC Framework

b. Data Quality Assessment

i. Data on the number of tests and location of testing labs comes directly from the team responsible for managing this function.

ii. The variable of mileage has a low uncertainty, especially given that the routing between points is generally consistent over time and Google Maps route selection controls for uncertainty by selecting the longest route when multiple are presented and using road directions to ensure a conservative estimate of removals

28. Disposal of Process Waste
    a. Fuel Emissions from Waste Transport to Disposal Facility
        i. Data Sources
            1. Cargo mass and destination are taken from manifests provided by EPA-regulated waste disposal providers
            2. The vehicle routing is assessed using Google Maps, using the longest given route in order to ensure a conservative estimate of removals
            3. Vehicle Type is always a Heavy Duty Vehicle due to the quantity and nature of cargo – A flatbed truck is used for solid waste, and a tanker truck is used for liquid waste
            4. This information allows for the calculation of WTW emissions from fuel use using GLEC emission factors
        ii. Data Quality Assessment
            1. Disposal manifests and providers are subject to regulation and review by USEPA, qualifying them as a reputable and reliable source of input data
            2. Google Maps routing is a reliable source of mileage information and is controlled for uncertainty as described in the Uncertainty Analysis section
            3. GLEC qualifies as a “reputable source” for emission factors given the Isometric Protocol’s definition
    b. Vehicle Embodied Emissions from Waste Transport to Disposal Facility
        i. Data Sources
            1. For all truck transport, the Vehicle Embodied Emissions are calculated using the GREET 2023 emissions factors for Medium-and-Heavy-Duty (MHDV) Trucks and Trailers. The total value for the embodied emissions is divided by the expected useful life in miles for a Heavy Duty compression ignition (diesel) engine established by US EPA Office of Transportation and Air Quality in 2016 for engines manufactured after 2004. This yields a quantity of embodied carbon per mile traveled.
                a. The trucks being used in this project are not used by Charm, but by a third-party transport provider contracted by Charm. Therefore, a method in which only the mileage driven while under contract is considered is necessary.
                b. The emissions factor is modified to include the average “deadhead” or unloaded journey distance for a vehicle of this type according to American Transportation Research Institute (ATRI) analysis
        ii. Data Quality Assessment
            1. GREET, ATRI, and the US EPA qualify as “reputable sources”

given the Isometric Protocol's definition

2. The variable of mileage has a low uncertainty, especially given that the routing between points is generally consistent over time and Google Maps route selection controls for uncertainty by selecting the longest route when multiple are presented to ensure a conservative estimate of removals

c. Emissions from Material Disposal and Decomposition

i. Data Sources

1. The quantity of material disposed is taken from manifests provided by EPA-regulated waste disposal providers
2. The emission factor(s) quantifying the emissions impact of the material disposed are sourced from reputable sources including but not limited to the GREET database, the EcoInvent database, Open LCA, USLCI database, ICE database, and/or the Athena Impact Estimator
    a. Emission factors are used as-is where feasible. To account for the disposal of less-studied material such as bio-oil, a reasonable estimate may be made using proxy values. In this case, references to baseline values and sources, as well as any additional assumptions and justifications will be documented by the project

ii. Data Quality Assessment

1. Disposal manifests and providers are subject to regulation and review by USEPA, qualifying them as a highly reputable and reliable source of input data
2. The LCA database sources listed above qualify as "reputable sources" given the Isometric Protocol's definition

GHG Statement Methodology - Calculations
Provide information on the GHG Statement approach and methodology in relation to calculations. *This should include the following information as a minimum:* ● *General description of the methodology, criteria and procedures used as a basis for the assessment, including reference to any documentation (including Isometric protocols and modules), guidance, industry standards and best practice that were followed* ● *Information on calculation procedures followed* ● *Information on any tools used as part of the assessment* ● *Procedure for selecting emission factors including how age (age of data, and the period over which they have been collected), geography (the region or country from where the data have originated), technology (whether the data are specific to a particular technology or mix of many), methodology (the approach applied to gather or calculate the data) and competency (proficiency of entity that developed the data) were considered.* ● *Assumptions and limitations - provide full transparency in terms of value-choices, rationales and expert judgements*

- *Details of a sensitivity analysis, how parameters were assessed and evidence behind choices*
- *Details of uncertainty analysis*
- *Details of any uncertainty adjustments (e.g. %) applied in instances of high uncertainty*

29. Fuel Use from Feedstock Transport – AECN
    a. Calculation methodology
        i. Value in Vendor LCA is applied as a constant deduction against each ton of bio oil purchased
    b. Justification of Variables and Emissions factors used
        i. Vendor has provided an LCA compiled using GHGenius software
        ii. Charm must rely on vendor inputs in order to account for the LCA associated with vendor production activities
            1. Data for this LCA was collected in Spring/Summer of 2023. The expectation is that the vendor will update the LCA annually, or in light of major operational changes, whichever comes first
        iii. GHGenius software is a long-standing and well reputed LCA software whose data is primarily sourced from the Canadian government
        iv. Values related to process emissions have been modified to ensure a conservative estimate of removals as detailed in the Uncertainty Analysis
    c. Unit/Frequency of Calculation
        i. This will be applied as a static value against each MT of bio-oil delivered
    d. Modifications to Address Uncertainty
        i. This value is part of the overall "Process Emissions" for the purchased bio oil. Process Emissions are treated as an emissions factor. An Uncertainty Factor has been applied to this value due to showing a greater than 1% effect on net removal quantity in the Sensitivity Analysis. Details regarding the Uncertainty Factor can be found in the Uncertainty Assessment section here and in the GHG Statement Supplemental Doc
30. Fuel Use from Feedstock Transport – Charm
    a. Calculation Methodology
        i. Feedstock is handled using a self-loading truck to facilitate removal from the collection area – Fuel used in handling accounted for by the total weight of biomass transported and a standard emission factor for lumber handling published by Auburn University Forestry and Wildlife Sciences. Emissions are quantified using the GLEC 3.0 emissions factor for diesel combustion
            1. Conversion from volume to mass measurements is based on the standard density of diesel fuel published in the Chevron Diesel Fuels Technical Review
        ii. Fuel use from the feedstock transport itself is calculated according to the Distance-Based Method outlined in Section 3.3 of the Isometric Transportation Emissions Accounting module
            1. Cargo weight according to Vendor-provided Bill of Lading, scale ticket and invoice docs and point-to-point distance from Google

Maps are used to calculate the WTW emissions for the journey by applying the appropriate GLEC V3.0 emissions factor (flatbed truck) to the tons-miles value

iii. Biomass is often stockpiled and progressively used over a long period of time. Biomass transport emissions may be allocated to bio-oil based on the quantity of raw biomass (logs) consumed during a given period – Usage rates are monitored by measuring the linear feet of raw logs before and after processing. The appropriate proportion of total transport emissions will then be allocated to the oil produced during that period

b. Justification of Variables and Emissions Factors Used
   i. Vehicle Type is always a Heavy Duty Vehicle (flatbed truck) due to the quantity and nature of cargo
   ii. Google Maps routing is a reliable source of mileage information and is controlled for uncertainty as described in the Uncertainty Analysis section
   iii. Truck scales are calibrated and the result is adjusted for the maximum allowable uncertainty for a calibrated legal-for-trade truck scale to ensure a conservative estimate of removals as detailed in the Uncertainty Analysis
   iv. GLEC and Auburn University qualify as a "reputable source" for emissions factors given the Isometric Protocol's definition
   v. Use of linear feet to monitor the progressive consumption of a biomass stockpile is preferable to mass measurements as it avoids the uncertainty involved in estimating moisture and process losses during the processing steps required to prepare biomass for pyrolysis
c. Unit/Frequency of Calculation
   i. Transport emissions are calculated for each delivery of biomass to Charm's facility
      1. When biomass is stockpiled, all deliveries composing the stockpile are summed to compute total transportation emissions for the stockpile
      2. The biomass team also measures and records to total linear feet of logs composing the biomass stockpile once fully delivered
   ii. Once a biomass stockpile begins supplying production, the remaining linear feet are measured and recorded for each Production Period – The proportion of total transport emissions corresponding to the proportion of biomass consumed is allocated across the bio oil produced during that period
      1. When a biomass stockpile is shared between Production and other functions outside of the system boundary (R+D), usage will be allocated appropriately as evidenced by pyrolyzer up-time
   iii. Delivery emissions may also be allocated to pyrolysis products as the deliveries occur
   iv. Emissions are allocated to pyrolysis products produced during the Production Period during which the delivery occurs
d. Modifications to Address Uncertainty
   i. The mass used to determine delivery emissions is adjusted for the maximum allowable uncertainty for a legal-for-trade truck scale to

ensure a conservative estimate of removals as detailed in the Uncertainty Analysis

31. Embodied Emissions from Feedstock Transport – AECN
    a. See "Embodied Emissions from Equipment and Facilities for Biomass Pyrolysis – AECN" below
32. Embodied Emissions from Feedstock Transport – Charm
    a. Calculation Methodology
        i. Per-mile embodied emissions are calculated based on GREET emissions levels for the manufacture of medium-and-heavy-duty trucks and trailers and divided by EPA expected useful life in mileage for heavy-duty diesel engines. This calculation yields a per-mile embodied emissions level that can be applied to Charm-specific journeys using vehicles owned by the vendor
        ii. Per-mile embodied emissions are applied to the feedstock being transported according to journey distance
    b. Justification of Variables and Emissions Factors Used
        i. GREET, ATRI, and the US EPA qualify as "reputable sources" given the Isometric Protocol's definition
        ii. Google Maps routing is a reliable source of mileage information and is controlled for uncertainty as described in the Uncertainty Analysis section
    c. Unit/Frequency of Calculation
        i. Transport emissions are calculated for each delivery of biomass to Charm's facility
            1. When biomass is stockpiled, all deliveries composing the stockpile are summed to compute total transportation emissions for the stockpile
            2. The biomass team also measures and records to total linear feet of logs composing the biomass stockpile once fully delivered
        ii. Once a biomass stockpile begins supplying production, the remaining linear feet are measured and recorded for each Production Period – The proportion of total transport emissions corresponding to the proportion of biomass consumed is allocated across the bio oil produced during that period
            1. When a biomass stockpile is shared between Production and other functions outside of the system boundary (R+D), usage will be allocated appropriately as evidenced by pyrolyzer up-time
        iii. Delivery emissions may also be allocated to pyrolysis products as the deliveries occur
        iv. Emissions are allocated to pyrolysis products produced during the Production Period during which the delivery occurs
    d. Modifications to Address Uncertainty
        i. The mass used to determine delivery emissions is adjusted for the maximum allowable uncertainty for a legal-for-trade truck scale to ensure a conservative estimate of removals as detailed in the Uncertainty Analysis
33. Fuel Use from Biomass Pyrolysis – AECN
    a. Calculation methodology

i. Value in Vendor LCA is applied as a constant deduction against each ton of bio oil purchased

b. Justification of Variables and Emissions factors used
   i. Vendor has provided an LCA compiled using GHGenius software
   ii. Charm must rely on vendor inputs in order to account for the LCA associated with vendor production activities
      1. Data for this LCA was collected in Spring/Summer of 2023. The expectation is that the vendor will update the LCA annually, or in light of major operational changes, whichever comes first
   iii. GHGenius software is a long-standing and well reputed LCA software whose data is primarily sourced from the Canadian government
   iv. Values related to process emissions have been modified with an Uncertainty Factor to ensure a conservative estimate of removals as detailed in the Uncertainty Analysis

c. Unit/Frequency of Calculation
   i. This will be applied as a static value against each MT of bio-oil delivered

d. Modifications to Address Uncertainty
   i. This value is part of the overall "Process Emissions" for the purchased bio oil. Process Emissions are treated as an emissions factor. An Uncertainty Factor has been applied to this value due to showing a greater than 1% effect on net removal quantity in the Sensitivity Analysis. Details regarding the Uncertainty Factor can be found in the Uncertainty Assessment section here and in the GHG Statement supplemental doc

34. Fuel Use from Biomass Pyrolysis – Charm

a. Calculation Methodology
   i. Emissions impact of fuel usage is calculated based on GLEC models for fuel combustion in terms of TCO2e/kg fuel multiplied by the quantity of fuel purchased for use on site as documented by purchase invoices. This quantity is corrected for fuel pump uncertainty as detailed in the Uncertainty Analysis
      1. Conversion from volume to mass measurements is based on the standard density of the applicable fuel as documented by an appropriate, reputable source
   ii. Fuel that is shared between Production and R+D processes will be allocated based on each process's proportion of total pyrolyzer up-time during a given Production Period

b. Justification of Variables and Emissions Factors Used
   i. All fuel used on site will be purchased and an invoice will be issued to record that purchase which is presumed to be reflective of the amount dispensed by a gas meter. Because the meter is used to determine billing, the fuel vendor has a direct financial incentive to ensure that it is accurate.
   ii. GLEC V3.0 is a reputable source for emissions factors according to the definition provided in the Isometric Protocol and is the most recent version of the framework

c. Unit/Frequency of Calculation
   i. Emissions are assigned to pyrolysis products produced during the

Production Period during which the delivery occurs

d. Modifications to Address Uncertainty
    i. The invoiced amount is adjusted for the maximum allowable error for a calibrated legal-for-trade fuel meter to ensure a conservative estimate of removals as detailed in the Uncertainty Analysis

35. Electricity Use from Biomass Pyrolysis – AECN
    a. Calculation methodology
        i. Value in Vendor LCA is applied as a constant deduction against each ton of bio oil purchased
    b. Justification of Variables and Emissions factors used
        i. Vendor has provided an LCA compiled using GHGenius software
        ii. Charm must rely on vendor inputs in order to account for the LCA associated with vendor production activities
            1. Data for this LCA was collected in Spring/Summer of 2023. The expectation is that the vendor will update the LCA annually, or in light of major operational changes, whichever comes first
        iii. GHGenius software is a long-standing and well reputed LCA software whose data is primarily sourced from the Canadian government
        iv. Values related to process emissions have been modified with an Uncertainty Factor to ensure a conservative estimate of removals as detailed in the Uncertainty Analysis
    c. Unit/Frequency of Calculation
        i. This will be applied as a static value against each MT of bio-oil delivered
    d. Modifications to Address Uncertainty
        i. This value is part of the overall "Process Emissions" for the purchased bio oil. Process Emissions are treated as an emissions factor. An Uncertainty Factor has been applied to this value due to showing a greater than 1% effect on net removal quantity in the Sensitivity Analysis. Details regarding the Uncertainty Factor can be found in the Uncertainty Assessment section here and in the GHG Statement supplemental doc

36. Electricity Use from Biomass Pyrolysis – Charm
    a. Calculation Methodology
        i. Electricity emissions are calculated by kWh usage multiplied by the per kWh emissions for the RMPA grid subregion as reported by the EPA eGrid emissions factors and adjusted by Green-e to show the residual grid mix. This is supplemented by a per-kWh addition to account for the embodied emissions associated with grid infrastructure as quantified by a weighted average of power generation sources based on the fuel mix for the subregion reported by the EPA and the applicable electricity infrastructure emissions factors reported by GREET
        ii. Estimated general and administrative electricity use, based on historical usage rates, is subtracted from the total facility electricity usage. The difference between total facility usage and G+A is then allocated between production and R+D pyrolysis based on each process's proportion of total pyrolyzer up-time during a given Production Period
        iii. Where RECs are used, the grid emission factor is replaced by an emission factor for the renewable generation type sourced from GREET

for every kWh matched by a REC

1. The volume of consumption matched by each REC is reduced by a derating factor which is equal to the grid-region T+D losses quantified by USEPA eGrid

b. Justification of Variables and Emissions Factors Used
   i. Electricity use is measured using a calibrated instrument from the local utility provider calibrated and purpose-built for measuring power usage over time. Given that the readings are the basis for billing for power usage, the utility has a direct incentive to ensure their accuracy
   ii. Subregional grid emissions factors and generation technology mix are provided by the US EPA and grid infrastructure emissions factors from GREET, which are reputable sources under the Isometric Protocol's definition
   iii. Estimating general and administrative power consumption based on historical estimates is inherently conservative given that the number of general and administrative functions supported by the facility has grown over time
   iv. Pyrolyzer up-time is logged by pyrolyzer operators as part of their standard operating procedure – Pyrolyzer operators are best positioned to log the start and end time of production runs
   v. The RECs calculation approach used is aligned with the approach provided in the Isometric Energy Use Accounting module
c. Unit/Frequency of Calculation
   i. At the close of a Production Period, total facility usage during the Period is pulled from the utility's customer portal and uploaded to the Ledger where the historical usage estimate and allocation between Production and R+D is computed
      1. Up-Time is logged for each production run and regularly uploaded to the Ledger
   ii. These emissions are assigned to bio oil produced during the Production Period during which the delivery occurs
d. Modifications to Address Uncertainty
   i. Meter readings are adjusted for uncertainty based on the maximum allowable uncertainty for a utility-provided electricity meter provided by the Code of Colorado Regulations

37. Stack Emissions from Biomass Pyrolysis – AECN
   a. Calculation Methodology
      i. The vendor has not performed an analysis of stack emissions arising from their pyrolysis process
      ii. In lieu of a direct analysis, the vendor provided an analysis of the specific technology in use prepared for Natural Resources Canada detailing the addition of the technology to the GHGenius software used for their process LCA
      iii. The values for Greenhouse Gases were converted from emissions per GJ of feedstock to emissions per MT of oil based on the input and output data from the Vendor LCA and added to the overall process emissions associated with their pyrolysis process
   b. Justification of Variables and Emissions Factors Used
      i. The object of the analysis provided is the specific technology and

feedstock used to produce the bio oil used for this project

ii. Charm must rely on vendor inputs in order to account for the LCA associated with vendor production activities

iii. GHGenius software is a long-standing and well reputed LCA software whose data is primarily sourced from the Canadian government

iv. Values related to process emissions have been modified with an Uncertainty Factor to ensure a conservative estimate of removals as detailed in the Uncertainty Analysis

c. Unit/Frequency of Calculation

i. This will be applied as a static value against each MT of bio-oil delivered

d. Modifications to Address Uncertainty

i. This value is part of the overall "Process Emissions" for the purchased bio oil. Process Emissions are treated as an emissions factor. An Uncertainty Factor has been applied to this value due to showing a greater than 1% effect on net removal quantity in the Sensitivity Analysis. Details regarding the Uncertainty Factor can be found in the Uncertainty Assessment section here and in the GHG Statement supplemental doc

38. Stack Emissions from Pyrolysis – Charm

a. Calculation Methodology

i. The GHG composition and flow rate of pyrolysis tailgas is characterized for each type of feedstock processed using the results of compliance emissions testing performed by a qualified emissions testing company

ii. This data is used by Charm's engineering team to calculate a quantity of tailags emitted per kg of biomass processed. This quantity is then scaled based on the biomass throughput recorded for production

iii. The total methane released as a result of these emissions is converted to CO2e using the 100-year GWP of Methane quantified by IPCC AR-6

b. Justification of Variables and Emissions Factors Used

i. The testing company's approval by an appropriate oversight authority to perform compliance emissions testing qualifies it as an acceptable source of such data per the Isometric Bio-Oil Geological Storage Protocol Section 7.4.3.5

ii. Charm's engineering team has unique process-specific knowledge required for a reasonable estimate of the relationship of the tailgas data provided by the emissions testing company to the emissions rate of a pyrolyzer as tested

iii. 100-year GWP based on the most recent volume of the IPCC Assessment Report is specified by the Isometric Standard for CO2e conversions

iv. The Isometric Bio-Oil Geological Storage Protocol Section 7.4.3.5 specifies that tailgas emissions should be calculated in terms of the total release of methane

c. Unit/Frequency of Calculation

i. Tailgas emissions will be computed based on the quantity of biomass consumed by production pyrolyzers during a given Production Period.

ii. Emissions are allocated to pyrolysis products produced during the Production Period when production occurs

d. Modifications to Address Uncertainty
    i. Total tailgas emissions are adjusted for uncertainty based on the measurement uncertainty reported for the emissions testing performed on the type of feedstock processed

39. Embodied Emissions from Equipment and Facilities for Biomass Pyrolysis – AECN
    a. Calculation Methodology
        i. The bio oil vendor has not performed an evaluation of the embodied emissions associated with the construction of their building or manufacture of their equipment. Therefore, the best option available to Charm for determining these impacts is to use a cost-based calculation. This is achieved by applying per-dollar-spent supply chain emissions factors to the cost of construction, equipment, and installation and dividing that total by the expected lifetime production of the facility to arrive at a quantity of CO2e per MT of oil produced associated with facility embodied emissions.
    b. Justification of Variables and Emissions factors used
        i. The Vendor is the best available source for cost information on the build of their facility. The figure provided exceeds what has been described in the public record, ensuring a conservative estimate of impacts using a cost-based model
        ii. NAICS Supply Chain Greenhouse Gas Emissions Factors are provided by the US EPA which qualifies as a "reputable source" given the Isometric Protocol's definition
    c. Unit/Frequency of Calculation
        i. This will be applied as a static value against each MT of bio-oil delivered
    d. Modifications to Address Uncertainty
        i. This value is part of the overall "Embodied Emissions" for the purchased bio oil. This value is being used as is due to demonstrating a less than 1% change in the Net Removal in the Sensitivity Analysis. See the "Sensitivity Analysis" sheet in the supplemental GHG Statement doc

40. Embodied Emissions from Equipment and Facilities for Biomass Pyrolysis – Charm
    a. Calculation Methodology
        i. Total embodied emissions associated with equipment and facilities improvements specific to the pyrolysis processes required for carbon removal are quantified using LCA as described in the Data section of this GHG Statement report
        ii. Total embodied emissions are amortized over the Estimated Useful Life of the equipment based on the quantity of biomass processed during a certain period as a proportion of the total expected processing capacity for the facility over time
        iii. Amortized embodied emissions are allocated to pyrolysis products produced during an Production Period based on the quantity of biomass processed during the period
    b. Justification of Variables and Emissions Factors Used
        i. Embodied emissions are quantified using emissions factors from reputable sources including but not limited to the GREET database, the EcoInvent database, NAICS supply Chain Emission Factors, USLCI

database, ICE database, and/or the Athena Impact Estimator

ii. Biomass throughput is the clearest metric to quantify equipment utilization – Output is problematic because pyrolyzers produce 5 distinct products at rates that vary depending on operating conditons

c. Unit/Frequency of Calculation

i. Each Production Period, embodied emissions for the pyrolysis facility are totaled based on the biomass processed during the period and allocated across the pyrolysis products produced during that period

d. Modifications to Address Uncertainty

i. Embodied emissions are scaled by biomass throughput mass which is modified for instrument uncertainty as detailed in the Uncertainty Analysis

41. Fuel Use from Injectate Transport to Pre-Processing

a. Truck

i. Calculation Methodology

1. Cargo weight according to Vendor-provided Bill of Lading, scale ticket and invoice docs and point-to-point distance from Google Maps are used to calculate the WTW emissions for the journey using a Tanker Truck by applying the appropriate GLEC V3.0 emissions factor (tanker truck) to the tons-miles value

ii. Justification of Variables and Emissions Factors Used

1. Vehicle Type is always a Heavy Duty Vehicle (tanker truck) due to the quantity and nature of cargo
2. Google Maps routing is a reliable source of mileage information and is controlled for uncertainty as described in the Uncertainty Analysis section
3. Vendor scales are calibrated and the result is adjusted for the maximum allowable uncertainty for a calibrated legal-for-trade truck scale to ensure a conservative estimate of removals as detailed in the Uncertainty Analysis
4. GLEC qualifies as a “reputable source” for emissions factors given the Isometric Protocol’s definition

iii. Unit/Frequency of Calculation

1. Transport emissions are calculated and assigned to each truckload of bio oil loaded into Transport Provider vehicles for delivery by vendor
2. Ledger software associates these values with the oil as it is divided and recombined for transport and processing until its final form as an injection batch

iv. Modifications to Address Uncertainty

1. Uncertainty considerations have been applied to this value due to showing a greater than 1% effect on net removal quantity in the Sensitivity Analysis. Details can be found in the Uncertainty Assessment section here and in the GHG Statement supplemental doc

b. Rail

i. Calculation Methodology

1. Cargo weight according to Vendor-provided Bill of Lading, scale ticket and invoice docs and point-to-point distance the rail

provider are used to calculate the WTW emissions for the journey using the GLEC V3.0 US rail emissions factor

ii. Justification of Variables and Emissions Factors Used
    1. GLEC provides a single unified emissions factor for North American rail transport
    2. The rail transport provider is a reliable source of mileage information
        a. The variable of mileage has a low uncertainty due to the number of alternative routes for rail transport being very limited relative to highway travel
    3. Vendor scales are calibrated and the result is adjusted for the maximum allowable error for a calibrated legal-for-trade truck scale to ensure a conservative estimate of removals as detailed in the Uncertainty Analysis
    4. GLEC qualifies as a "reputable source" for emissions factors given the Isometric Protocol's definition

iii. Unit/Frequency of Calculation
    1. Transport emissions are calculated and assigned to each railcar of bio oil loaded into Charm-leased railcars for delivery by vendor
    2. Ledger software associates these values with the oil as it is divided and recombined for transport and processing until its final form as an injection batch

iv. Modifications to Address Uncertainty
    1. Uncertainty considerations have been applied to this value due to showing a greater than 1% effect on net removal quantity in the Sensitivity Analysis. Details can be found in the Uncertainty Assessment section here and in the GHG Statement supplemental doc

42. Vehicle Embodied Emissions from Injectate Transport to Pre-Processing
    a. Truck
        i. Calculation Methodology
            1. Per-mile embodied emissions are calculated based on GREET emissions levels for the manufacture of medium-and-heavy-duty trucks and trailers and divided by EPA expected useful life in mileage for heavy-duty diesel engines. This calculation yields a per-mile embodied emissions level that can be applied to Charm-specific journeys using vehicles owned by a transport contractor.
            2. The emissions factor is modified to include the average "deadhead" or unloaded journey distance for a vehicle of this type according to American Transportation Research Institute analysis
            3. Per-mile embodied emissions are applied to the oil being transported according to journey distance.
        ii. Justification of Variables and Emissions factors used
            1. GREET, ATRI, and the US EPA qualify as "reputable sources" given the Isometric Protocol's definition of "A source that would be widely considered trustworthy based on the process

undertaken (e.g., peer review) or origin of the information (e.g., government body)."

2. Google Maps routing is a reliable source of mileage information and is controlled for uncertainty as described in the Uncertainty Analysis section
3. Calculation of embodied emissions by mile is necessary due to the fact that the trucks may be used by the transport provider for different routes and applications over time. Therefore, a calculation method is appropriate that only takes into account the mileage associated with Charm journeys.

iii. Unit/Frequency of Calculation

1. Transport emissions are calculated and assigned to each truckload of bio oil loaded into Transport Provider vehicles for delivery by vendor
2. Ledger software associates these values with the oil as it is divided and recombined for transport and processing until its final form as an injection batch

iv. Modifications to Address Uncertainty

1. This value is being used as is due to demonstrating a less than 1% change in the Net Removal in the Sensitivity Analysis. See the "Sensitivity Analysis" sheet in the supplemental GHG Statement doc

b. Rail

i. Calculation Methodology

1. Per-mile embodied emissions are calculated based on DOT's reported average purchase cost of a DOT-111 tank car, which is the type used by Charm for bio oil transport. This cost is multiplied by the NAICS Supply Chain Emissions Factor (cost-based) for the manufacture of railroad rolling stock.
2. Per-mile embodied emissions are applied to the oil being transported according to journey distance. Emissions associated with transport once the vehicle is empty are calculated based on the assumption of a round trip due to the cars only being used for Charm bio oil transport while under lease by Charm

ii. Justification of Variables and Emissions factors used

1. Both DOT and the NAICS qualify as "reputable sources" given the Isometric Protocol's definition of "A source that would be widely considered trustworthy based on the process undertaken (e.g., peer review) or origin of the information (e.g., government body)."
    a. GREET emissions data for the manufacture of railcars is not available. The most expedient, reliably sourced data to calculate embodied emissions was cost-based, sourced from DOT and NAICS
2. Calculation of embodied emissions by mile is necessary due to the fact that the railcars are only leased by Charm for a portion of their useful life. Therefore, journey mileage is the most representative way to quantify the portion of that life associated with Charm's use.

iii. Unit/Frequency of Calculation
    1. Transport emissions are calculated by mile and assigned to each truckload of bio oil loaded into Charm railcars for delivery by vendor
    2. Ledger software associates these values with the oil as it is divided and recombined for transport and processing until its final form as an injection batch
iv. Modifications to Address Uncertainty
    1. This value is being used as is due to demonstrating a less than 1% change in the Net Removal in the Sensitivity Analysis. See the "Sensitivity Analysis" sheet in the supplemental GHG Statement doc

43. Electricity Use from Injectate Pre-Processing
    a. Calculation Methodology
        i. Electricity emissions are calculated by kWh usage multiplied by the per kWh emissions for the SPP North grid subregion as reported by the EPA eGrid emissions factors and modified by Green-e to show the residual grid mix. This is supplemented by a per-kWh addition to account for the embodied emissions associated with grid infrastructure as quantified by a weighted average of power generation sources based on the fuel mix for the subregion reported by the EPA and the applicable electricity infrastructure emissions factors reported by GREET
    b. Justification of Variables and Emissions Factors Used
        i. Electricity use is measured using a calibrated instrument from the local utility provider calibrated and purpose-built for measuring power usage over time. Given that the readings are the basis for billing for power usage, the utility has a direct incentive to ensure their accuracy
        ii. Subregional grid emissions factors and generation technology mix are provided by the US EPA and grid infrastructure emissions factors from GREET, which are reputable sources under the Isometric Protocol's definition
    c. Unit/Frequency of Calculation
        i. Electricity use is treated as a "site emission"
            1. Metered use rates are reported monthly by site staff
            2. Reported use-rates are multiplied by listed emissions factors to calculate emissions impact of usage
            3. Calculated impact is deducted from net CDR for oil processed on site
    d. Modifications to Address Uncertainty
        i. Meter readings are adjusted for uncertainty based on the maximum allowable uncertainty for a utility-provided electricity meter provided by the Code of Colorado Regulations

44. Fuel Use from Injectate Pre-Processing
    a. Calculation Methodology
        i. Emissions impact of fuel usage is calculated based on GLEC models for fuel combustion in terms of TCO2e/kg fuel multiplied by the quantity of fuel purchased for use on site as documented by purchase invoices. This quantity is corrected for fuel pump uncertainty as detailed in the Uncertainty Analysis

b. Justification of Variables and Emissions Factors Used
    i. All fuel used on site will be purchased and an invoice will be issued to record that purchase which is presumed to be reflective of the amount dispensed by a gas meter. Because the meter is used to determine billing, the fuel vendor has a direct financial incentive to ensure that it is accurate.
    ii. GLEC V3.0 is a reputable source for emissions factors according to the definition provided in the Isometric Protocol and is the most recent version of the framework
c. Unit/Frequency of Calculation
    i. Fuel use is treated as a "site emission"
        1. Invoiced use rates are reported monthly by site staff
        2. Reported use-rates are multiplied by listed emissions factors to calculate emissions impact of usage
        3. Calculated impact is deducted from net CDR for oil processed on site
d. Modifications to Address Uncertainty
    i. Uncertainty considerations have been applied to this value due to showing a greater than 1% effect on net removal quantity in the Sensitivity Analysis. Details regarding these considerations can be found in the Uncertainty Assessment section here and in the GHG Statement supplemental doc

45. Embodied Emissions from Equipment for Injectate Pre-Processing
    a. Calculation Methodology
        i. Total embodied emissions associated with equipment and facilities improvements specific to the pyrolysis processes required for carbon removal are quantified using LCA as described in the Data section of this GHG Statement report
        ii. Total embodied emissions are amortized over the Estimated Useful Life of the equipment as quantified by internal engineering resources and/or relevant literature. Where relevant, this may be modified by expected use rates for the equipment (e.g. hours of use per week)
        iii. Amortized embodied emissions are allocated to pyrolysis products produced during an Production Period based on the length of the period, since the amortization is time-based
    b. Justification of Variables and Emissions Factors Used
        i. Embodied emissions are quantified using emissions factors from reputable sources including but not limited to the GREET database, the EcoInvent database, NAICS supply Chain Emission Factors, USLCI database, ICE database, and/or the Athena Impact Estimator
        ii. Industry standards and the estimation of internal engineering resources provide a reasonable expectation regarding the overall service life of a given piece of equipment
    c. Unit/Frequency of Calculation
        i. Embodied emissions of equipment are calculated and automatically amortized over the expected useful life of the equipment and deducted from net CDR associated with removals processed at the facility during each reporting period based on time elapsed
    d. Modifications to Address Uncertainty

i. This value is being used as is due to demonstrating a less than 1% change in the Net Removal in the Sensitivity Analysis. See the "Sensitivity Analysis" sheet in the supplemental GHG Statement doc

46. Embodied Emissions from Consumables for Injectate Pre-Processing
    a. Calculation Methodology
        i. Liquid Caustic Soda (50%) is added to bio oil at a rate measured by a flow meter and recorded by on-site staff
        ii. Quantity of LCS add is multiplied by the GREET emissions factor for LCS in order to calculate embodied emissions
        iii. Salt added to the bio oil is a waste product and is not included in the calculation.
    b. Justification of Variables and Emissions Factors Used
        i. Volume measured by a flow meter provides an acceptably accurate measurement of the volume of LCS. The result is adjusted for the typical uncertainty for a flow meter to ensure a conservative estimate of removals as detailed in the Uncertainty Analysis
        ii. GREET 2023 is a reputable source for emissions factors according to the definition provided in the Isometric Protocol and is the most recent version of the framework
    c. Unit/Frequency of Calculation
        i. Embodied emissions of LCS are deducted from net CDR for each MT of oil processed
    d. Modifications to Address Uncertainty
        i. Uncertainty considerations have been applied to this value due to showing a greater than 1% effect on net removal quantity in the Sensitivity Analysis. Details regarding the Uncertainty Factor can be found in the Uncertainty Assessment section here and in the GHG Statement supplemental doc

47. Fuel Use from Consumables Transport to Pre-Processing Site
    a. Calculation Methodology
        i. Cargo weight according to Vendor-provided Bill of Lading and point-to-point distance from Google Maps are used to calculate the WTW emissions for the journey using a flatbed truck by applying the appropriate GLEC V3.0 emissions factor (flatbed truck) to the tons-miles value
    b. Justification of Variables and Emissions Factors Used
        i. Vehicle Type is always a Heavy Duty Vehicle (flatbed truck) due to the quantity and nature of cargo
        ii. Google Maps routing is a reliable source of mileage information and is controlled for uncertainty as described in the Uncertainty Analysis section
        iii. Vendor scales are calibrated and the result is adjusted for the maximum allowable error for a calibrated legal-for-trade truck scale to ensure a conservative estimate of removals as detailed in the Uncertainty Analysis
        iv. GLEC qualifies as a "reputable source" for emissions factors given the Isometric Protocol's definition
    c. Unit/Frequency of Calculation
        i. Transport emissions are calculated and assigned to each delivery of

consumables to the pre-processing site

ii. Delivery transport emissions are treated as a “site emission”

1. Delivery source and weight are recorded by site staff for each delivery
2. Reported deliveries are multiplied by listed emissions factors to calculate emissions impact of deliveries
3. Calculated impact is deducted from net CDR for oil processed on site

d. Modifications to Address Uncertainty

i. Uncertainty considerations have been applied to this value due to showing a greater than 1% effect on net removal quantity in the Sensitivity Analysis. Details regarding the Uncertainty Factor can be found in the Uncertainty Assessment section here and in the GHG Statement supplemental doc

48. Embodied Emissions from Consumables Transport to Pre-Processing Site

a. Calculation Methodology

i. Per-mile embodied emissions are calculated based on GREET emissions levels for the manufacture of medium-and-heavy-duty trucks and trailers and divided by EPA expected useful life in mileage for heavy-duty diesel engines. This calculation yields a per-mile embodied emissions level that can be applied to Charm-specific journeys using vehicles owned by a vendor.

ii. Actual journey distance is extended by the average proportion of “deadhead” or unloaded miles traveled by a vehicle of this type as reported by the American Transportation Research Institute analysis of the trucking industry

iii. Per-mile embodied emissions are applied to the product being transported according to journey distance.

b. Justification of Variables and Emissions factors used

i. GREET, ATRI, and the US EPA qualify as “reputable sources” given the Isometric Protocol’s definition.

ii. Google Maps routing is a reliable source of mileage information and is controlled for uncertainty as described in the Uncertainty Analysis section

iii. Calculation of embodied emissions by mile is necessary due to the fact that the trucks may be used by the transport provider for different routes and applications over time. Therefore, a calculation method is appropriate that only takes into account the mileage associated with Charm journeys.

c. Unit/Frequency of Calculation

i. Transport emissions are calculated and assigned to each delivery of consumables to the pre-processing site

ii. Delivery transport emissions are treated as a “site emission”

1. Delivery source and weight are recorded by site staff for each delivery
2. Reported deliveries are multiplied by listed emissions factors to calculate emissions impact of deliveries
3. Calculated impact is deducted from net CDR for oil processed on site

d. Modifications to Address Uncertainty
    i. This value is being used as is due to demonstrating a less than 1% change in the Net Removal in the Sensitivity Analysis. See the "Sensitivity Analysis" sheet in the supplemental GHG Statement doc

49. Fuel Use from Processed Injectate Transport to Injection Site
    a. Calculation Methodology
        i. Cargo weight according to a scale ticket from Well Operator and point-to-point distance from Google Maps are used to calculate the WTW emissions for the journey using a Tanker Truck by applying the appropriate GLEC V3.0 emissions factor (tanker truck) to the tons-miles value
    b. Justification of Variables and Emissions Factors Used
        i. Vehicle Type is always a Heavy Duty Vehicle (tanker truck) due to the quantity and nature of cargo
        ii. Google Maps routing is a reliable source of mileage information and is controlled for uncertainty as described in the Uncertainty Analysis section
        iii. Well Operator scales are calibrated and the result is adjusted for the maximum allowable error for a calibrated legal-for-trade truck scale to ensure a conservative estimate of removals as detailed in the Uncertainty Analysis
        iv. GLEC qualifies as a "reputable source" for emissions factors given the Isometric Protocol's definition
    c. Unit/Frequency of Calculation
        i. Transport emissions are calculated and assigned to each truckload of bio oil delivered to the injection site
        ii. Ledger software associates these values with the specific injection batch and deducts them from net CDR
    d. Modifications to Address Uncertainty
        i. Uncertainty considerations have been applied to this value due to showing a greater than 1% effect on net removal quantity in the Sensitivity Analysis. Details regarding the Uncertainty Factor can be found in the Uncertainty Assessment section here and in the GHG Statement supplemental doc

50. Vehicle Embodied Emissions for Processed Injectate Transport to Injection Site
    a. Calculation Methodology
        i. Per-mile embodied emissions are calculated based on GREET emissions levels for the manufacture of medium-and-heavy-duty trucks and trailers and divided by EPA expected useful life in mileage for heavy-duty diesel engines. This calculation yields a per-mile embodied emissions level that can be applied to Charm-specific journeys using vehicles owned by a transport contractor.
        ii. Per-mile embodied emissions are applied to the oil being transported according to journey distance
        iii. Actual journey distance is extended by the average proportion of "deadhead" or unloaded miles traveled by a vehicle of this type as reported by the American Transportation Research Institute analysis of the trucking industry
    b. Justification of Variables and Emissions factors used

i. GREET, ATRI, and the US EPA qualify as “reputable sources” given the Isometric Protocol’s definition.
ii. Google Maps routing is a reliable source of mileage information and is controlled for uncertainty as described in the Uncertainty Analysis section
iii. Calculation of embodied emissions by mile is necessary due to the fact that the trucks may be used by the transport provider for different routes and applications over time. Therefore, a calculation method is appropriate that only takes into account the mileage associated with Charm journeys.

c. Unit/Frequency of Calculation
    i. Transport emissions are calculated and assigned to each truckload of bio oil loaded into Transport Provider vehicles for delivery by vendor
    ii. Ledger software associates these values with the oil as it is divided and recombined for transport and processing until its final form as an injection batch
d. Modifications to Address Uncertainty
    i. This value is being used as is due to demonstrating a less than 1% change in the Net Removal in the Sensitivity Analysis. See the “Sensitivity Analysis” sheet in the supplemental GHG Statement doc

51. Final Injectate Mass
    a. Calculation Methodology
        i. Injectate Mass for each injection batch is determined by scale tickets produced by the calibrated truck scale at the Well Operator site
    b. Justification of Variables and Emissions Factors Used
        i. The directly-measured mass of the injectate delivered to the site is the most reliable measure of this parameter
    c. Unit/Frequency of Calculation
        i. Weights are recorded by the Well Operator, delivered to Charm, and entered into the Ledger for each injection batch
    d. Modifications to Address Uncertainty
        i. Uncertainty considerations have been applied to this value due to showing a greater than 1% effect on net removal quantity in the Sensitivity Analysis. Details can be found in the Uncertainty Assessment section here and in the GHG Statement supplemental doc
52. Mass of Bio-Oil Injected
    a. Calculation Methodology
        i. Mass of Bio-Oil Injected is computed by subtracting the mass of salt and LCS added to a given batch of bio-oil during pre-processing from the Final Injectate Mass of that batch
    b. Justification of Variables and Emissions Factors Used
        i. The directly-measured mass of the injectate delivered to the site is the most reliable measure of this parameter
        ii. The consumables subtracted from this quantity are also measured directly by site personnel
    c. Unit/Frequency of Calculation
        i. Mass of Bio-Oil Injected is computed for each injection batch
    d. Modifications to Address Uncertainty
        i. Uncertainty considerations have been applied to this value due to

showing a greater than 1% effect on net removal quantity in the Sensitivity Analysis. Details can be found in the Uncertainty Assessment section here and in the GHG Statement supplemental doc

53. Electricity Use from Injection
    a. Calculation Methodology
        i. Well Operator is required to report energy usage per Service Agreement with Charm based on the number of hours spent emplacing Charm material and the overall usage by the site during the same period
        ii. Electricity emissions are calculated by kWh usage multiplied by the per kWh emissions for the SPP North grid subregion as reported by the EPA eGrid emissions factors modified by Green-e to show the residual grid mix. This is supplemented by a per-kWh addition to account for the embodied emissions associated with grid infrastructure as quantified by a weighted average of power generation sources based on the fuel mix for the subregion reported by the EPA and the applicable electricity infrastructure emissions factors reported by GREET
    b. Justification of Variables and Emissions Factors Used
        i. Electricity use is measured using a calibrated instrument from the local utility provider calibrated and purpose-built for measuring power usage over time. Given that the readings are the basis for billing for power usage, the utility has a direct incentive to ensure their accuracy
        ii. Subregional grid emissions factors and generation technology mix are provided by the US EPA and grid infrastructure emissions factors from GREET, which are reputable sources under the Isometric Protocol's definition
    c. Unit/Frequency of Calculation
        i. Electricity use is treated as a "site emission"
            1. Metered use rates are reported monthly by site staff
            2. Reported use-rates are multiplied by listed emissions factors to calculate emissions impact of usage
            3. Calculated impact is deducted from net CDR for oil processed on site
    d. Modifications to Address Uncertainty
        i. This value is being used as is due to demonstrating a less than 1% change in the Net Removal in the Sensitivity Analysis. See the "Sensitivity Analysis" sheet in the supplemental GHG Statement doc

54. Embodied Emissions from Injection Equipment
    a. Calculation Methodology
        i. Vaulted calculated the embodied site emissions in their own PDD in terms of the tonnage of steel and concrete per metric tonne of material emplaced in their cavern in light of a conservative assumption of the caverns' overall capacity.
        ii. This allows Charm to calculate embodied emissions per ton of oil emplaced by multiplying those values by the GREET 2023 emissions factors for steel and concrete in order to calculate total embodied CO2e assignable to each injected metric tonne. See "Vaulted Emissions Calc Sheet" in GHG Statement supplemental document
    b. Justification of Variables and Emissions Factors Used
        i. Embodied emission calculations provided by Vaulted have already been

validated by Isometric and a selected third-party verifier, therefore should be considered a reliable metric for calculating embodied emissions for operations occurring at Vaulted

ii. GREET 2023 is a reputable source for emissions factors according to the definition provided in the Isometric Protocol and is the most recent version of the framework

c. Unit/Frequency of Calculation

i. This will be applied as a static value to each MT of bio oil injected

d. Modifications to Address Uncertainty

i. This value is being used as is due to demonstrating a less than 1% change in the Net Removal in the Sensitivity Analysis. See the "Sensitivity Analysis" sheet in the supplemental GHG Statement doc

55. Carbon Content of Injected Bio Oil

a. Calculation Methodology

i. The injectate C content is computed by multiplying the Mass of Bio Oil Injected by the average C content of the samples as established by lab testing. This is then multiplied by 44/12 to convert the C content to the equivalent in CO2e

b. Justification of Variables and Emissions Factors Used

i. ISO-certified analytical labs provide a high degree of certainty in analytical measurements

ii. Results are modified for uncertainty based on the reported relative standard deviation reported by the manufacturer of comparable analytical equipment

c. Unit/Frequency of Calculation

i. Tested Carbon content is applied to each Injection Batch in order to complete removal accounting

d. Modifications to Address Uncertainty

i. Uncertainty considerations have been applied to this value due to showing a greater than 1% effect on net removal quantity in the Sensitivity Analysis. Details can be found in the Uncertainty Assessment section here and in the GHG Statement supplemental doc

56. Emissions from Sample Transport to Lab

a. Calculation Methodology

i. Emissions are calculated for air shipment of samples based on a conservative estimate of the point-to-point distance between the sampling site and labs in Google Maps, and the weight and number of samples according to the standard testing battery for each injection batch of bio oil established by the Charm Research and Chemistry teams who manage the testing program

ii. Weight and distance are used to calculate emissions using the GLEC emissions factor for air freight

iii. See "Emissions from Samples Calcs" in GHG Statement supplemental document

b. Justification of Variables and Emissions Factors Used

i. The number of tests, samples, and sample weights are internally standardized for each injection batch and calculated as reported by the Charm Chemistry and Research teams

ii. Distance between the sampling site and labs is determined according to

Google Maps routing. Road distances are used despite the transport assumption being air freight. This is due to the fact that road distances will reliably be longer than the more direct route allowed by air travel, ensuring a conservative estimate of removals

iii. GLEC qualifies as a "reputable source" for emissions factors given the Isometric Protocol's definition

c. Unit/Frequency of Calculation

i. Emissions are applied as a static value to each injection batch

d. Modifications to Address Uncertainty

i. This value is being used as is due to demonstrating a less than 1% change in the Net Removal in the Sensitivity Analysis. See the "Sensitivity Analysis" sheet in the supplemental GHG Statement doc

57. Disposal of Process Waste

a. Fuel Emissions from Waste Transport to Disposal Facility

i. Calculation Methodology

1. Cargo weight and destination are documented according to manifest from EPA-regulated waste disposal provider and point-to-point distance between origination and disposal facilities from Google Maps are used to calculate the WTW emissions for the journey using the appropriate vehicle type by applying the appropriate GLEC emissions factor (flatbed or tanker truck) to the tons-miles value

ii. Justification of Variables and Emission Factors Used

1. Vehicle Type is always a Heavy Duty Vehicle due to the quantity and nature of cargo – A flatbed truck is used for solid waste, and a tanker truck is used for liquid waste
2. Google Maps routing is a reliable source of mileage information and is controlled for uncertainty as described in the Uncertainty Analysis section
3. Disposal manifests and providers are subject to regulation and review by USEPA, qualifying them as a reputable and reliable source of input data
4. GLEC qualifies as a "reputable source" for emissions factors given the Isometric Protocol's definition

iii. Unit/Frequency of Calculation

1. Transport emissions are calculated for each manifested waste disposal required by project activities

iv. Modifications to Address Uncertainty

1. Masses are transmitted to Isometric DMRV with the maximum error of a legal-for-trade truck scale for variance propagation, as is the case for other project transport emissions as discussed in the Uncertainty Assessment section of the PDD
2. When Google Maps routing provides multiple options, the longest routing option is chosen as discussed in the Uncertainty Assessment section of the PDD

b. Vehicle Embodied Emissions from Waste Transport to Disposal Facility

i. Calculation Methodology

1. Per-mile embodied emissions are calculated based on GREET emissions levels for the manufacture of medium-and-heavy-duty

trucks and trailers and divided by EPA expected useful life in mileage for heavy-duty diesel engines. This calculation yields a per-mile embodied emissions level that can be applied to Charm-specific journeys using vehicles owned by a transport contractor.

2. The emissions factor is modified to include the average "deadhead" or unloaded journey distance for a vehicle of this type according to American Transportation Research Institute analysis
3. Per-mile embodied emissions are applied to the waste being transported according to journey distance.

ii. Justification of Variables and Emission Factors Used

1. GREET, ATRI, and the US EPA qualify as "reputable sources" given the Isometric Protocol's definition of "A source that would be widely considered trustworthy based on the process undertaken (e.g., peer review) or origin of the information (e.g., government body)."
2. Google Maps routing is a reliable source of mileage information and is controlled for uncertainty as described in the Uncertainty Assessment section
3. Calculation of embodied emissions by mile is necessary due to the fact that the trucks may be used by the transport provider for different routes and applications over time. Therefore, a calculation method is appropriate that only takes into account the mileage associated with Charm journeys.

iii. Unit/Frequency of Calculation

1. Transport emissions are calculated for each manifested waste disposal required by project activities

iv. Modifications to Address Uncertainty

1. When Google Maps routing provides multiple options, the longest routing option is chosen as discussed in the Uncertainty Assessment section of the PDD

c. Emissions from Material Disposal and Decomposition

i. Calculation Methodology

1. Total mass of material disposed is scaled by appropriate emission factor as indicated by the disposal method and material

ii. Justification of Variables and Emission Factors Used

1. The mass, material, and the disposal method are documented in the manifest and disposal documents provided by the EPA-regulated waste disposal provider, a reputable and reliable source for these datapoints
2. The emission impact of disposal is quantified using emissions factors from reputable sources including but not limited to the GREET database, the EcoInvent database, NAICS supply Chain Emission Factors, USLCI database, ICE database, and/or the Athena Impact Estimator, reputable sources based on the Isometric Protocol's definition
    a. Emission factors are used as-is where feasible. To account for the disposal of less-studied material such as

bio-oil, a reasonable estimate may be made using proxy values. In this case, references to baseline values and sources, as well as any additional assumptions and justifications will be documented by the project

iii. Unit/Frequency of Calculation
    1. Disposal emissions are calculated for each manifested waste disposal required by project activities at the time that the disposal activity is confirmed by relevant documentation

iv. Modifications to Address Uncertainty
    1. Masses are transmitted to Isometric DMRV with the maximum error of a legal-for-trade truck scale for variance propagation, as is the case for other project transport emissions as discussed in the Uncertainty Assessment section of the PDD

Additional Summary: Lot Emissions vs. Site Emissions
Overview: Emissions associated with bio-oil sequestration are calculated one of two ways.

1. *Lot Emissions* are emissions that can be clearly associated with a particular lot of bio oil. These emissions are associated with a specific lot of oil in the ledger and deducted from the net CDR associated with that specific lot or injection batch of bio-oil.
2. *Site Emissions* are emissions that cannot be associated with a specific lot of oil, but are associated with a particular processing site. These emissions are accrued to the site and deducted from removals associated with the site each reporting period. Site Emissions include, but are not limited to, emissions from energy used to power the site overall as these emissions cannot be reliably mapped to a particular processed lot.

For clarity and transparency, all emission calculations discussed in this PDD are sorted into Lot Emission and Site Emission categories below

1. Lot Emissions
    a. Pyrolysis Process Emissions
        i. Fuel Use from Feedstock Transport
        ii. Fuel Use from Biomass Pyrolysis
        iii. Electricity Use from Biomass Pyrolysis
        iv. Pyrolysis Tailgas
    b. Pyrolysis Embodied Emissions
        i. Embodied Emissions from Feedstock Transport
        ii. Embodied Emissions from Equipment and Facilities for Biomass Pyrolysis
    c. Fuel Use from Injectate Transport to Pre-Processing
    d. Vehicle Embodied Emissions from Injectate Transport to Pre-Processing
    e. Embodied Emissions from Liquid Caustic Soda for Injectate Pre-Processing
    f. Fuel Use from Processed Injectate Transport to Injection Site
    g. Vehicle Embodied Emissions from Processed Injectate Transport to Injection Site
    h. Embodied Emissions from Injection Equipment
    i. Carbon Content of Injected Biomass (contra-emission)
    j. Emissions from Sample Transport to lab
2. Site Emissions
    a. Electricity Use from Injectate Pre-Processing

b. Fuel Use from Injectate Pre-Processing
c. Fuel Use from Consumables Delivery Transport
d. Embodied Emissions from Consumables Delivery Transport
e. Electricity Use from Injection
f. Fuel Use from Injection

3. All emissions from Charm bio-oil production
    a. All emissions (process and embodied) are aggregated at the site level and allocated across all pyrolysis products produced during a given Production Period
    b. This means that they are calculated at the site level during production. However, when the emissions associated with a particular lot of Charm-produced bio-oil are used in the net CDR calculation of a particular removal, they are treated as lot emissions because they are associated with a particular bio-oil lot
    c. This distinction also relates to periodization
        i. A Production Period designates all of the CO2e emitted and pyrolysis products produced during a particular period at Charm's pyrolysis facility produced between a start date and end date
            1. All emissions produced during a given Production Period are allocated to the injectable pyrolysis products produced during that period
            2. As injectable pyrolysis products are sequestered, the allocated emissions are deducted from the Net CDR associated with the removal. This calculation is made during the Reporting Period during which the injection occurs
        ii. A Reporting Period designates the reporting of all removals completed between a start and end date and the reporting of associated CDR calculations
            1. Emissions associated with the production of pyrolysis products are netted out against the Net CDR claimed during the Reporting Period during which they are injected

GHG Statement Results

The following information should be provided in this section:

- *Baseline results reported in t CO_2e for the Reporting Period*
- *Net CO_2e removals results reported in t CO_2e for the Reporting Period. These must be aggregated and also broken down into GHG SSRs*
- *If it has not already been covered in the* 'Uncertainty assessment' section of the PDD, provide *outcomes of sensitivity and uncertainty analyses including a statement of how uncertainty affects the results and how it has been addressed to minimize misrepresentation*
- *A statement of how the assumptions and choices made in the assessment are conservative*

The GHG Statement calculations must be provided separately, including all raw data and evidence. The GHG Statement calculations must be clearly referenced with a transparent audit

trail of evidence, decision making, assumptions, explanations.
This PDD is being submitted for the purposes of Project Validation. GHG Statement Results will be updated with each subsequent verification.

Appendix 6: BCU Calculation Methodology

As a relatively small, distributed operation, Charm relies on multiple operating activities which require the use of diesel fuel:

1. Transportation via road and rail
2. Other rolling stock such as telehandlers and front-end loaders
3. Onsite generators where grid power is not available

Our options for addressing the emissions impact of this fuel usage are limited. For transportation, our scale requires us to rely on third-party transportation providers who use fossil diesel. Our ability to use renewable diesel (RD) for other applications is also limited by availability as none of our operations occur within an LCFS jurisdiction. As a result, supply of renewable diesel is not practically accessible for our on-site applications.

The most expedient means of addressing the impact of diesel required by our operations, then, is the procurement of renewable diesel emissions attributes via Book and Claim Units (BCUs). By purchasing credibly additional BCUs, we finance the production of RD that would not be economically feasible to produce absent the BCU purchase. This is delivered outside of an LCFS jurisdiction, displacing volumes of fossil diesel that would have been used in a counterfactual scenario.

The BCUs generated by the production and delivery of the fuel are associated with a specific quantity of avoided emissions estimated based on the volume of fuel delivered. This is documented by a Proof of Sustainability (PoS) certificate provided for the volumes delivered. The PoS certificate accounts for the Well-to-Tank (WTT) emissions impact of the fuel production from feedstock through production and ultimately delivery. This assessment is conducted in accordance with and certified by a reputable existing framework (RSB Global). The Tank-to-Wheel (TTW) emissions of the fuel usage are quantified using the GLEC emission factor for the appropriate fuel associated with the BCUs. This allows us to account for the full Well-to-Wheel (WTW) emissions for the production, distribution, and use of the RD whose attributes are purchased as BCUs.

As already noted in the PDD, our fossil diesel emissions are also calculated using WTW emission factors. Transport emissions are calculated using the Distance Based Method outlined in the existing Transportation Emissions Accounting Module, while onsite usage is calculated based on the total quantity of fuel delivered to the site as reflected by invoices. These values are scaled by the applicable WTW emission factor from GLEC in order to quantify the total emissions associated with production, distribution, and use of the fuel.

In each case, the use of WTW emission factors allows for a straightforward comparison between total emissions generated as a consequence of fossil diesel usage and those generated by renewable diesel. The renewable diesel displaces the use of fossil diesel, and the demonstration of additionality required by the Transportation Emissions Accounting Module

for BCUs establishes that displacement would not have occurred in a counterfactual scenario where the BCU was not purchased by the project. Therefore, the difference between the WTW emissions of a given quantity of fossil diesel and renewable diesel constitute the quantity of emissions avoided as a result of the purchase of BCUs associated with the renewable diesel.

With this as the background, Charm is using a calculation method that is different from the equations in 4.2.1 of the Transportation Emissions Accounting Module. The Transportation Emissions Accounting Module requires LCA calculations to be done in a manner that simulates the direct replacement of FD by RD by volume for operations, which means the LCA acknowledges the emissions associated with the RD value chain. The alternate calculation used by Charm results in BCUs being retired to address emissions from the RD volume used, and the associated value chain emissions, in the LCA calculation**.** We can document that our BCU purchases are additional and can account for the WTW emissions of both fossil and renewable diesel using emission factors from reputable sources. Therefore, we have a reliable quantity of avoided emissions associated with our BCU purchases that will allow us to transparently quantify the impact of avoided emissions from the RD value chain.

Our methodology is to match the emissions impact in terms of kg CO2e emitted with the avoided emissions of BCUs purchased and retired. We will purchase and retire a quantity of BCUs whose avoided emissions will be less than or equal to the emissions generated as a result of fossil diesel usage. This enables a reduction of diesel emissions that are netted against removals in a GHG statement by the quantity of avoided emissions attributable to the RD associated with purchased and retired BCUs.

This is accounted for using the following set of equations:

Equation for Well-to-Tank Emissions for Renewable Diesel

$EP_{RD,\ WTT} = CI_{RD,\ WTT} * LHV_{RD,\ WTT}$

Where:

1. $EP_{RD,WTT}$ is the WTT emission profile for each kilogram of a given batch or delivery of RD
2. $CI_{RD,\ WTT}$ is the WTT carbon intensity given in the PoS denominated in grams of CO2e per MJ of fuel
3. $LHV_{RD,\ WTT}$ is the lower heating value given in the PoS denominated in MJ per kg of fuel

Equation for Well-to-Wheel Emissions for Renewable Diesel

$EP_{RD,\ WTW} = (EP_{RD,\ WTT} + EF_{RD,\ TTW})$

Where:

1. $EP_{RD,\ WTW}$ is the WTW emission profile for each kilogram of a given batch or delivery of RD
2. $EP_{RD,WTT}$ is the WTT emission profile for each kilogram of a given batch or delivery of RD

3. $EF_{RD, TTW}$ is the TTW emission factor for the applicable type of fuel (currently sourced from GLEC)

Equation for Renewable Diesel WTW Avoided Emissions

$AE_{RD} = EF_{FD, WTW} - EP_{RD, WTW}$

Where:

1. AE_{RD} is the avoided emissions from the use of a given quantity of RD
2. $EF_{FD, WTW}$ is the WTW emission factor quantifying the emission impact in CO2e for each kilogram of fossil diesel (currently sourced from GLEC)
3. $EP_{RD, WTW}$ is the WTW emission profile quantifying the emission impact in CO2e for each kilogram of renewable diesel associated with a given batch or delivery

Equation for Netting Fossil Diesel Emissions against Avoided Emissions from Renewable Diesel BCUs

Net $CO2e_{FD} = (CO2e_{Transport} + CO2e_{On\text{-}Site}) - (AE_{RD} * Mass_{RD})$

Where:

1. Net $CO2e_{FD}$ is the net CO2e attributable to the use of fossil diesel which will be deducted from the gross sequestration in a given GHG Statement
2. $CO2e_{Transport}$ is the total CO2e attributable to transport activities, typically calculated using the Distance-Based Method provided in the Transportation Emissions Accounting module
3. $CO2e_{On\text{-}Site}$ is the total CO2e attributable to on-site (non-transport) activities, typically calculated by applying a WTW emission factor to the quantity of fuel delivered to the site for project activities
4. AE_{RD} is the avoided emissions from the use of a given quantity of RD
5. $Mass_{RD}$ is the mass of RD whose attributes have been procured and are being retired by the project in order to offset FD emissions in a given GHG Statement

Note: Net $CO2e_{FD}$ may only be a positive value or zero. It may never be a negative value.

Appendix 7 – Basco 6 EW Bio-oil Sequestration Site Permanence Monitoring and Permit Compliance Plan

Supporting data here

The Basco 6 EW is a former saltwater disposal well located in Evangeline Parish, Louisiana. It was drilled in 2011, and while active, injected nearly 20,000 bbls of produced water before transitioning to orphan well status in 2016. Charm gained permission to test the well's integrity in February 2024, and finding the well in suitable condition, requested liability from the State of Louisiana to repurpose the well for permanent sequestration of bio-oil.

The data acquisition and monitoring plan framed below highlights how Charm will fulfill Isometric's Bio-oil Storage in Permeable Reservoirs ("Protocol") module with a single injection asset ("Well") in addition to those anticipated from a Class V UIC permit issued by the Louisiana Department of Energy and Natural Resources ("LDENR").

The following deviations from the Protocol are required on account of the project design; see discussion below for alternative methods, where applicable:

- While the Protocol accommodates the presence of a monitoring well, the Basco 6 EW project consists only of a single Well for injection. As a result, *formation fluid analytes (i.e., pH, density, anions/cations, etc.) will not be available* after bio-oil injection operations commence, as doing so would jeopardize previously sequestered bio-oil. Bio-oil density will be referenced relative to that of the in-situ formation fluid at baseline conditions. In addition, formation fluid will not be available for corrosion monitoring purposes. See Section 3.1.1 and Section 3.2.

- Bio-oil analytes at each injection batch will not include isotopes or geochemical characterization ("fingerprint") of bio-oil compounds unless mandated by LDENR, as referenced in Section 3.1.1. This data is both time and cost prohibitive to conduct routinely, and does not inform project permanence.

- Corrosion monitoring as described in Section 3.1.2 will be continuously monitored by tubing and annulus pressure, with visual inspection performed on tubing and/or packer only if and when the tubing and/or packer is run out of the hole for maintenance. Pulling the equipment from the well for the purpose of visually monitoring this equipment every 6 months unnecessarily jeopardizes the well's completion and the equipment's condition. It is also time and cost prohibitive, requiring LDENR approval before the work is scheduled with an on site MIPT to be witnessed by LDENR staff upon completion of the work.

  Charm continues to explore and mature material compatibility testing with a wide range of steel alloys and bio-oils to inform longevity of J-55 and other well-specific materials.

- Direct measurement of bio-oil migration as described in Section 3.1.3 is not possible without use of a monitoring well(s). Plume development will be inferred from the Warner-Lehr model, but is not expected to be required by LDENR. Vertical migration will be monitored annually via radioactive tracer survey.

- Groundwater monitoring as described in Section 3.1.3 is not anticipated to be required by LDENR.
- Continuous inline gas detectors equipped with sensors for GHG and VOCs specified in the Protocol (Section 3.1.3) are prohibitively expensive for incorporating into Charm's project. The unlikely occurrence of gas to surface will be monitored as described below:
    - Continuous annulus pressure monitoring will detect any subsurface tubing or casing leakage. In the event of an unexplained annulus pressure reading, gas detection sensors and/or cameras will be used to determine the presence of any gas leakage and corrective action applied where root cause analysis justifies.
    - Every six months, a wellhead gas measurement will be conducted and compared with baseline readings as described in the Protocol.
    - Annual mechanical integrity testing will confirm that the well construction materials are sound as described in the Protocol.
- A seismic monitoring program as described in Section 3.1.3 is not expected to be required by LDENR.
- Charm's understanding of bio-oil compatibility has been enhanced by external research which identified calcium as a common reservoir mineral most sensitive to bio-oil; work is underway to extend this research to the Lower Miocene formation sampled from the Basco 6 EW well. In addition, we are leveraging this learning in our candidate selection process as a primary means of mitigating, if not eliminating, bio-oil reactivity. Monitoring for potential reactivity will be inferred from continuously recorded surface pressure and injectivity trends. Periodic falloff tests will also inform reservoir quality changes resulting from either aging (e.g., increased "skin effect") or reactivity (decreasing "skin effect"). Even in the event of stimulation-scale dissolution within the wellbore, a wellbore maintained to regulatory standards will not generate leakage.

Permitting

2.1	*The injection site must have a current Class I or V UIC[10] well permit issued by the responsible authority for the location of the injection facility and reservoir. The permit must specifically identify bio-oil or an equivalent type of injectant, as acceptable injectants under the permit[11].*

A LDENR-issued Class V UIC permit is pending for Charm LA, LLC-operated asset, Basco 6 EW (S/N 974025), located in Evangeline Parish, Louisiana.

Site Characterization and Feasibility Requirements

2.2	*The injection site must have a current Class I or V UIC[10] well permit issued by the responsible authority for the location of the injection facility and reservoir. The permit must specifically identify bio-oil or an equivalent type of injectant, as acceptable injectants*

	under the permit[11].

The UIC permit application did not require demonstration of the geologic system for bio-oil sequestration, as the well was historically approved for salt water disposal operations.

Parameter	Result
Reservoir lithology and mineralogy	The Lower Miocene formation is an unconsolidated sandstone, consisting primarily of quartz as indicated by XRD of fill taken from the below the perforations.
Porosity and permeability of sequestration zone strata	The completed interval ranges from 25-27% effective porosity (as determined from cased hole logs), with an estimated permeability range from 465 mD to > 2D.
Confirmation of low permeability and structural integrity of confining layer/cap rock	The previous operator injected 17,700 bbls of brine in this formation, which remained geologically isolated from overlying aquifers by mechanical integrity and numerous geologic seals.
Temperature, pH, conductivity and fluid saturation of storage reservoir formation fluid/brine	Formation fluid properties here.
Dissolved gas, including dissolved inorganic carbon (DIC), composition in formation fluids and composition of any tracers being used (e.g., $\delta\ ^{13}C$ signature and/or major and minor ion).	Formation fluid composition here. No tracers being used.
Formation fluid stable oxygen $\delta^{18}O$ and hydrogen δD isotope signature, where required in the permit	Not required by the permit
$\delta^{13}C$ of the compounds of the bio-oil, where applicable	Not applicable
Quantity of hydrocarbons present in the reservoir prior to bio-oil injection	Trace (residual) hydrocarbons, likely entrained from produced water injection, are present but not recoverable

Well construction requirements

2.3	If pre-existing wells are being used for injection or monitoring, special considerations are necessary to ensure the integrity of the well and to prevent fluid migration along the borehole. These must be agreed and checked by the regulating body to determine construction and safety is consistent with well construction requirements. All checks and modifications must be recorded and all records kept. This must include the following: ● determining the integrity of the cement and casing. ● conducting any necessary action to repair defects. ● determining whether the existing well materials are adequate for the new function of the well. ● determining the diameter of the hole, any deviations from vertical, and any significant curvature or bends in the well should be compared with the size of the proposed monitoring equipment. ● existing well materials should be checked to ensure that they are compatible with fluids with which they will come into contact, such as carbon dioxide and carbon dioxide-rich brines if they are completed in the injection zone. ● any flaws in the casing or cement will need to be repaired.

All records confirming well integrity have been reviewed and approved by LDENR. Charm will maintain all well records documenting relevant activity on the location.

Injection and injectant operation & monitoring

3.1.1	• Maximum allowable surface injection pressure (MASIP) at the injection wellhead that is allowed during injection operations to prevent fracturing of the formation…
	  ○ Installation and use of continuous recording devices to monitor injection pressure and the pressure on the annulus between the tubing and the long string casing
	  ○ Monitoring and documentation of injection pressure must be performed and records maintained for review
	• Maximum bio-oil injection rate to monitor volumes injected, prevent induced seismicity or return of injectant
	  ○ Installation and use of continuous recording devices to monitor injection rate and volume
	  ○ Monitoring and documentation of injection rate must be performed and records maintained for review
	• Limitations on composition of the injected fluid, including, but not limited to pH, density, temperature or other parameters, if relevant, to ensure injectant does not negatively impact the formation via inducing dissolution, reaction, or other degradation pathways, resulting in increased potential for bio-oil and fluid migration. Density in particular must be greater than that of the formation water.
	  ○ Records of laboratory analyses and relevant permit limitations to demonstrate compliance must be maintained in accordance with the well permit and available for review.
	• Analysis of the bio-oil stream with sufficient frequency to yield data representative of its chemical and physical characteristics, including analysis of the following for each injection batch using industry standard or indicated methods and quality and properly calibrated equipment:
	  ○ pH
	  ○ density
	  ○ viscosity
	  ○ TAN
	  ○ carbon content (for further details see Section 7.3.3.1 of the main protocol)
	  ○ δ13C of the bio-oil compounds (where applicable)
	• Additional measurements of the bio-oil stream may include, oxygen content or a geochemical characterization/fingerprint of bio-oil (for example the major and minor ions).
	• Wells must have gas detectors (or equivalent sensors/imaging) with alarms and injection shut-off systems (e.g., automatic shut-off or procedures in place for manual shut off of injection/operation), including for gaseous release (CO_2, hydrocarbons or other GHGs) and injection pump shutoff when maximum pressure is reached or maximum flow rate is exceeded. If activated the operator must immediately investigate and identify as expeditiously as possible (or in accordance with permit requirements) the cause of the alarm or shutoff, and report the instance to the validation and verification body (VVB).

LDENR issued the MASIP at 455 psi. Continuous recording devices will capture flowrate, injection volume, tubing, annulus and bottomhole pressures, and bottomhole temperature for regulatory reporting. Equipment list found in the site Embodied Emissions Inventory submitted with this document.

Operations will be conducted according to any other operational conditions (e.g., flow rate, injectant composition or properties) cited on the approved UIC permit.

Injectant samples will be taken at a frequency mandated by LDENR to confirm pH, density, viscosity, TAN and carbon content and/or as required by the approved UIC permit.

Production batches, representing a change in biomass feedstock or pyrolysis process, will extend the above analysis to include major anions and cations.

Wellhead gas detection will be performed every 6 mos to confirm no gas leakage at surface. Setpoints on the injection pump and flowmeter will trigger equipment alarm and automatic shutoff.

System Integrity Monitoring

3.1.2	• Corrosion monitoring of well materials, upon well completion and performed and reported every 6 months thereafter, for loss of mass, thickness, cracking, pitting, and other signs of corrosion, to ensure that well components meet the minimum standards for material strength and performance set by API, ASTM International, or equivalent. If injection frequency is less than twice per week in that timeframe, then corrosion monitoring requirements are not mandatory, but are recommended. Alternatively, corrosion performance of well materials may be evaluated using the Corrosivity Screening of Pyrolysis Bio-oils by Short Term Alloy Exposures developed by NREL[14] • A demonstration of external mechanical integrity on a cadence specified by the UIC permit annually until the injection well is plugged • A pressure fall-off test, annually

Corrosion monitoring will be performed at a 6 month frequency. Witnessed mechanical integrity tests will be conducted annually per LDENR. A pressure falloff test will be performed annually.

Migration and Storage Reversal Monitoring

3.1.2	• As applicable based on specific site conditions, formation type, and permit class, monitoring to ensure bio-oil migration beyond the AOR has not occurred. Such approaches may include periodic monitoring of pressure and/or water chemistry in overlying formations above the storage complex, noting changes versus baseline conditions that may indicate bio-oil related migration. • As applicable based on specific site conditions, formation type, permit class, and as required by the UIC permit, analysis of groundwater for bio-oil related parameters on an annual basis, to include at a minimum, analysis for bio-oil fingerprint. Results must be compared to baseline values obtained prior to bio-oil injection[15]. • Monitoring of the composition of any gas recovered from well(s) or representative sampling locations where detection of gases from the reservoir may be completed (such as a recovered brine sealed holding tank). Gas composition monitoring shall include CO_2, CH4 and VOCs emissions from the storage reservoir via CO_2, CH4 and VOCs gas monitors with a resolution of at least 0.01 vol% or lab analysis, if sampled. Results must be compared to baseline values obtained prior to bio-oil injection and any differences must be assessed by the Project Proponent to see if they are attributable and material to the project.

	Sampling frequency should be aligned with other well and reservoir testing (e.g., corrosion monitoring) every 6 months.
	● As applicable based on specific site conditions, formation type, and permit class, monitoring of wellbores within the AOR that intersect the storage complex at depth, annually while operating and every 5 years post-closure. Monitoring should include direct observation of the wells, if possible, and surface air monitoring around the wellbore. Monitoring should focus on pressure measurements and identifying gaseous degradation products (CO_2, CH4, VOCs) in the vicinity of the wellbore that may indicate bio-oil degradation and/or migration. Results must be compared to baseline values obtained prior to bio-oil injection.
	● A seismic monitoring program may be suggested at the discretion of the UIC director in areas of increased seismic risk where seismicity may have an impact on the formation and the durability of the bio-oil being stored.The area within the AOR of the injection facility and the area of the storage complex must be monitored for magnitude 2.7[12] or greater seismic events to determine the presence or absence of any induced micro-seismic activity associated with the wells and near any discontinuities, faults, or fractures in the subsurface.
	
	For all injection and injectant (bio-oil) monitoring and analyses, sufficient samples must be analyzed to determine that the composition of the bio-oil is within specified parameters in the UIC permit, where required. Each individual batch of bio-oil that is injected should be analyzed and characterized to ensure variations in pH, density, viscosity, TAN and δ13C of the bio-oil compounds (where applicable) from batch to batch are accounted for. Samples should be from a well mixed and representative container of the bio-oil.
	Any major consistent deviations could result in an AOR re-evaluation, which may be performed based upon injection operations at the UIC Program Director's discretion/request.
	
	If any CO_2 leaks are detected from the targeted reservoir or there are significant irregularities from the used model(s), the Project Proponent/operators must undertake corrective measures as set out in their monitoring plan submitted and approved by the UIC Program Director. For a loss of conformance with models, the Project Proponent must halt injection whilst they identify the cause of this loss, and then revise the monitoring plan to account for this. If the caprock is found to have lost integrity, the Project Proponent must halt injection whilst they conduct an assessment to determine whether there is any leakage and whether the loss of containment and/or well mechanical integrity can be repaired prior to injection beginning again. If loss of caprock integrity is found to be the cause, injection must cease. The amount of CO_2 lost must also be quantified and subtracted from the overall total of CO_2 stored.
	
	Re-evaluations of the bio-oil fluid plume extent must also be implemented when warranted based on observational or quantitative changes of the monitoring parameters of the storage reservoir, including but not limited to:
	● observed migration of the bio-oil fluid plume is unexpected and suggests potential movement of bio-oil outside the intended formation
	● bio-oil migration into a zone above the storage complex
	● actual bio-oil fluid plume or elevated pressure extend beyond analytical model expectation because any of the following has occurred:
	  ○ any seismic activity in the area of magnitude 2.7[16] or greater within the AOR;
	  ○ new site characterization data change the model inputs to such an extent that the predicted bio-oil fluid and/or pressure plume extends vertically or horizontally beyond what was originally predicted.

Charm does not operate an offset well in the immediate vicinity of the Basco 6 EW. As a result, Charm will estimate the radius of the pressure front throughout operations.

Charm will conduct grab samples of injectant as required by the approved UIC permit.

Gas detection at surface will be conducted at a 6 month frequency. Any deviations from baseline measurements attributed to Charm's operations will be reported to LDENR with plans for corrective action, with operations resuming with LDENR approval.

Charm's Class V UIC permit is not expected to require groundwater or seismic monitoring.

Post-injection Monitoring Plan

3.2	The aim of this post-injection monitoring plan is to put in place scientific and/or operational monitoring practices which go beyond requirements of the UIC injection permits specified under the UIC well classes allowed under this protocol, in order to prove beyond reasonable doubt that bio-oil storage will be durable for the expected 100,000-1,000,000 year timescales. As outlined in Section 1, bio-oil injection is a nascent area of research, and so addressing potential risks to durability is important for ensuring robust and diligent CO_2 removals. Direct imaging of bio-oil using seismic surveys will be challenging due to the density contrast between bio-oil and the surrounding formation fluid/brine being minimal[2]. Therefore, post-injection monitoring must focus on using both injection and monitoring wells, if applicable (the plans for which are addressed by the UIC permit) to measure presence and characteristics of bio-oil directly. Please note, the requirements in this section should be followed prior to closure of the injection well (see Section 3.5). The Project Proponent must follow any post-injection requirements of the UIC permit for the specified project, in addition to the following: Before polymerization (solidification) of bio-oil: ● Density contrast is the key mechanism that provides long-term durability of bio-oil storage. The density of bio-oil samples must be measured prior to each injection and compared to formation fluid values to ensure that bio-oil will sink upon injection. These measurements should be repeated before each new injection of bio-oil. Formation fluid samples will be acquired prior to bio-oil injection and compared to historical production data from the reservoir or nearby representative analogues. Density contrast is demonstrated by: ○ Calculation according to the density contrast equation below: ○ $\rho_{bio-oil} > \rho_{formationfluid}$ where ρ is density at reservoir conditions calculated ○ $\rho = m/V$ where m is mass and V is volume, calculated according to reservoir-specific conditions, such as pressure, temperature and salinity (and any other reservoir conditions relevant to a density calculation). ○ Bio-oil density will be calculated according to reservoir conditions (as above) using the measured surface density (measured following ASTM D 4052-22 or ASTM D7042-21a) before commencing injection. Formation fluid density will be calculated or measured using samples of the formation fluid collected prior to injection. ● Bio-gas monitoring: ○ Intermittent gas sampling for emissions of CO_2, CH4 and VOCs is required, when gas is detected, during injection of bio-oil until closure of the well. If any emissions outside of the normal average baseline range

for any of these gases occur, then further monitoring of gases should take place. Further monitoring entails measurements of:

- concentrations (e.g., CO_2, CH4 VOCs), usually measured by GC
- stable isotope compositions of CO_2, CH4 and VOCs ($\delta 13C$ of CO_2, CH4 , VOCs and δD of CH4), usually measured by an isotope ratio mass spectrometer (IRMS) or cavity ring down mass spectrometry, to determine the source of the bio-gas emissions.
- $\delta 13C$ must be measured against the standard reference material Vienna Peedee Belemnite (VPDB).
- δD must be measured against the standard reference material Vienna Standard Mean Ocean Water (VSMOW).

- Monitoring for interaction with surrounding reservoir rocks:
  - Intermittent gas sampling every 6 months of CO_2, CH4 and VOCs, as above, with the same trigger conditions for further measurements of CO_2, CH4 and VOCs.
  - Baseline formation fluid composition, to include temperature and major cations and anions, other characterization may include water isotope ratios ($\delta 18O$ and δD)
  - Formation fluid composition samples taken where possible during corrosion monitoring every 6 months (see Section 3.1.2), to include major cations and anions and temperature as well as if possible characterization of the water isotope ratios ($\delta 18O$ and δD)
- System Integrity monitoring should continue, with frequencies set based on information gathered during the operational phase
- The spread of the plume post-injection should be monitored to ensure that it is within laterally contained land and/or pore space ownership limits and vertically protected from the defined lowermost USDW using:
  - Monitoring via monitoring wells, where possible
  - Modeling (or alternative methods)

At a time period expected between 2-15 years, sampling and/or other monitoring data from the injection well and/or monitoring wells at the time intervals specified here may demonstrate that bio-oil has polymerized (solidified)[17] for example by obtaining a core from the reservoir. After the given time period, the following monitoring may be required under certain site conditions:

- Periodic biogas monitoring every 6 months (exactly as in Section 3.1.3)
- It is possible that before polymerization there will be separation of a carbon-rich aqueous phase from bio-oil. This is expected to be denser than formation fluids and will therefore remain unable to migrate upwards towards the surface[2] [18] [19]. Monitoring of the aqueous phase will be challenging due to it remaining deep in the reservoir and being indistinguishable from formation brines on geophysical logs. Any migration of fluids in an unexpected manner observed at the surface prior to closure of the site should be sampled and measured for (i) carbon content and density (as above), and any reversals in storage accounted for as outlined in Section 3.3.

During operations phase

Density contrast

In the absence of a monitoring well, formation fluid samples cannot be collected from the Basco 6 EW without jeopardizing previously sequestered volumes. All density values will be referenced to baseline formation fluid analysis results.

Biogas monitoring

Intermittent gas sampling will be performed if detected at surface.

Monitoring for interaction with reservoir rocks

Pressure falloff tests conducted throughout the operations phase that signal increase in pore space will be investigated with wellhead samples to corroborate the source.

System integrity monitoring

System integrity monitoring throughout the operations phase will continue as per UIC requirements.

Post-injection monitoring plan

Upon ceasing injection, Charm will issue a post-injection monitoring plan leveraging appropriate technologies for confirming permanence. A summary of closing activities is included in the following Appendix

Appendix 8: Basco 6 Bio-Oil Sequestration Site Well Closure and Remediation Plan

Location: Basco 6 EW, Evangeline Parish, Louisiana

Preliminary Work and Notifications

Prior to initiating the closure of the injection well, the operator shall notify the Louisiana Department of Natural Resources (LDNR) Office of Conservation by submitting a completed Form UIC-23. This notification must be provided at least ten days in advance of the intended start date of plugging and abandonment (P&A) operations. As part of the preparation, a mechanical integrity test (MIT) will be conducted to evaluate the condition of the well and confirm the integrity of the casing and tubing. If issues are detected, they will be documented and incorporated into the final plugging design.

A review of all available well records will be undertaken to inform the P&A approach. This review will include analysis of drilling and completion reports, cement bond logs (CBL), casing and tubing schematics, and any historical workover documentation. These data will guide decisions about plug placement and the required cement volumes to ensure a secure and permanent abandonment.

After the well has reached the end of injection, but prior to commencing P&A operations, the well will be shut-in and monitored to ensure bio-oil is stable. Once stability is verified, the well will be cleared for preparation to start plugging operations.

Wellbore Preparation

The plugging operations will begin with the removal of any downhole equipment, including injection tubing and the packer, to ensure clear access to the wellbore. Once these components are extracted, the well will be circulated with clean fluid to displace residual injection fluids and suspended solids from the wellbore. Wireline logs will be run in accordance with the UIC permit, including corrosion inspection and final vertical saturation. A caliper log will also be run to assess the internal diameter and condition of the casing to verify it can support planned plug placements without risk of washout or collapse.

Plugging Operations

The plugging plan for this well includes the strategic placement of cement plugs throughout the wellbore to isolate any perforated zones, protect underground sources of drinking water (USDW), and secure the surface. At the base of the well, a cement plug will be set to cover the bottom perforation or the base of the injection zone, with a typical plug length of at least 50 feet. A second plug will be placed across the casing shoe or the base of the USDW, extending for approximately 100 feet to ensure zonal isolation.

A surface plug will then be placed across the casing shoe, generally from 50 feet below to 50 feet above the shoe, for a total length of 100 feet. Finally, a shallow surface seal plug will be installed within the top 25 feet of the well to provide a protective barrier near the ground

surface. All cement used will conform to API Class A specifications and may be modified with additives such as retarders or accelerators to accommodate specific well conditions, such as temperature or pressure.

Casing and Surface Equipment

Following the installation of cement plugs, the production casing will be cut approximately five feet below ground level using mechanical cutting tools. All surface equipment, including the wellhead and associated valves, will be removed to eliminate any above-ground hazards. A steel plate will be welded over the exposed casing stub to serve as a permanent barrier. In cases where venting is necessary, a vented cap may be installed instead. The site will then be secured, and appropriate signage will be installed to mark the well location for future reference.

Site Remediation

Upon completion of plugging activities, the well pad and surrounding site will be reclaimed. Grading operations will restore the terrain to its original contours or to a condition compatible with surrounding land use. All debris and materials from the P&A operation will be removed from the location. To encourage environmental recovery, the site will be reseeded with an approved native vegetation mix.

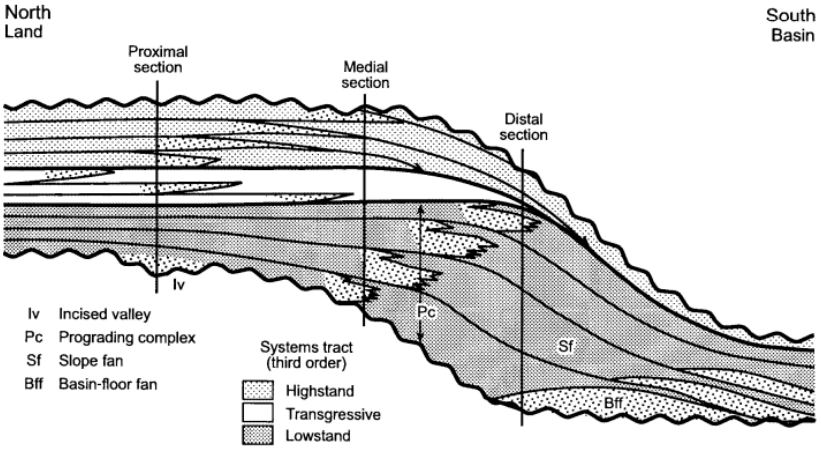
If required by site history or proximity to sensitive receptors, soil and groundwater samples may be collected to test for residual hydrocarbons, chlorides, or other contaminants associated with historical injection activities. These samples will be analyzed and results submitted to the relevant regulatory agencies to confirm that environmental remediation standards have been met.

Final Reporting

Following the successful completion of well plugging and site remediation, a Plug and Abandonment Report will be submitted to the LDNR using Form UIC-24. This report will include details such as the depth and composition of each cement plug, the results of pressure testing, and any supporting logs or diagrams generated during the operation. The report will also include photographic documentation and post-operation site descriptions as necessary. The operator will request that the well status be formally updated to "P&A" in the SONRIS database maintained by the LDNR.

Appendix 9: Basco 6 Supplementary Subsurface Characterization

Basco 6 EW



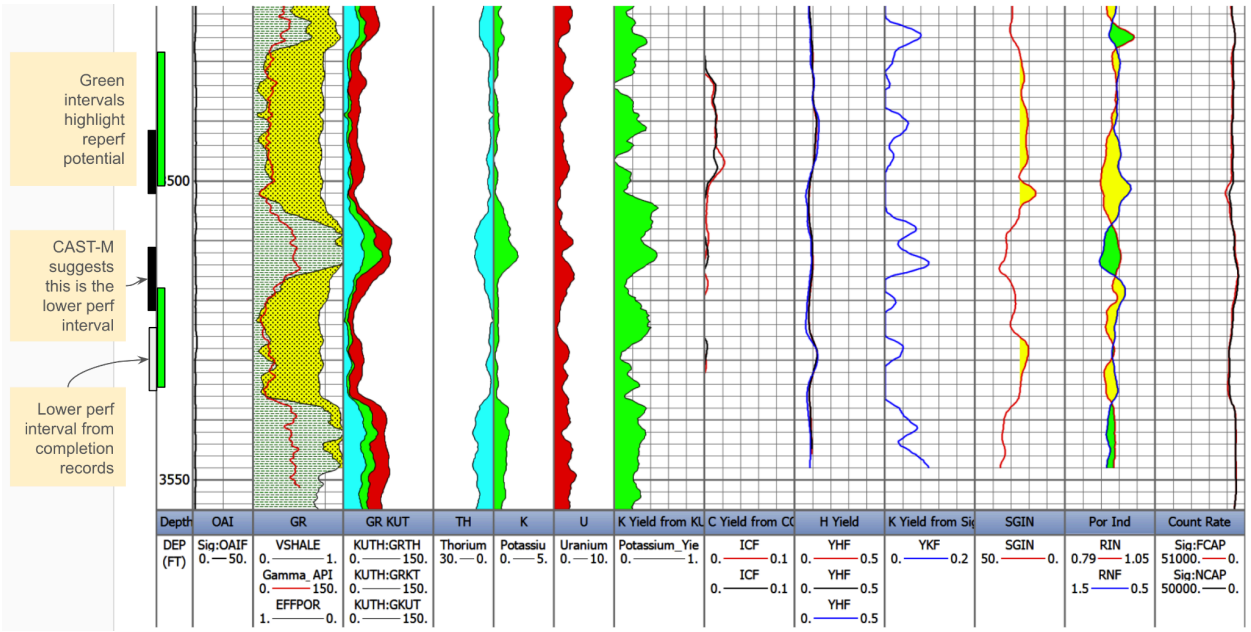
The sequestration interval is within the onshore Lower–Middle Miocene deltaic system sourced by the ancestral Mississippi. At the Basco 6 EW location, the section comprises progradational shoreface, or delta front sands separated by marine mudstones. The overall dip is gently basinward (south), and accommodation increased downdip during Miocene high sedimentation rates, yielding thicker, cleaner sand bodies basinward with intraformational shales that act as baffles [1]. Regionally, Miocene reservoirs in south Louisiana are quartz-rich and moderately texturally mature, with diagenetic overprint (calcite cementation, compaction) increasing downdip [2], consistent with our XRD from Basco fill (≈84% SiO_2 with minor feldspar, calcite, clays, magnetite). This fabric supports good injectivity in the perforated sands and predictable vertical heterogeneity due to shale partings and flooding surfaces.

Confining layer(s) characterization

The perforated Miocene sands are underlain and overlain by thick marine mudstones and intraformational shales that provide low-transmissivity barriers. Log character (GR/spectral GR and density-neutron) indicates laterally continuous shale packages bracketing each sand lobe; these, together with multiple intra-sand baffles, hydraulically isolate the injection interval from the lowest USDW (~1,170’ MD). No free gas saturation is indicated on cased-hole logs, and baseline gases are trace-level only, consistent with aqueous solution gas rather than mobile gas.

Faults and transmissivity

There is no mapped faulting at the well or immediate lease based on available well control; however, seismic control was not licensed. Regionally, Louisiana’s onshore growth-fault trends are best developed toward the coast (e.g., Baton Rouge–Denham Springs–Scotlandville system) and are associated with Miocene–Pliocene strata at ~1,200–2,000 ft in the Baton Rouge aquifer system; inland “central Louisiana” occurrences such as at the Basco 6 EW location are fewer and smaller in throw than coastal growth faults [3, 4]. In the absence of site seismic, we assume no transmissive fault intersecting the interval. Given bio-oil’s density and limited mobility, leakage through a fault system is unlikely.



Pressure margin / MASIP

The Maximum Authorized Surface Injection Pressure (MASIP) is set by the regulator as a function of depth and formation fracture resistance to protect USDWs. In Louisiana (Class V primacy), IMD/Office of Conservation establishes MASIP using formulas or fracture-gradient/step-rate testing; if a recompletion moves the interval shallower, a new MASIP is issued for that depth [5]. Our operations remain well below MASIP, providing a safety margin to avoid fracture initiation or propagation of the confining units.

The Basco 6 SWD's currently perforated interval (black bars, far left) within the Miocene Formation, likely isolated from one another. The carbon:oxygen response, "sigma", and spectral gamma ray values (blue, green and red colorfill) will inform the presence of bio-oil post-injection.

Dissolved Inorganic Carbon (DIC) and Baseline Geochemistry

Formation water chemistry provides a quantitative basis for DIC concentrations in the Basco 6 SWD reservoir interval (see table below). Alkalinity and bicarbonate were measured at 502 mg/L as HCO_3^-, with carbonate below detection (< 2 mg/L) and pH at 7.7. At this pH, the carbonate system is dominated by bicarbonate. Converting the bicarbonate concentration to a carbon-equivalent yields a DIC concentration of approximately 100 mg/L as C.

Complementing this, isotopic analysis of dissolved inorganic carbon reported $\delta^{13}C$ of DIC = +5.4‰ (VPDB). Together, these results establish both concentration and isotopic baseline conditions for dissolved inorganic carbon in the formation fluid.

Baseline hydrocarbon gas composition was also determined by ASTM D1946 analysis of downhole gas samples, with methane ranging from 0.045–1.2% v/v, carbon dioxide at ~0.06–0.08% v/v, and nitrogen and oxygen/argon comprising the balance. These data provide a quantitative hydrocarbon baseline for the reservoir environment and reflect trace amounts of hydrocarbon saturation. Cased hole logging (pulsed neutron) showed no

significant gas saturation, consistent with these results. The concentrations are too low to indicate mobile or volumetrically meaningful hydrocarbons in the pore system.

Matrix	Parameter	Result / Range	Units	Method
Formation fluid	Bicarbonate (HCO_3^-)	502	mg/L	EPA 2320B-2011
Formation fluid	Carbonate (CO_3^{2-})	< 2	mg/L	EPA 2320B-2011
Formation fluid	Alkalinity (as $CaCO_3$)	502	mg/L	EPA 2320B-2011
Formation fluid	pH	7.7	s.u.	EPA 4500H+B
Formation fluid	DIC (as C, inferred)	~100	mg/L	Calculated from HCO_3^-
Formation fluid	$\delta^{13}C$ of DIC	+5.4	‰ VPDB	Isotech Lab (2024)
Formation fluid	δD of water	-16.4	‰ VSMOW	Isotech Lab (2024)
Formation fluid	$\delta^{18}O$ of water	+9.78	‰ VSMOW	Isotech Lab (2024)
Gas produced from wellbore	Methane (CH_4)	0.045 – 1.2	% v/v	ASTM D1946
Gas produced from wellbore	Carbon Dioxide (CO_2)	0.061 – 0.083	% v/v	ASTM D1946
Gas produced from wellbore	Nitrogen (N_2)	77 – 78	% v/v	ASTM D1946
Gas produced from wellbore	Oxygen/Argon (O_2/Ar)	20 – 22	% v/v	ASTM D1946

Attachments

- Halliburton Wireline Services, cased hole logging reservoir evaluation (link)
- Stratum Reservoir (Isotech), formation fluid (solids) isotopes (link)
- Stratum Reservoir (Isotech), formation fluid isotopes (link)
- Green Analytical Laboratories, formation fluid DOC (link)
- Air Technology Laboratories, gas sample analysis (link)

References

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2. S. P. Dutton, R. G. Loucks, and R. J. Day-Stirrat, “Impact of regional variation in detrital mineral composition on reservoir quality in deep to Ultradeep Lower Miocene sandstones, Western Gulf of Mexico," Marine and Petroleum Geology, vol. 35, no. 1, pp. 139–153, Aug. 2012. doi:10.1016/j.marpetgeo.2012.01.006

3. Tsai, F.T.-C., & Li, X. (2013). *Groundwater modeling with geological uncertainty from fault and aquifer architecture in the Baton Rouge area, Louisiana (USA).* Hydrogeology Journal, 21, 959–972. doi:10.1007/s10040-013-1037-5.
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5. U.S. Environmental Protection Agency & Ground Water Protection Council (2024). *Class II UIC Program: Technical and Regulatory Practices.* Final Webinar Slides, November 2024. Available at: https://www.gwpc.org/wp-content/uploads/2024/11/Final-Slides-Class-II-UIC-Webinar.pdf (Accessed: 21 September 2025).